

FORGE: REACTION-GUIDED GENERATIVE DESIGN OF IONIZABLE LIPIDS

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ABSTRACT

Ionizable lipids are central components of lipid nanoparticles for RNA delivery. Their discovery often relies on libraries assembled from predefined reactants, restricting the precursor structures available for design. We introduce FORGE (Flow-matched, Open-ended, Route-resolved Generation and Exploration), which conditions whole-molecule generation on precursor roles, reaction-core positions, transformation order and molecular architecture. Reverse decomposition and exact forward replay verify the implied assembly. One shared model covers 22 assembly families, including multicomponent condensations, repeated additions, ring openings and acylations. On 1,408 TRAIN-derived development requests from one checkpoint, the complete generation and selection procedure achieves 98.5% exact level-one (L1) consistency, with every family above 90%. Exact L1 together with qualified limited design checks holds for 94.0%; 74.6% of requests produce exact assemblies containing a recovered component absent from TRAIN. Structural assessment and synthetic-reference comparisons reveal residual topology defects, uneven component diversity and family-dependent distribution shifts. Direct listings or recursive computational routes to vendor-listed terminal materials support makeability for 43.75% of requests; the separate strict exact-source/current-item dossier criterion closes 1.9%. These development results distinguish assembly consistency, structural quality and computational route evidence. Independent-seed held-out comparisons and independent empirical lipid-reference validation remain unresolved; the reported routes do not establish experimental synthesis or biological activity.

1 INTRODUCTION

Lipid nanoparticles (LNPs) deliver siRNA, mRNA vaccines and genome-editing reagents (Adams et al., 2018; Polack et al., 2020; Baden et al., 2021; Gillmore et al., 2021), with ionizable lipids supporting nucleic-acid complexation, particle assembly and endosomal escape (Cullis & Hope, 2017; Hou et al., 2021). Discovery of these lipids commonly relies on combinatorial synthesis and screening, in which modular precursors are assembled into libraries and the resulting formulations are evaluated (Akinc et al., 2008; Love et al., 2010; Li et al., 2024; Xu et al., 2024). This approach uses established reactions to combine known building blocks, which limits the lipids it can generate.

Generative methods incorporate assembly chemistry at different stages of molecular design. SynNet, RGfN and SynFlowNet retain reaction routes by selecting reactants from a supplied inventory and applying predefined transformations (Gao et al., 2022; Koziarski et al., 2024; Cretu et al., 2025), a strategy also used by lipid-specific synthesis-DAG and tree-search methods (Ou et al., 2024; Zhao et al., 2024). SynCoGen couples precursor design to assembly by jointly generating building-block graphs, reaction assignments and atomic coordinates (Rekesh et al., 2026). Whole-molecule models such as DeFoG and GenMol generate structures without a prescribed precursor inventory (Qin et al., 2025; Lee et al., 2025), leaving assembly and synthesizability to separate assessment (Gao & Coley, 2020).

FORGE learns a conditional distribution over complete lipid graphs from reaction programs that specify roles, reaction-core positions, transformation order and coarse architecture. It receives no precursor identifiers, stored component graphs or fragment tokens. Reverse decomposition and exact forward replay verify each generated assembly. A bounded recursive search then assesses whether

the implied precursors have preparation routes supported by reaction and supplier evidence, returning a computational dossier or an explicit unresolved outcome.

Our contributions are:

1. **Reaction-program-conditioned whole-molecule generation.** We developed a conditional discrete flow that generates complete lipid graphs with precursor identities left open, coupled to reverse decomposition and exact forward replay for assembly verification.
2. **Evaluation across 22 assembly families.** We evaluate one shared model with paired program-identity, readout and training-continuation diagnostics. Family-wise assembly, quality and route outcomes distinguish their contributions. Matched trained-null and component-based generator comparisons, decomposition precision and held-component recovery remain unresolved.
3. **Structural quality and synthesis evidence.** We assess headgroup retention, ring structure, chemical alerts, lipid-reference realism and product and precursor diversity alongside exact assembly. Recursive route assessment records precursor-preparation evidence, terminal-material support and unresolved outcomes for the generated identities, separating L1 consistency from complete synthesis dossiers.

2 PRELIMINARIES AND PROBLEM FORMULATION

2.1 IONIZABLE LIPIDS AND MOLECULAR ARCHITECTURE

Ionizable lipids combine protonatable amine-containing headgroups, linkers and hydrophobic tails (Semple et al., 2010; Jayaraman et al., 2012). Protonation supports nucleic-acid association during acidic formulation and membrane interactions involved in endosomal release (Xu et al., 2024). LNPs commonly combine an ionizable lipid with helper lipids, cholesterol and PEGylated lipids. Lipid structure, stereochemistry and formulation composition affect delivery (Han et al., 2021; Hatit et al., 2023; Cheng et al., 2020); molecular constitution and formulated-particle performance are therefore distinct evaluation levels.

2.2 REACTION FAMILIES AND MODULAR ASSEMBLY

We study 22 registered assembly families (Table 21). These include multicomponent condensations, repeated amine additions, ring-opening reactions, esterification and amidation. Ugi 3-CR combines an amine, aldehyde and isocyanide into an α -amino amide (Xu et al., 2024). Repeated aza-Michael addition attaches acrylate-bearing components to amine sites (Li et al., 2023); repeated reductive amination couples aldehyde-bearing components to amines through condensation and reduction (Xue et al., 2024). The Ugi variant has no carboxylic-acid reactant component and uses an acidic phosphorus catalyst; its ester functionality comes from the aldehyde precursor. Repeated assemblies require both transformation order and participating sites. Precursor-origin boundaries can span functional headgroup, linker and tail regions. These distinctions motivate role-assigned reaction programs (Appendix D.1, Table 21).

2.3 MOLECULAR GRAPHS AND REACTION PROGRAMS

A lipid constitution is a nonempty, connected, simple molecular graph $G = (V, E, \ell_V, \ell_E)$, with atom labels $\ell_V : V \rightarrow \Sigma_{\text{atom}}$ and bond labels $\ell_E : E \rightarrow \Sigma_{\text{bond}}$ in finite vocabularies. Bond order is an edge label. We compare graphs up to label-preserving isomorphism after removing atom-map and stereochemical labels, retaining formal charges and bond orders. We write G for a graph or its constitutional isomorphism class when the distinction is immaterial. The bounded graph space $\mathfrak{G}_{N, K_{\text{cl}}}$ contains these classes with $1 \leq |V| \leq N$ and cycle rank $\beta(G) = |E| - |V| + 1 \leq K_{\text{cl}}$. This is a structural support definition; molecular validity additionally requires valence and sanitization checks. Here $N = N_{\text{max}} = 254$ and $K_{\text{cl}} = 12$.

A reaction program specifies the assembly constraints,

$$P = (c, d, \mathbf{r}, \mathbf{k}, M). \quad (1)$$

Here c identifies a registered assembly chemistry with its admitted transformation sequences, d is the requested depth, r assigns precursor roles to atom positions, and k marks reaction-core positions. The four morphology fields in M record each role’s exterior atom count, junction budget, closure count and core-attachment count. These quantities constrain coarse architecture without specifying precursor identities. Precursor origin and core membership are separate: an atom can belong to a precursor and participate in the core; a template-introduced atom belongs to the core without belonging to a precursor. Appendix A.1 defines the morphology counts.

2.4 EXACT ASSEMBLY, NOVELTY AND ROUTE EVIDENCE

For registered chemistry c , a precursor input $\mathbf{C} \in \mathcal{C}_c$ is an ordered tuple of constitutions, roles and transformation steps. Its registered forward products within the bounded support form $\mathcal{F}_c(\mathbf{C}) \subseteq \mathcal{G}_{N, K_{cl}}$. Reverse search returns a finite set $\mathcal{D}_c(G)$ of precursor inputs and assembly traces $(\mathbf{C}, \tau)$. Exact forward replay retains

$$\hat{\mathcal{E}}_c(G) = \{(\mathbf{C}, \tau) \in \mathcal{D}_c(G) : G \in \mathcal{F}_c(\mathbf{C})\}. \quad (2)$$

Membership uses constitutional identity. A valid graph passes verified exact level-one assembly (L1) when $\hat{\mathcal{E}}_c(G) \neq \emptyset$. Bounded reverse search can miss assemblies, and recovered traces need not match every sampled depth, role-position or morphology field in P . Assembly existence and conditioning adherence are distinct checks, as are decomposition coverage and replay precision (Appendix A.5).

A recovered precursor is component-novel when its constitution is absent from its role’s catalogue. Strong product-level catalogue escape requires *every* exact precursor input to contain an outside component. We measure recovered component novelty, which does not establish that universal claim (Appendix C.2). A complete computational synthesis dossier also requires evidence-qualified precursor preparation (L2) and terminal-material availability (L3). Incomplete assessments remain unresolved or search-censored; these checks do not establish experimental synthesis success or delivery activity.

3 FORGE

3.1 OVERVIEW

FORGE learns the distribution $p_\theta(G | P)$ of complete lipid molecules G given a reaction program P . The model receives no precursor identifiers, stored component graphs or fragment tokens. The program specifies how the molecule is assembled without selecting the precursor structures. FORGE has four main steps (Figure 1):

1. **Specify the reaction program.** Choose assembly chemistry, roles, reaction-core positions, transformation order and morphology, leaving precursor identities open.
2. **Generate the whole graph.** A conditional discrete flow generates atom, bond and topology states while preserving program-fixed coordinates.
3. **Verify exact assembly.** Recover role-assigned precursors and require their forward transform to reconstruct the generated molecule.
4. **Construct precursor routes.** Search for preparation routes to evidence-qualified starting materials and abstain when a complete dossier cannot be constructed.

3.2 CONDITIONING ON REACTION PROGRAMS

An admissible complete program is sampled from the training-fold prior,

$$p_\theta(G, P | c) = p_{\text{prog}}(P | c) p_\theta(G | P). \quad (3)$$

Learned embeddings combine chemistry, depth, precursor role, core status and role-local morphology at each graph position. The adapter determines active positions and fixed or variable coordinate masks. Appendix A.1 specifies the embeddings and morphology counts.

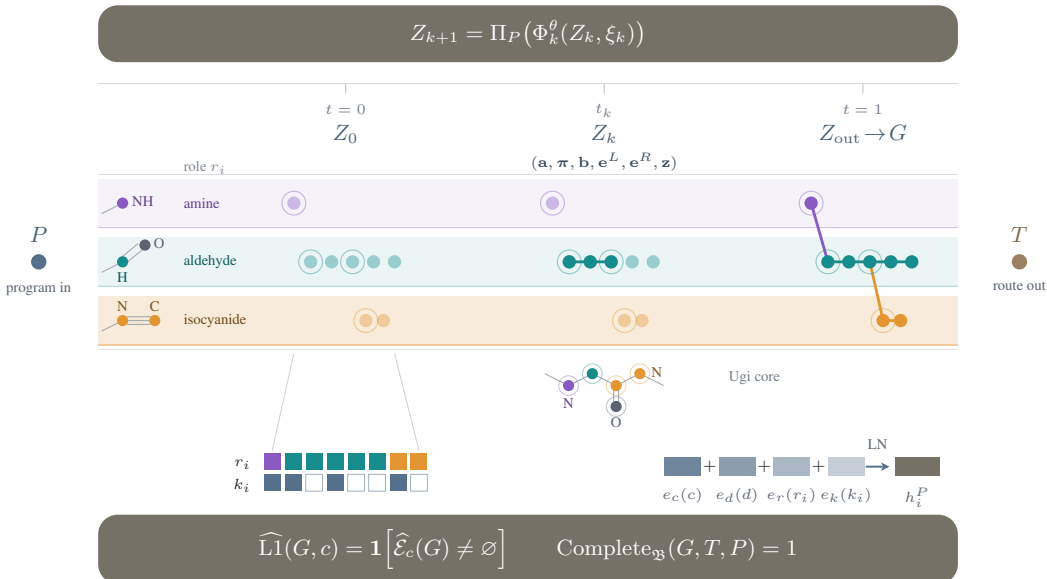


Figure 1: **FORGE generation and route assessment.** A reaction program fixes assembly roles and core coordinates without selecting precursor identities. FORGE generates the whole graph, verifies exact assembly replay, and recursively traces its implied precursors to evidence-qualified materials or an explicit abstention. Colours mark precursor origin; grey denotes the Ugi template-introduced oxygen. Replay and route closure are computational checks, not experimental synthesis evidence.

3.3 GRAPH REPRESENTATION AND DISCRETE FLOW

FORGE represents each molecule as a spanning tree with additional bonds for ring closures. A deterministic encoder maps the molecular graph G and its selected atom-mapped assembly annotation α to

$$X_{\star} := \text{Enc}_P(G, \alpha) = (\mathbf{a}, \boldsymbol{\pi}, \mathbf{b}, \mathbf{e}^L, \mathbf{e}^R, \mathbf{z}), \quad (4)$$

where $\mathbf{a}$ records atom states, $\boldsymbol{\pi}$ identifies each atom’s parent in the tree, and $\mathbf{b}$ records the tree bonds. The arrays $\mathbf{e}^L$ and $\mathbf{e}^R$ identify the endpoints of additional bonds, whose states are stored in $\mathbf{z}$.

The padded state space $\bar{\mathfrak{X}}_{N, K_{\text{cl}}}$ is finite. Its valid encodings form $\mathfrak{X}_{N, K_{\text{cl}}}$: each contains a connected tree and at most K_{cl} distinct closure edges. The decoder Dec reconstructs the graph from a valid encoding and returns $\perp_{\text{dec}}$ for a malformed state. Every graph within the specified atom, bond, size and closure limits has an encoding (Theorem C.3).

During sampling, we also restrict the allowed parents and number of children for each atom. These constraints limit which valid encodings the sampler can use.

For a given P , let $\mathcal{I}(P)$ index the categorical coordinates, each with finite alphabet $\mathcal{X}_i(P)$. Pointer alphabets contain only positions permitted by the layout. The adapter partitions coordinates into fixed positions $\mathcal{F}(P)$ with values $x_{P,i}$ and variable positions $\mathcal{V}(P)$. The set Ω_P contains raw states within the allowed alphabets that satisfy the fixed values. It can contain malformed or chemically invalid states. The source law q_0^P fixes $\mathcal{F}(P)$ and uses independent, full-support marginals $q_{0,i}^P$ on variable alphabets. For a clean endpoint $x_1 \in \Omega_P$, the conditional corruption law is

$$q_{t|1}^P(x \mid x_1) = \mathbf{1}[x_{\mathcal{F}(P)} = x_{P,\mathcal{F}(P)}] \prod_{i \in \mathcal{V}(P)} [(1-t)q_{0,i}^P(x_i) + t\mathbf{1}\{x_i = x_{1,i}\}], \quad 0 \leq t \leq 1. \quad (5)$$

Each coordinate interpolates between its source and clean value. Their conditional joint law is a product, so it generally differs from a linear mixture of the joint source and endpoint laws.

The clean-coordinate predictor $d_{\theta,i}(a \mid x, t, P)$ assigns a probability to each $a \in \mathcal{X}_i(P)$. A continuous-time Markov chain (CTMC) uses these predictions to average conditional one-coordinate

rates,

$$R_{t,i}^\theta(u, v \mid x, P) = \sum_{a \in \mathcal{X}_i(P)} d_{\theta,i}(a \mid x, t, P) R_{t,i}^*(u, v \mid a, P), \quad u \neq v, \quad t < 1. \quad (6)$$

The conditional rate $R_{t,i}^*$ realizes the corresponding factor in Equation 5; Appendix C.3 defines it and proves the path identity. The joint generator permits one coordinate to change at an infinitesimal time. Coordinate posterior marginals therefore suffice to specify its rates, even when endpoint coordinates are statistically dependent.

3.4 PROGRAM-CONDITIONED TRANSFORMER AND TRAINING

A graph Transformer predicts clean coordinates from (X_t, t, P) . Each layer combines graph self-attention with parent, child and closure relation biases and cross-attention to program, depth, role and core tokens. Softly program-routed residual adapters add family-specific parameters. Qualified atom-mapped traces supervise role and core annotations; unresolved source-trace ambiguity excludes a record from semantic training.

The source-balanced training law μ_{bal} covers annotated triples (G, α, P) with equal mass across the 22 qualified families and uniform sampling over unique constitutions within each family, with replacement. No second loss reweighting changes this measure. The model combines masked coordinate cross-entropy, role-balanced chemistry, role/core supervision, and repeat and topology consistency terms. We combined chemistry-task gradients with PCGrad (Yu et al., 2020). Appendix A.3 specifies the full objective and training-time distribution.

To isolate the statistical property of coordinate denoising, define the reference population loss

$$\mathcal{L}_{\text{ref}}(\theta) = \mathbb{E}_{(G, \alpha, P) \sim \mu_{\text{bal}}, t \sim \mathcal{U}(0,1)} \left[\sum_{\substack{X_t \sim q_{t|1}^P(\cdot | X_*) \\ i \in \mathcal{V}(P)}} w_i(P) [-\log d_{\theta,i}(X_{*,i} \mid X_t, t, P)] \right], \quad w_i(P) > 0. \quad (7)$$

The weights depend only on the program; they can normalize coordinate counts without depending on the unknown clean category. This reference loss supports the population analysis. It is not the complete production training objective.

The reference objective has an ideal population interpretation. For a program with positive training mass, let q_1^P be the law of $\text{Enc}_P(G, \alpha)$ under $\mu_{\text{bal}}(\cdot \mid P)$, and let μ_G^P be the corresponding graph law.

Theorem 3.1 (Ideal conditional generation). *Fix such a program P , with registered chemistry c , finite coordinate alphabets and a full-support source on each variable alphabet. Assume every training endpoint belongs to Ω_P , decodes to its source graph, and passes molecular validity and verified exact LI. Assume an unrestricted measurable coordinate predictor attains a population minimum of Equation 7. Starting from q_0^P , exactly simulate the local CTMC of Equation 6 on $0 \leq t < 1$. Then its marginal law converges to q_1^P as $t \uparrow 1$, and*

$$\text{Dec}_\# q_1^P = \mu_G^P, \quad \Pr_{G \sim \mu_G^P} [\text{Valid}(G) = 1, \widehat{\text{LI}}(G, c) = 1] = 1. \quad (8)$$

Here $\text{Dec}_\#$ denotes the distribution obtained by decoding.

Appendix C.3 proves this result by recovering the clean-coordinate posteriors and verifying the CTMC forward equation. The theorem concerns the reference population objective and the limit of exact continuous-time sampling. It establishes neither generalization beyond q_1^P nor correctness of finite training, PCGrad, capped Euler updates or constrained terminal readout.

3.5 SAMPLING AND EXACT ASSEMBLY VERIFICATION

The numerical sampler uses states Z_k , distinct from the ideal CTMC X_t . Starting from $Z_0 \sim q_0^P$, it applies

$$Z_{k+1} = \Pi_P(\Phi_k^\theta(Z_k, \xi_k)), \quad k = 0, \dots, L_{\text{samp}} - 1, \quad (9)$$

where Φ_k^θ is a capped independent-coordinate Euler update, ξ_k contains its random draws and Π_P restores program-fixed coordinates. Fixed coordinates have zero rate and remain constant (Proposition C.6); chemical consistency requires verification. A fresh prediction at $t = 1$ drives the constrained argmax readout $Z_{\text{out}} = H_P^\theta(Z_{L_{\text{samp}}})$, including family-specific core and reactive-site constraints, or a typed failure. The checked inference arm augments candidate graphs with qualified structural proposals and selects under explicit product and precursor diversity constraints. We reassess the final graph and retain raw outputs as a separate arm; Table 9 distinguishes these inference contributions from the learned generator. We then decode, sanitize and apply Equation 2 to each valid graph. Molecular validation, decomposition, exact replay and root ambiguity are recorded separately. Appendix A.4 specifies the numerical operations and their distinction from the ideal CTMC.

3.6 RECURSIVE ROUTE ASSESSMENT

For a molecule with exactly one verified precursor decomposition $(\mathbf{C}_T, \tau_T)$, we search for routes to prepare its precursors within fixed depth and query limits:

$$Y = \text{Assess}_{\mathfrak{B}}(G, P, \mathbf{C}_T, \tau_T) \in \mathfrak{D} \cup \mathfrak{F}_{\mathfrak{B}}. \quad (10)$$

Here $\mathfrak{B}$ is the frozen route contract, $\mathfrak{D}$ the dossier space and $\mathfrak{F}_{\mathfrak{B}}$ its finite typed outcome set. Ambiguous roots cause abstention. A complete dossier requires molecular validity, exact L1, admitted and verified upstream reactions, and available terminal materials. Other outcomes are unresolved or search-censored. Assessment operates on the completed graph without modifying generation. Appendix A.6 and Algorithm 1 specify admission and inference; Appendix C states the formal guarantees and their assumptions.

4 RESULTS

4.1 EVALUATION POPULATIONS AND COMPUTATIONAL BUDGETS

One shared model was trained across 22 families with 402,336 presentations per family (8,851,392 total), using 2,794 optimizer updates and deterministic FP32 execution on eight H100s. Each family participated in 381 updates. The complete execution took 1,377.7 seconds, including preparation and checkpoints (Table 5). The principal development population contains 1,408 TRAIN-derived requests, 64 per family, evaluated using the original checkpoint and the developed assembly, proposal and selection procedure. All requests, including failures, remain in the denominator. This population was used during method development and is not an independent held-out evaluation. Independent training-seed estimates and final held-out results remain [\[independent-seed held-out evaluation\]](#).

Separate paired diagnostics use 352 requests (16 per family), one draw per request and 64 sampling steps. They compare raw flow endpoints, terminal argmax and the bounded D1 completion procedure, without the later full proposal and selection procedure used for the 1,408-request population. These populations and inference budgets are not interchangeable. Means and sample standard deviations across independent fits are not estimable from one original fit and its continuations. Post-hoc rejection cannot increase a proposal’s per-attempt acceptance probability (Proposition C.10); larger candidate pools and changed selection are therefore reported separately from learned-model comparisons.

4.2 SHARED GENERATION ACROSS 22 ASSEMBLY FAMILIES

The selected development outputs achieve exact L1 in 1,387/1,408 requests (98.5%), with all 22 families above 90% (Table 1). The lowest family yield is 58/64 (90.6%) for aryl reductive amination. Exact L1 requires molecular validation, an admitted decomposition and exact forward replay. The equal-family and pooled yields coincide because every family has 64 requests. These counts describe the combined model and inference procedure; the paired raw, null and cyclic results for this same population remain unmeasured. Decomposition coverage, candidate precision, ambiguity and the matched three-family comparison remain [\[independent decomposition and common-program comparisons\]](#). Acceptance checks alone do not provide the precision of all attempted decompositions.

Table 1: **Exact assembly across 22 ionizable-lipid reaction families.** Selected exact-L1 percentages for one original checkpoint and 64 TRAIN-derived development requests per family. Final includes the developed checked assembly, proposal and selection procedure. Dashes denote unavailable paired measurements for this population, not zero. The separate 352-request cyclic intervention is reported in Table 8. Independent-fit uncertainty is not estimated.

Family	Raw L1 (%)	Final L1 (%)	Shared null (%)	Cyclic label (%)	Decomp. cov. / candidate prec.
A3 amine–aldehyde–alkyne	—	95.3	—	—	—
Acid–epoxide diester	—	98.4	—	—	—
AEMA	—	98.4	—	—	—
Aldehyde Ugi 3-CR	—	98.4	—	—	—
Aldehyde Ugi 4-CR	—	100.0	—	—	—
α -Isocyanooester	—	100.0	—	—	—
dihydroimidazole	—	—	—	—	—
Amine alkylation	—	96.9	—	—	—
Amine–epoxide	—	100.0	—	—	—
Aryl reductive amination	—	90.6	—	—	—
Aza-Michael acrylamide	—	100.0	—	—	—
Aza-Michael acrylate	—	100.0	—	—	—
Disulfide Michael	—	98.4	—	—	—
Epoxide O-acylation	—	98.4	—	—	—
iPhos	—	100.0	—	—	—
Ketone–isocyanide amide	—	100.0	—	—	—
Ketone Ugi 4-CR	—	96.9	—	—	—
Maleate	—	100.0	—	—	—
O-esterification	—	100.0	—	—	—
Passerini	—	100.0	—	—	—
Preassembled thiol-yne	—	100.0	—	—	—
Reductive amination	—	95.3	—	—	—
Thiolactone	—	100.0	—	—	—
Equal-family macro	—	98.5	—	—	—
Pooled all requests	—	98.5	—	—	—
Worst-family result	—	90.6	—	—	—

4.3 PROGRAM CONDITIONING AND TRAINING DIAGNOSTICS

On the 352-request paired population, changing only the program identity at inference reduces D1 exact L1 from 140/352 to 105/352, a 9.94 percentage-point decrease. Raw endpoint exact counts are 80 versus 71, and terminal-argmax counts are 91 versus 85. The cyclic intervention retains the true reaction core, roles, layout and depth. It therefore measures sensitivity to the program identity within otherwise correct structural context, not the full effect of synthesis information. The independently trained shared-null contrast remains [\[trained null comparison\]](#).

Additional training did not improve D1 exact yield. With another 31,680 presentations per family, 220 large-batch updates and 2,640 small-batch updates produced 135 and 122 exact outputs, respectively, versus 140 for their common initializer (all /352). Hardware, computation and PCGrad grouping differed, so matched exposure does not imply matched cost. A second experiment compared two 66-update continuations, each adding 9,504 presentations per family. Ordinary categorical and coarse parent-group training gave 136 and 135 exact outputs, respectively. Although coarse training reduced nonconsecutive-parent CAL errors from 1,164 to 1,026 of 7,936, it did not preserve the original generation floors. Neither continuation replaced the original checkpoint (Figure 2; Table 18).

4.4 DECODER CONTRIBUTIONS AND LIMITATIONS

For the original checkpoint on the 352-request diagnostic, raw endpoint, terminal-argmax and D1 readouts yield 80, 91 and 140 exact products, respectively. A fixed fallback among these already

computed candidates increases D1 exact yield to 145/352 and limited-design yield from 111 to 116/352, without new model or chemistry calls. This is a selection gain. In a separate paired 192-request repair diagnostic covering A3 and the two Ugi 4-CR families, exact L1 remains 187/192 while limited-design passes increase from 140 to 146. Additional attachment proposals change three structures but add no design passes. These repairs were not substituted into the 1,408-request routing population. Table 9 retains the separate denominators and adverse results. Matched whole-graph versus factorized capacity, cross-attention and gradient-control contrasts remain [architecture comparisons].

4.5 PRECURSOR NOVELTY AND DIVERSITY

The selected population contains 1,407 distinct products among 1,408 requests. Product novelty relative to TRAIN occurs in 1,216 observations (1,215 distinct identities). Exact products with at least one globally TRAIN-novel recovered component occur in 1,050/1,408 requests (74.6%), representing 1,049 distinct products. This gives 745.0 distinct component-novel products per 1,000 requests. Role-relative novelty occurs in 1,081 observations (1,080 distinct products). Component counts remain uneven: the pooled isocyanide role contains 29 identities among 190 observations, with Shannon and inverse-Simpson effective counts of 10.21 and 6.09 (Table 15). High product uniqueness therefore does not ensure broad component use.

TRAIN-relative novelty does not establish catalogue escape, recovery of deliberately withheld components or source-disjoint generalization. Only 15/2,059 guarded CAL molecules are wholly component-disjoint. The newer 1,859-molecule exact-source and representation-qualified CAL union contains 12 such cases, all in two aza-Michael families; it differs from the older 1,821-reference distribution cohort below. Final held-component and source-disjoint generation results remain [generalization evaluation] (Table 16).

4.6 STRUCTURAL QUALITY AND AGREEMENT WITH LIPID REFERENCES

All 1,408 selected structures are valid and connected, but only 1,324/1,408 (94.0%) satisfy exact L1 together with all applicable limited design checks (Table 2). There are 69 failures of the combined ring predicate. The scoped retained-head predicate fails in 5/64 aldehyde Ugi 4-CR requests; the other 1,344 structures are outside its qualified scope and are not counted as passes. Chemistry-context review is flagged for 41/1,408 structures (2.9%). The overlapping flags include hydrated or hemiacetal carbons (24), peroxide groups (8), carbons bearing at least three oxygens (6), other allenes (5), a small unsaturated ring (1) and a heteroatom cumulene (1). Limited design without context flags holds for 1,292/1,408 (91.8%).

Table 2: **Structural quality of the selected development population.** Counts retain all 1,408 requests except the explicitly scoped head predicate. The limited-design conjunction includes exact L1 and the qualified chemical, head and ring checks; it is not a broad empirical lipid-realism assessment. Dashes retain unavailable paired or separately resolved metrics.

Metric	Selected count	(%)
Valid molecules / requests	1,408/1,408	100.0
Connected molecules / requests	1,408/1,408	100.0
Exact L1 + qualified limited design / requests	1,324/1,408	94.0
Limited design without context flags / requests	1,292/1,408	91.8
Justified narrow chemical alerts / requests	0/1,408	0.0
Chemistry-context flags / requests	41/1,408	2.9
Required head retained / scoped requests	59/64	92.2
Combined ring predicate / requests	1,339/1,408	95.1
Separate role-local cycle allocation	—	—
Separate requested ring-size agreement	—	—
TRAIN atom-environment coverage	—	—
Ring-system TRAIN support	—	—
Exact L1 + product/component features observed in TRAIN	974/1,408	69.2

Source-positive controls pass the narrow chemical-alert screen in 29/29 cases, but cover only two families and two studies, with zero individually characterized products in that control set. The other 20 families lack these controls. Absence of an alert does not establish stability, pKa or delivery activity. Definitions and family-level counts: Tables 10–13.

The distribution comparison uses 1,821 exact-source-qualified *synthetic CAL structures*; these are not an independent empirical lipid reference. Equal-family fingerprint precision is 56.9%, coverage is 49.1%, and mean within-family ECFP4 diversity is 0.552 (Table 3). Precision varies from 1/64 for ketone Ugi 4-CR to 64/64 for aza-Michael acrylate. Aryl reductive amination has 2/63 supported distinct outputs. These failures remain visible despite high exact L1. A3 has 36/64 supported outputs; its normalized Wasserstein distances for cLogP and rotatable bonds are 1.133 and 1.658. Ketone Ugi 4-CR has an sp^3 -carbon fraction distance of 1.082. All 24 descriptor panels are reported by family (Table 14). Independent empirical-reference comparisons, grouped two-sample classification and blinded structural review remain unmeasured.

Table 3: **Distributional agreement with synthetic CAL references.** Values are equal-family means of 22 within-family statistics, not pooled molecular statistics or independent-seed means. Each arm has 64 requests per family; fingerprint calculations deduplicate connected products within family. The predecessor is the saved selection preceding the current context-preserving selection, not raw model output. TRAIN controls are a reference diagnostic, not independent quality evidence.

Metric	Selected	Predecessor	TRAIN control
Fingerprint precision (%)	56.9	57.1	75.1
Fingerprint coverage (%)	49.1	49.4	59.0
Fingerprint recall (%)	82.1	81.4	88.4
Distinct supported products / requests (%)	56.9	57.1	75.1
Nearest-reference Tanimoto similarity	0.638	0.641	0.693
Within-family ECFP4 diversity	0.552	0.550	0.529
Descriptor distances (24 features, per family)	Table 14		
Independent descriptor precision / coverage	—	—	—
Independent source-grouped classifier AUC	—	—	—

ECFP4 uses Morgan radius 2 and 2,048 bits; neighborhood metrics use five neighbors. Similarity to this synthetic reference is not a probability of chemical realism. Family counts, estimators and limitations: Tables 12 and 13.

4.7 RECURSIVE COMPUTATIONAL MAKEABILITY AND STRICT DOSSIERS

The primary route criterion requires every assembly component to be directly vendor-listed or to have a complete computational path to vendor-listed terminal materials. Planner-generated paths and forward-applied published transforms may contribute without exact-substrate experimental precedent.

On the unchanged 1,408-product population, 616/1,408 (43.75%) meet this criterion, including 606 satisfying limited design and 604 also free of context flags (Table 17). L2 readiness is 1,189/1,408 (84.4%); it does not require terminal listing closure. Direct-only L3 covers 119/1,408 (8.5%). Strict exact-source/current-item dossiers remain a separate secondary outcome, closing 27/1,408 requests (1.9%).

Makeability ranges from 0/64 for reductive amination and 2/64 for ketone Ugi 4-CR to 58/64 for ketone-isocyanide amide. The complete denominator includes 3,723 required component branches. The latest listing evidence is dated September 25, 2026 (local date), with a 30-day policy window. A vendor-directory listing does not establish stock, price or an experimentally executed route. The primary count rose from 581 to 616 through additional evidence, without changing molecules, L1 checks or structural predicates. Missing listings and bounded route searches remain unresolved, not evidence of unsynthesizability. Neither outcome establishes experimental synthesis or delivery.

5 DISCUSSION

FORGE combines whole-molecule generation with checked assembly across 22 reaction families. The 98.5% exact-L1 result describes one checkpoint together with its developed inference procedure on TRAIN-derived requests. A paired cyclic program-ID intervention reduces exact yield in a separate, smaller inference diagnostic, establishing sensitivity to identity conditioning within correct structural context. Independently trained null and factorized controls remain necessary to attribute gains to the full conditioning and joint-generation architecture. The unsuccessful continuation experiments further show that lower teacher-forced parent error does not by itself predict improved completed molecules.

The generation procedure produces exact assemblies containing TRAIN-novel components in 1,050/1,408 requests, but novelty and product uniqueness do not guarantee broad component use. The concentration of the isocyanide role illustrates this distinction. TRAIN-relative novelty also differs from recovery of withheld components or exclusion from every catalogue-reachable assembly; those comparisons remain unresolved.

Exact assembly is one level of evidence. Limited design passes decrease to 94.0%, and to 91.8% when chemistry-context flags are also excluded. Ketone Ugi 4-CR, aldehyde Ugi 4-CR and A3 retain substantial topology or distributional deficiencies. The synthetic CAL reference reveals these shifts but cannot establish independent lipid realism. Source-positive structural controls cover only two families, leaving broader control coverage and blinded review necessary. TRAIN-derived priors and support checks cannot independently validate the same features they helped select.

Computational makeability closes 616/1,408 products, while strict source-supported dossiers close 27/1,408. The distinction separates computational route construction and dated vendor-directory listings from exact experimental precedent and current-item evidence. An unresolved branch is not a negative synthesis result. These computational checks do not establish stock availability, experimental synthesis success, ionization or delivery performance. Independent fits, held-out requests, matched inference budgets and qualified external reference populations are required to assess how the observed results generalize.

AI USE STATEMENT

Generative AI tools assisted with conceptual and methodological development, software implementation and debugging, experimental design and diagnostic analysis, interpretation of computational results, literature search and summarization, and drafting and editing the manuscript. The authors are responsible for the final code, reported results, citations and conclusions, including material produced with AI assistance.

ETHICS STATEMENT

This work evaluates ionizable-lipid generation and synthesis evidence computationally. Molecular design systems can be misused. Chemical validity, exact assembly, synthesis-route evidence and predicted properties do not establish safety or biological efficacy. Generated structures require appropriate evaluation before experimental use.

REPRODUCIBILITY STATEMENT

The Methods and Appendix specify the molecular representation, conditional discrete flow, numerical sampler, exact assembly verification, route assessment, benchmark protocols and evaluation metrics. Appendix A.8 records the source, configuration, split, checkpoint and evaluation artifacts. Per-seed results appear in Table 7; assumptions and proofs appear in Appendix C.

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A METHOD AND IMPLEMENTATION DETAILS

This appendix specifies program and graph encoding, model training, sampling, verification and route assessment. Algorithm 1 combines the inference steps.

A.1 REACTION-PROGRAM ENCODING

The program tuple in Equation 1 uses the ordered atom-position set $\mathcal{I}_{\text{atom}}(P) = \{1, \dots, n(P)\}$. The arrays $\mathbf{r}$ and $\mathbf{k}$ assign precursor roles and core positions to these atoms. This atom-position index differs from $\mathcal{I}(P)$, which indexes all six families of categorical coordinates.

Morphology is computed from the selected atom-mapped trace and its sparse serialization. For role r , let V_r^{ext} contain its exterior atoms, excluding reaction-core atoms, and let $o_r(v)$ count the same-role exterior children of v in the serialized tree. The four fields are

$$M_r = (n_r, j_r, \gamma_r, a_r), \quad n_r = |V_r^{\text{ext}}|, \quad j_r = \sum_{v \in V_r^{\text{ext}}} \max\{o_r(v) - 1, 0\}. \quad (11)$$

Here γ_r counts closure edges with both endpoints in V_r^{ext} , and a_r counts same-role bonds between exterior and core atoms. When the serialized tree restricts to a spanning forest of the exterior subgraph, γ_r equals that subgraph’s cycle rank. The junction budget is a tree-branching count; it does not count degree contributed by closure bonds. Consequently these fields depend on the selected annotation and serialization, not on the unannotated graph alone. Fixed-coordinate equality alone does not establish agreement with every morphology field. Positions without precursor origin use the designated absent morphology state. Integer morphology values are encoded with an offset of one, reserving zero for that absent state.

The training-fold program prior is defined in Section 3.2. The memory encoder uses separate program-identity and depth tokens, $m_c = e_c(c) + e_{\text{type}}(0)$ and $m_d = e_d(d) + e_{\text{type}}(1)$, together with one token per active atom position,

$$m_i = e_r(r_i) + e_k(k_i) + e_{\text{pos}}(i) + e_{\text{rep}}(g_i) + e_{\text{comp}}(u_i) + \sum_{v=1}^4 e_{M,v}(M_{r_i,v}) + e_{\text{type}}(2). \quad (12)$$

Here g_i is the repeat-group state and u_i the position within the precursor-origin component; they are structural conditioning variables, not precursor identity labels. All embeddings share the model hidden dimension. Layer normalization precedes memory self-attention, followed by residual attention and feed-forward updates. The resulting program, depth and atom-position tokens provide the memory for cross-attention in the graph denoiser. The morphology fields determine active positions and fixed or variable masks. Sampling from the prior specifies these constraints; generation assigns the atom, bond and topology states within each role.

A chemistry-specific adapter determines the precursor roles, reaction core, reverse decomposition and forward transform associated with c . Its sampling layout determines active positions and fixed or variable coordinates; the embeddings condition neural prediction on the remaining program fields.

A.2 GRAPH SERIALIZATION AND SOURCE ANNOTATIONS

A spanning-tree representation with at most twelve residual closure edges preserves the full 254-heavy-atom support without an $O(N_{\text{max}}^2)$ edge state. The discrete flow generates the atom, bond and topology states in this representation. Let $\alpha = (\mathbf{C}, \tau, \mathbf{r}, \mathbf{k})$ contain the selected exact precursor tuple, registered assembly trace, atom-wise precursor-role assignment and reaction-core annotation. The deterministic program-aware encoder

$$\text{Enc}_P : \{(G, \alpha) : G \in \mathfrak{G}_{N_{\text{max}}, K_{\text{cl}}} \text{ and } \alpha \text{ is admitted for } P\} \longrightarrow \mathfrak{X}_{N_{\text{max}}, K_{\text{cl}}} \quad (13)$$

constructs the categorical graph state

$$X_\star := \text{Enc}_P(G, \alpha) = (\mathbf{a}, \boldsymbol{\pi}, \mathbf{b}, \mathbf{e}^L, \mathbf{e}^R, \mathbf{z}) \quad (14)$$

where $\mathbf{a}$ contains atom states, $\boldsymbol{\pi}$ parent pointers, $\mathbf{b}$ spanning-tree bond states, $\mathbf{e}^L$ and $\mathbf{e}^R$ closure-endpoint pointers, and $\mathbf{z}$ closure-bond states.

The encoder first partitions the atoms into connected precursor-origin components under α . It selects as root the reaction-core atom with the smallest tie-broken canonical RDKit rank, and orders the origin-component graph by breadth-first traversal from that root component, breaking ties by role index, component canonical ranks and original atom index. Within each component it uses a deterministic depth-first preorder from the selected attachment atom, with neighbors ordered by the same canonical ranks. The traversed bonds define the spanning tree. Every residual bond is represented once as an ordered endpoint pair with its bond label, and residual triples are sorted lexicographically. Role, core, fixed-coordinate and morphology arrays follow the same atom order. Each admitted source record supplies a selected qualified exact assembly trace and its precursor-origin assignment. Alternative source-role bindings remain explicit annotations; they do not receive additional graph sampling mass when their product constitutions coincide. Thus $\text{Dec}(\text{Enc}_P(G, \alpha)) \cong G$ for every admitted record.

The raw padded categorical product space $\bar{\mathcal{X}}_{N_{\max}, K_{\text{cl}}}$ also contains malformed states. Its well-formed subset $\mathcal{X}_{N_{\max}, K_{\text{cl}}}$ enforces an active node prefix, earlier-parent pointers, distinct valid closure endpoints, no duplicate edges, and null values in unused closure slots. The total decoder, including its failure value, is typed as

$$\text{Dec} : \bar{\mathcal{X}}_{N_{\max}, K_{\text{cl}}} \longrightarrow \mathcal{G}_{N_{\max}, K_{\text{cl}}} \cup \{\perp_{\text{dec}}\}. \quad (15)$$

For this study, $N_{\max} = 254$ and $K_{\text{cl}} = 12$. The atom vocabulary includes qualified bromine states; bond states encode single, double and triple bonds, with aromatic structures represented through Kekulé bond assignments. These bounds specify architectural support. Reaction consistency, nonzero probability under a trained model and synthesis-program completeness require separate conditions.

The structural decoder specifies graph reconstruction from an already chosen state. The production terminal readout additionally chooses that state from neural predictions (Appendix A.4). These are separate operations.

The chemistry adapter marks any atom, pointer, or bond states fixed by P . FORGE holds these states constant while corrupting the remaining states along the linear probability path and predicting their clean values from (X_t, t, P) . Let Ω_P be the subset of the raw state space satisfying the allowed coordinate alphabets and fixed values, and define the admissible program set

$$\mathcal{P}_{\text{adm}} = \{P : \mathcal{X}_{N_{\max}, K_{\text{cl}}} \cap \Omega_P \neq \emptyset\}. \quad (16)$$

The morphology prior is normalized only on $\mathcal{P}_{\text{adm}}$. Each chemistry adapter is required to satisfy the program-compatible encoding property: every admitted annotated record obeys $\text{Enc}_P(G, \alpha) \in \Omega_P$. Corpus admission requires this property, an exact constitutional encode–decode round trip and source-consistent origin/core assignments. Records failing admission are excluded. The frozen split is [split manifest]; noise marginals, program priors and motif references are fitted only to its training partition. These checks do not establish the property for unregistered chemistry.

A.3 TRANSFORMER ARCHITECTURE AND TRAINING OBJECTIVES

The denoiser is a graph Transformer over the complete sparse state. Each layer applies graph self-attention with relation biases for parent, child and closure edges, then cross-attends to memory tokens representing program identity, program depth, precursor role and reaction-core position. Cross-attention is repeated in every layer so that reaction semantics can influence both local state updates and long-range coordination among precursor-derived regions. The backbone has six layers, hidden dimension 192, eight attention heads and dropout 0.1. Each routed adapter mixes three experts with bottleneck dimension 64 using weights computed from the program summary. Additional routed adapters feed atom/parent-bond predictions and closure predictions. Auxiliary heads predict precursor roles, core positions and exterior child counts. The shared backbone receives repeat-group, component-position and role-morphology conditioning.

Training assigns equal mass to the 22 admitted formal families and uniform mass to qualified unique product constitutions within each family. Sampling is with replacement; multiple source bindings do not multiply a constitution’s probability, and sampling weights are not applied again to the loss. Each optimizer update contains 3,168 examples from three distinct families, with 1,056 examples per selected family. A seeded permutation defines a balanced 22-update cycle in which every family appears three times. Family batches are sharded across eight H100 GPUs, and their coordinate reductions are accumulated before family-gradient projection. Padding is trimmed to the longest molecule within a family batch while retaining the full closure support.

The optimizer is AdamW with learning rate 10^{-4} , weight decay 5×10^{-5} , $(\beta_1, \beta_2) = (0.9, 0.999)$, $\epsilon = 10^{-8}$ and global gradient-norm clipping at 1. The original fit uses 2,794 updates with seed 2026092401 and its fixed final checkpoint. Independent replication seeds remain [independent training seeds]. Model and gradient computations use deterministic FP32 arithmetic. Checkpoints retain model, optimizer, random-number streams and the committed sampler state; uncommitted prefetch work is discarded on restart. Training times are sampled uniformly, $t \sim \mathcal{U}(0, 1)$, without endpoint clipping.

In the task losses below, c indexes a formal family; multiple registered program variants in that family share its task reduction. For formal family c and state family $q \in \mathcal{J} = \{\mathbf{a}, \boldsymbol{\pi}, \mathbf{b}, \mathbf{e}^L, \mathbf{e}^R, \mathbf{z}\}$, let $\mathcal{S}_{c,q}$ be the selected variable coordinates across that family’s minibatch. With coordinate loss $\ell_i = -\log d_{\theta,i}(X_{\star,i} \mid X_t, t, P)$, the pooled reduction is

$$L_{c,q}^{\text{pool}} = \frac{\sum_{i \in \mathcal{S}_{c,q}} \ell_i}{\max\{1, |\mathcal{S}_{c,q}|\}}. \quad (17)$$

Coordinates here include their example index. Chemistry states (atoms, parent bonds and closure bonds) additionally use equal mass over roles present in the selected coordinates. If $\mathcal{S}_{c,q,r}$ contains coordinates assigned to role r and $\mathcal{R}_{c,q} = \{r : |\mathcal{S}_{c,q,r}| > 0\}$, this reduction is

$$L_{c,q}^{\text{role}} = \frac{1}{|\mathcal{R}_{c,q}|} \sum_{r \in \mathcal{R}_{c,q}} \frac{\sum_{i \in \mathcal{S}_{c,q,r}} \ell_i}{|\mathcal{S}_{c,q,r}|}, \quad (18)$$

with zero loss if no role is present. Atom and parent-bond groups use their position’s role; closure-bond groups use the role at the clean left endpoint. Pointer losses use the pooled reduction and mask logits to the permitted layout positions. The flow term $L_{c,\text{flow}}$ sums these six reduced losses, with unit weights.

For each family, the task loss is

$$L_c = L_{c,\text{flow}} + 0.25L_{c,\text{role}} + 0.25L_{c,\text{core}} + 0.25L_{c,\text{repeat}} + L_{c,\text{child}} + 0.5L_{c,\text{junction}} + L_{c,\text{chem|top}}. \quad (19)$$

The role and core terms are cross-entropies averaged equally over present target states. The repeat term averages squared differences between predicted atom distributions, and separately parent-bond distributions, over exact retained-fragment correspondences in repeated precursor instances; it sums the two means. Atom correspondence requires a unique labelled fragment match that preserves core annotations. Parent-bond correspondence matches both edge endpoints rather than equating serialized child positions. Unequal fragments or ambiguous matches contribute no repeat pairs; these target-derived alignments supervise the loss and are not supplied at inference. The child term is state-balanced cross-entropy for same-component exterior child counts in $\{0, 1, 2, 3\}$. The junction term applies smooth-L1 loss to predicted and target role-local junction totals, each divided by the number of exterior atoms in that role, then averages over eligible examples within a role and equally across present roles. Predicted junction totals sum the expected child-count contribution $\max\{o - 1, 0\}$ across exterior atoms. The topology-conditioned chemistry term repeats the role-balanced atom and bond losses in a second forward pass with clean parent and closure endpoints and corrupted atom and bond states. Each ablation changes only the declared objective or architecture terms under its matched training contract.

PCGrad combines the task gradients in integer formal-family-state order. For $g_c = \nabla_{\theta} L_c$, it initializes $\tilde{g}_c = g_c$ and, for each other task h in that order, applies

$$\tilde{g}_c \leftarrow \tilde{g}_c - \mathbf{1}\{\langle \tilde{g}_c, g_h \rangle < 0\} \frac{\langle \tilde{g}_c, g_h \rangle}{\max\{\|g_h\|_2^2, 10^{-12}\}} g_h \quad (20)$$

on parameter coordinates available to both tasks. Gradients are not norm-normalized, and the optimizer receives the arithmetic mean of the projected task gradients. This update is not generally the gradient of a scalar sum of task losses. No population optimality claim is made for the full procedure. In particular, target-dependent group reductions and the auxiliary objectives require analysis beyond Proposition C.7.

Layerwise cross-attention, auxiliary losses, routed adapters and gradient-conflict control are separate interventions in the architecture comparison (Table 18).

A.4 NUMERICAL SAMPLING AND TERMINAL DECODING

The saved diagnostic sampler uses 64 steps, denoted L_{samp} . Its strictly positive atom and bond source marginals are indexed by registered program and precursor role and fitted to TRAIN data. Pointer marginals are restricted and normalized on their allowed layout positions. The finite sampler evaluates the denoiser at $t_k = k/L_{\text{samp}}$, for $k = 0, \dots, L_{\text{samp}} - 1$, and uses $\Delta t = 1/L_{\text{samp}}$. Conditional on $Z_k = x$, it draws one clean category $A_{k,i} \sim d_{\theta,i}(\cdot | x, t_k, P)$ independently for each variable coordinate. Write $s_{t,i}^P(u | a) = (1 - t)q_{0,i}^P(u) + t\mathbf{1}\{u = a\}$ for its conditional path factor, $\dot{s}_i^P(u | a) = \mathbf{1}\{u = a\} - q_{0,i}^P(u)$ for the derivative, and $[z]_+ = \max\{z, 0\}$. The code uses the conditional off-diagonal rate

$$R_{t,i}^{*,\varepsilon}(u, v | a, P) = \frac{[\dot{s}_i^P(v | a) - \dot{s}_i^P(u | a)]_+}{|\mathcal{X}_i(P)| \max\{s_{t,i}^P(u | a), \varepsilon\}}, \quad u \neq v, \quad \varepsilon = 10^{-8}. \quad (21)$$

Full source support is required only on the allowed coordinate alphabet, including the masked candidate set for a pointer. Thus that alphabet is the support for every $t_k < 1$. The denominator clamp in Equation 21 is a numerical safeguard absent from the ideal rate in Appendix C.3.

For the sampled endpoint $a = A_{k,i}$, let $h_{k,i} = \Delta t \sum_{v \neq x_i} R_{t_k,i}^{*,\varepsilon}(x_i, v | a, P)$ and set $\kappa_{k,i} = \min\{1, 0.999/h_{k,i}\}$ when $h_{k,i} > 0$, with $\kappa_{k,i} = 1$ otherwise. The implemented coordinate kernel is

$$\hat{K}_{k,i}(v | x, a) = \begin{cases} \kappa_{k,i} \Delta t R_{t_k,i}^{*,\varepsilon}(x_i, v | a, P), & v \neq x_i, \\ 1 - \kappa_{k,i} h_{k,i}, & v = x_i. \end{cases} \quad (22)$$

Its off-diagonal masses sum to at most 0.999; the diagonal is the remaining mass. Consequently all entries are nonnegative and each row sums to one. Coordinates are drawn independently from these kernels, followed by the projection Π_P .

Without the clamp or cap, and when every conditional Euler row is nonnegative, averaging over $A_{k,i}$ gives the Euler kernel built from Equation 6 exactly. One sampled endpoint per coordinate therefore does not by itself bias that uncapped one-step law. The nonlinear cap generally changes the averaged kernel. Independent coordinate updates also permit simultaneous finite-step changes, so their product is a numerical approximation to the local CTMC, not its exact transition kernel. No discretization error bound is asserted near the terminal singularity.

After all steps, the model evaluates $(Z_{L_{\text{samp}}}, 1, P)$ once more. The production terminal readout H_P^θ selects coordinate values using constrained argmax, restores fixed states and returns either a raw state or a typed terminal failure, denoted collectively by $\perp_{\text{term}}$. The atom-aware readout first chooses atom states subject to unavoidable incident-bond demands from the fixed core and non-root tree attachments; subsequent variable bonds must fit those atom capacities. It abstains when this bounded assignment cannot be completed. It does not exchange an element for a higher-valence element merely to accommodate a chosen topology. The registered core condition reserves the required bond and hydrogen states for each program. In the AGILE-type Ugi 3-CR, these include one hydrogen on the aldehyde-derived α carbon and one on the isocyanide-derived amide nitrogen; these conditions are not imposed on other Ugi variants. The selected state is then reconstructed and checked for connectedness, valence and sanitization. A readout failure retains its detailed reason in the ledger; for graph-law notation we extend $\text{Dec}(\perp_{\text{term}}) = \perp_{\text{dec}}$. If $q_{\theta,\text{out}}^P$ is the raw law after terminal readout, including failure, then

$$p_\theta(\cdot | P) = \text{Dec}_\# q_{\theta,\text{out}}^P, \quad p_\theta(G | P) = \sum_{z: \text{Dec}(z)=G} q_{\theta,\text{out}}^P(z). \quad (23)$$

This law retains structural failures and chemically invalid graphs before the validity and L1 filters; it is not conditioned on acceptance. The ideal theorem does not include the learned $t = 1$ readout, which is unconstrained by a loss integrated over $t \in (0, 1)$. Decoder and source-prior comparisons use their declared fixed-weight or retrained contrasts (Table 9).

A.4.1 CHECKED CONSTRUCTION AND SELECTION

Let Π specify a bounded inference policy: the draw count, ordered proposal operators, TRAIN references, search limits, source verifier, selection objective and selection-group size. Its version is

[inference policy]; the draw count D is [draws per request], the stage caps are [proposal and search caps], and the selection-group size is [selection batch size]. For request b , retain all raw trajectories and form

$$\mathcal{C}_b = \{G_{b,d}^{\text{raw}} : 1 \leq d \leq D\} \cup \mathcal{R}_{\Pi}(P_b, \{Z_{b,d}^{\text{out}}, \lambda_{b,d}\}_{d=1}^D), \quad (24)$$

where $\lambda_{b,d}$ denotes saved terminal predictions and $\mathcal{R}_{\Pi}$ is the bounded construction map. Original graphs, failed trajectories, rejected proposals and exceptions remain in the attempt ledger. A request is not removed when its candidate set has no eligible graph.

Construction can tie repeated components, restore source-required component constraints, propose qualified retained-head sites, and assign TRAIN-supported joint atom/bond or ring-system motifs. The operators act on generated atom slots while preserving the declared atom budget, precursor origins and fixed reaction core. Atom relabelling and topology edits are recorded separately. Ring repair is applied after component and head construction, and all checks are evaluated again on the final graph. Repeated arms receive tied edits when their established correspondences permit this; every proposed product must still pass the unchanged source verifier. A TRAIN motif is an explicit inference prior and does not itself certify chemical plausibility or a complete route. An unsupported or context-dependent ring fragment is unassessed rather than assigned a fabricated known identity. Graph cycle rank, ring systems, SSSR rings and serialization-dependent fundamental cycles remain distinct quantities.

The selector maps the request-indexed pools to at most one graph per request,

$$(G_1^{\text{sel}}, \dots, G_B^{\text{sel}}) = \mathcal{S}_{\Pi}((P_b, \mathcal{C}_b)_{b=1}^B), \quad G_b^{\text{sel}} \in \mathcal{C}_b \cup \{\perp_{\text{dec}}\}. \quad (25)$$

Source-exact eligibility, applicable design predicates and product/role-component diversity and novelty constraints are explicit parts of Π . A combined design pass requires every applicable qualified predicate to be assessed and pass; unassessed checks cannot establish that endpoint. The selection objective and constraint reference are [selector objective and floors]. Each output retains its request, draw and proposal lineage. Raw and selected yields both use the original B requests as denominator, while cost accounting includes BD trajectories, all construction attempts, source checks and selection work.

The selected joint law is the pushforward of the raw trajectory law through $\mathcal{R}_{\Pi}$ and $\mathcal{S}_{\Pi}$. It is distinct from Equation 23 and can couple different requests. The ideal flow theorem does not certify these inference maps; independent-draw accepted-count formulas apply only when their independence assumptions hold. A graph-changing repair changes the proposal distribution, whereas rejecting completed graphs alone does not.

A.5 EXACT ASSEMBLY VERIFICATION

For the chemistry c specified by P , FORGE decomposes each valid graph into role-assigned precursors and applies the registered forward transform. Exact replay checks assembly existence under c ; it does not test full agreement with the sampled conditioning layout.

Definition A.1 (Exact assembly and recovered witnesses). For the registered forward operator $\mathcal{F}_c$ on precursor inputs $\mathcal{C}_c$, define

$$\mathcal{E}_c^*(G) = \{\mathbf{C} \in \mathcal{C}_c : G \in \mathcal{F}_c(\mathbf{C})\}, \quad \text{L1}^*(G, c) = \mathbf{1}\{\mathcal{E}_c^*(G) \neq \emptyset\}. \quad (26)$$

The bounded reverse procedure returns candidate pairs $(\mathbf{C}, \tau)$ in $\mathcal{D}_c(G)$. Equation 2 retains the pairs passing exact forward replay, and

$$\widehat{\text{L1}}(G, c) = \mathbf{1}\{\widehat{\mathcal{E}}_c(G) \neq \emptyset\}. \quad (27)$$

Both predicates describe assembly existence; molecular validity is checked separately.

The set $\mathcal{E}_c^*(G)$ contains precursor inputs, whereas $\widehat{\mathcal{E}}_c(G)$ contains input-trace pairs. Their compatible comparison is

$$\{\mathbf{C} : \exists \tau, (\mathbf{C}, \tau) \in \widehat{\mathcal{E}}_c(G)\} \subseteq \mathcal{E}_c^*(G), \quad \widehat{\text{L1}}(G, c) \leq \text{L1}^*(G, c). \quad (28)$$

The inclusion follows directly from exact replay. Its converse requires complete reverse search, which has not been established. Throughout the paper, verified exact-L1 yield counts valid attempts satisfying the recovered predicate. Candidate traces are compared after the registered role- and step-aware canonicalization, so duplicate encodings do not create additional witnesses.

Post-hoc filtering changes which completed samples are retained but cannot change the proposal distribution’s exact-L1 acceptance probability. Under independent sampling, this probability determines the complete distribution of accepted counts under a fixed attempt budget and the expected cost of obtaining a fixed number of accepted products (Proposition C.10). We distinguish decomposition coverage from forward-replay precision because a candidate disconnection may fail to reconstruct G . Candidate decomposition coverage is the fraction of valid graphs for which $\mathcal{D}_c(G) \neq \emptyset$. Candidate yield additionally uses all original requests as its denominator; undefined conditional ratios are not replaced by zero. For a declared sequence of valid evaluation graphs $G_1, \dots, G_m$, candidate replay precision is

$$\text{ReplayPrecision} = \frac{\sum_{b=1}^m |\hat{\mathcal{E}}_c(G_b)|}{\sum_{b=1}^m |\mathcal{D}_c(G_b)|}, \quad (29)$$

with N/E when the denominator is zero. Replay of tuples already in $\hat{\mathcal{E}}_c(G)$ has 100% success by construction. This invariant does not estimate candidate replay precision. We separately define

$$\text{CandidateAmbig}(G, c) = \mathbf{1}\{|\mathcal{D}_c(G)| > 1\}, \quad \text{ExactAmbig}(G, c) = \mathbf{1}\{|\hat{\mathcal{E}}_c(G)| > 1\}. \quad (30)$$

A distinct exact-L1 product is a deduplicated generated constitution. An unambiguous exact decomposition satisfies $|\hat{\mathcal{E}}_c(G)| = 1$. We distinguish product uniqueness from decomposition uniqueness throughout.

A.6 RECURSIVE ROUTE ASSESSMENT AND DOSSIER ADMISSION

A.6.1 PRIMARY AND SECONDARY EVIDENCE CONTRACTS

We evaluate two separately named contracts. The primary computational-makeability contract, $\mathfrak{B}_{\text{comp}}$, permits computationally proposed or forward-applied published-transform paths ending in exact vendor-listed constitutions. It does not require experimental precedent for every exact substrate. The secondary strict dossier contract, $\mathfrak{B}_{\text{strict}}$, retains its exact-source and current-item evidence requirements. A successful primary assessment does not imply strict dossier closure. The complete-tree equations below are parameterized by the contract; reaction admission and terminal qualification must always be interpreted under that same contract. For $\mathfrak{B}_{\text{comp}}$, the terminal availability predicate means a dated vendor-directory listing within the 30-day evidence window, not verified stock or price. L2 readiness is reported separately and can hold before terminal listing closure. Direct-only L3 requires all recovered components to qualify directly without recursive preparation. Neither contract establishes experimental execution or synthesis success.

A.6.2 ASSESSMENT INTERFACE

For a valid generated graph G and reaction program P with chemistry c , route assessment returns either a complete synthesis dossier or a typed abstention. Let $\mathfrak{B}$ denote the frozen route contract: depth and query budgets, registered transforms, evidence-admission rules and a dated supplier snapshot. Let $\mathfrak{D}$ be the space of candidate dossiers, each containing a synthesis tree and its reaction and terminal evidence, and let $\mathfrak{F}_{\mathfrak{B}}$ be the disjoint finite set of typed failures. Generation and assessment are defined by

$$Y = \text{Assess}_{\mathfrak{B}}(G, P, \mathbf{C}_T, \tau_T) \in \mathfrak{D} \cup \mathfrak{F}_{\mathfrak{B}}. \quad (31)$$

$\text{Assess}_{\mathfrak{B}}$ is invoked after the verifier returns one admitted root trace $(\mathbf{C}_T, \tau_T) \in \hat{\mathcal{E}}_c(G)$. It verifies the final assembly, searches for preparation routes to unavailable precursors and checks terminal supplier evidence. P and the assessor act at different stages: P guides generation, whereas the assessor constructs a dossier from the completed graph without modifying it. FORGE accepts (G, T) only when G passes molecular and exact-assembly verification and T traces every required precursor to an available starting material; otherwise, it abstains.

A.6.3 RECURSIVE EXPANSION AND COMPLETION CHECKS

Completing a synthesis dossier requires evidence for every recovered precursor. Exact L1 verification specifies the precursor structures; availability and preparation routes require additional evidence. Each precursor must therefore match an available material or have a supported route from available starting materials. Let T be a candidate dossier with bipartite synthesis tree (V_M, V_R, E_T) rooted at G , where each product molecule points to its preparation reaction and each reaction points to its required reactants. The assessor constructs these routes recursively and withholds a complete dossier when any branch remains unresolved. A molecule node terminates when a current, dated supplier record lists an exact constitutional match as available; otherwise, FORGE must expand that node through a supported, forward-verified reaction. FORGE uses a bounded level-two (L2) planner to propose each expansion. For each proposal, FORGE records the reaction, reactants, product, and supporting evidence, and accepts the step only when the forward transform reconstructs the target precursor exactly. Each accepted reaction adds its reactants as new molecule nodes, and FORGE applies the same termination-or-expansion rule recursively. The current assessor admits only an unambiguous verified exact-L1 root. Every dossier therefore records the sole selected root trace $(\mathbf{C}_T, \tau_T) \in \hat{\mathcal{E}}_c(G)$; an ambiguous exact set produces a typed abstention. For each upstream reaction node r , let $\text{Admitted}_{\mathfrak{B}}(r) = 1$ when its reaction and substrate scope satisfy the frozen evidence policy, and let $\text{Verify}_{\mathfrak{B}}(r) = 1$ when $\text{target}(r) \in \mathcal{F}_r(\text{reactants}(r))$. For each leaf molecule ℓ , let $\text{Available}_{\mathfrak{B}}(\ell) = 1$ when it satisfies the terminal supplier-evidence rule. Let $V_R^{\text{up}}(T)$ denote the upstream preparation-reaction nodes and $L(T)$ the leaf molecules. $\text{WellFormed}(G, T, P)$ requires that T is a connected, acyclic bipartite tree rooted at G ; each internal molecule has exactly one recorded producing reaction; no molecular constitution repeats along an ancestor path; each reaction’s molecule children equal its recorded reactants with their multiplicities; and T has no disconnected, cyclic or extraneous node. Its distinguished final-assembly record must have product G and precursor children exactly $\mathbf{C}_T$ under trace τ_T .

$$\begin{aligned} \text{Complete}_{\mathfrak{B}}(G, T, P) = & \text{Valid}(G) \text{WellFormed}(G, T, P) \\ & \times \mathbf{1} \left[\hat{\mathcal{E}}_c(G) = \{(\mathbf{C}_T, \tau_T)\} \right] \\ & \times \prod_{r \in V_R^{\text{up}}(T)} \text{Admitted}_{\mathfrak{B}}(r) \text{Verify}_{\mathfrak{B}}(r) \prod_{\ell \in L(T)} \text{Available}_{\mathfrak{B}}(\ell). \end{aligned} \quad (32)$$

All factors are binary and empty products equal one. Repeated use of a material is represented by separate tree occurrences; equality of their constitutions does not merge their nodes. Under finite search bounds and terminating evidence queries, the assessor terminates and returns a dossier only when $\text{Complete}_{\mathfrak{B}}(G, T, P) = 1$ (Lemma C.11 and Invariant C.12). An abstention means that FORGE did not complete a dossier under $\mathfrak{B}$; it does not imply that no synthesis route exists outside the frozen search space or evidence snapshot. The guarantee is computational and does not establish experimental synthesis success.

A.6.4 RECORDED EVIDENCE AND SUPPORT TIERS

For each accepted pair (G, T) , the synthesis dossier contains every molecule and reaction in T , the evidence for each reaction and a dated supplier record for each terminal material. We classify assembly support relative to the fixed reference space into four tiers: **E0 (enumerated)**: the generated lipid matches a product that was already constructed computationally from known components using a supported reaction. **E1 (known components)**: the generated lipid was not included in the bounded E0 enumeration, but a supported reaction assembles it entirely from known components. **E2 (generated precursor)**: a supported reaction assembles the lipid, but at least one precursor is newly generated and requires a complete L2 route to available starting materials. **E3 (new assembly family)**: the lipid requires a final assembly reaction outside the qualified reaction registry.

A.7 END-TO-END INFERENCE PROCEDURE

Algorithm 1 records every requested output and every trajectory used to construct it, including failures in molecular validation, exact replay or route completion.

Algorithm 1 FORGE generation and fail-closed synthesis-program construction.

```

1080 Require: trained coordinate predictors, chemistry  $c$ , program prior  $p_{\text{prog}}(\cdot \mid c)$ , integers  $L_{\text{samp}}, D \geq$ 
1081  $1$  and  $B \geq 0$ , recorded seed  $s$ , inference policy  $\Pi$ , route contract  $\mathfrak{B}$ 
1082 Ensure: attempt-indexed dossier sequence  $\mathcal{D}_{\text{out}}$ , exact-L1 ledger  $\mathcal{L}_{\text{out}}$ , and typed failure and pro-
1083 posal ledger  $\mathcal{H}$ 
1084 1: initialize the random-number streams from  $s$ ;  $\mathcal{D}_{\text{out}} \leftarrow []$ ;  $\mathcal{L}_{\text{out}} \leftarrow []$ ;  $\mathcal{H} \leftarrow []$ 
1085 2: for  $b = 1, \dots, B$  do
1086 3:   sample and record  $P_b \sim p_{\text{prog}}(\cdot \mid c)$ ; initialize raw draw ledger  $\mathcal{Z}_b$ 
1087 4:   for  $d = 1, \dots, D$  do
1088 5:     sample  $Z_0 \sim q_0^{P_b}$  and set  $Z \leftarrow \Pi_{P_b}(Z_0)$ 
1089 6:     for  $k = 0, \dots, L_{\text{samp}} - 1$  do
1090 7:        $t_k \leftarrow k/L_{\text{samp}}$ ;  $\Delta t_k \leftarrow 1/L_{\text{samp}}$ 
1091 8:        $Z \leftarrow \Pi_{P_b}(\text{CAPPEDCOORDINATEEULER}(Z, t_k, \Delta t_k, P_b))$ 
1092 9:     end for
1093 10:    obtain terminal predictions  $\lambda_{b,d}$  and  $Z_{b,d}^{\text{out}} \leftarrow H_{P_b}^{\theta}(Z)$ 
1094 11:    append the state or typed terminal failure, predictions and raw decoded graph to  $\mathcal{Z}_b$ 
1095 12:  end for
1096 13:   $\mathcal{C}_b \leftarrow \text{CHECKEDCONSTRUCTION}(P_b, \mathcal{Z}_b, \Pi)$ 
1097 14:  record every raw/proposed graph, check, abstention and cost in  $\mathcal{H}$ 
1098 15: end for
1099 16:  $(G_1, \dots, G_B) \leftarrow \mathcal{S}_{\Pi}((P_b, \mathcal{C}_b)_{b=1}^B)$ 
1100 17: for  $b = 1, \dots, B$  do
1101 18:    $G \leftarrow G_b$ ;  $P \leftarrow P_b$ 
1102 19:   if  $G = \perp_{\text{dec}}$  or  $\text{Valid}(G) = 0$  then
1103 20:     record  $(b, \text{INVALIDGRAPH}, \text{decode or validation reason})$  in  $\mathcal{H}$ 
1104 21:     continue
1105 22:   end if
1106 23:    $\hat{\mathcal{E}}_c(G) \leftarrow \{(\mathbf{C}, \tau) \in \mathcal{D}_c(G) : \exists H \in \mathcal{F}_c(\mathbf{C}), H \cong G\}$ 
1107 24:   if  $\hat{\mathcal{E}}_c(G) = \emptyset$  then
1108 25:      $\mathcal{H} \leftarrow \mathcal{H} \parallel [(b, \text{L1FAILURE})]$ 
1109 26:     continue
1110 27:   end if
1111 28:    $\mathcal{L}_{\text{out}} \leftarrow \mathcal{L}_{\text{out}} \parallel [(b, G, \hat{\mathcal{E}}_c(G))]$ 
1112 29:   if  $|\hat{\mathcal{E}}_c(G)| \neq 1$  then
1113 30:      $\mathcal{H} \leftarrow \mathcal{H} \parallel [(b, \text{AMBIGUOUSL1})]$ 
1114 31:     continue
1115 32:   end if
1116 33:   select the sole  $(\mathbf{C}_T, \tau_T) \in \hat{\mathcal{E}}_c(G)$ 
1117 34:    $Y \leftarrow \text{Assess}_{\mathfrak{B}}(G, P, \mathbf{C}_T, \tau_T)$ 
1118 35:   if  $Y$  is a dossier  $T$  and  $\text{Complete}_{\mathfrak{B}}(G, T, P) = 1$  then
1119 36:      $\mathcal{D}_{\text{out}} \leftarrow \mathcal{D}_{\text{out}} \parallel [(b, T)]$ 
1120 37:   else if  $Y \in \mathfrak{F}_{\mathfrak{B}}$  then
1121 38:      $\mathcal{H} \leftarrow \mathcal{H} \parallel [(b, Y)]$ 
1122 39:   else
1123 40:      $\mathcal{H} \leftarrow \mathcal{H} \parallel [(b, \text{INCOMPLETEDOSSIER})]$ 
1124 41:   end if
1125 42: end for
1126 43: return  $\mathcal{D}_{\text{out}}, \mathcal{L}_{\text{out}}, \mathcal{H}$ 

```

A.8 REPRODUCIBILITY AND ARTIFACT PROVENANCE

Each run is identified by its training seed, data and split manifests, atom/program vocabularies, source registries, fitted noise marginals, configuration, code snapshot and checkpoint SHA-256. The independent training-seed identifiers are [\[training seed identifiers\]](#). Each comparison retains its declared attempt budget, checkpoint rule and evaluator. Sampling seeds, request/layout identifiers and proposal lineage are recorded separately from training seeds. The manuscript provenance records identify the source artifacts for tables and figures through SHA-256 hashes.

The inference-policy identifier is [inference policy identifier]; its record includes all proposal caps, TRAIN motif references, admission order, selection-group boundaries and source checks. The route contract is [route contract identifier], including depth/query limits, evidence-admission policy and dated terminal-material snapshots. Restarts preserve committed checkpoints and completed request shards, and source-preserving fixes retain the exact earlier implementation and an equivalence receipt. A change to an inference map defines a distinct arm unless equivalence is demonstrated for the declared inputs.

Reproduction requires source code, configuration and environment manifests, model checkpoints, split/registry hashes, attempt-level output and verification ledgers, and figure/table exporters. These artifacts are identified by [reproduction archive identifier]. Corpus access conditions are [data availability]. Constitutional identity removes atom maps and stereochemical labels while retaining formal charge and bond order; no tautomer normalization is applied.

B EVALUATION ACROSS REACTION FAMILIES

We evaluate complete lipid graphs at three distinct levels: exact assembly consistency, structural and distributional properties, and synthesis-evidence completeness. Unless stated otherwise, N denotes all requested products. Conditional rates specify their assessed population and retain invalid, unsupported and abstained requests in the accompanying accounting.

B.1 COHORT, PROGRAM REGISTRY AND SPLIT ACCOUNTING

We associate each registered assembly program with its source evidence, reactive-site multiplicity, precursor roles and repeat/depth support. Admitted training records and qualified CAL molecules are counted separately; a complete source-row versus constitutional-product and study census remains unmeasured. The family census reports sampling mass, representation admission and exclusions; support tiers E0–E3 describe evidence depth rather than a probability of synthesis success.

Table 4: **Admitted training corpus and guarded CAL representation.** TRAIN records and CAL molecules are different units. A CAL molecule may have multiple qualified condition variants. These counts measure support and admission, not model success.

Family	TRAIN records	Sampling mass	Admitted CAL / guarded molecules	CAL variants	Independent study census
A3 amine–aldehyde–alkyne	34,104	1/22	27/36	27	—
Acid–epoxide diester	101,098	1/22	95/96	95	—
AEMA	5,428	1/22	33/39	59	—
Aldehyde Ugi 3-CR	2,283	1/22	12/17	12	—
Aldehyde Ugi 4-CR	63,616	1/22	61/66	61	—
α -Isocyanoester dihydroimidazole	35,728	1/22	103/135	103	—
Amine alkylation	92,697	1/22	171/171	171	—
Amine–epoxide	33,507	1/22	113/113	113	—
Aryl reductive amination	7,788	1/22	79/80	79	—
Aza-Michael acrylamide	10,127	1/22	33/33	33	—
Aza-Michael acrylate	10,270	1/22	33/57	33	—
Disulfide Michael	96,833	1/22	128/128	128	—
Epoxide O-acylation	21,462	1/22	70/70	70	—
iPhos	62,636	1/22	62/110	62	—
Ketone–isocyanide amide	12,482	1/22	30/60	30	—
Ketone Ugi 4-CR	27,027	1/22	29/29	29	—
Maleate	38,472	1/22	49/49	49	—
O-esterification	10,391	1/22	43/79	43	—
Passerini	62,188	1/22	68/68	68	—
Preassembled thiol-yne	260,470	1/22	379/382	379	—
Reductive amination	28,471	1/22	145/145	145	—
Thiolactone	174,987	1/22	96/96	96	—
Total	1,192,065	1	1,859/2,059	1,885	—

The training measure assigns equal family mass and uniform constitutional-product mass within each family. Pending support excludes 80,401 records; vitamin-B5 multistep has no admitted training support. Declared graph support is 254 atoms and 12 closures. The corpus contains 44,312 records above 96 atoms. Independent study counts, source-disjoint evaluation and a separate source-row versus constitution census remain unmeasured. TEST outcomes are unopened.

B.2 TRAINING, NUMERICAL VALIDATION AND COMPUTATIONAL COST

Table 5: Measured training configuration and execution. One complete fit initializes the continuation diagnostics. Hash-pinned inputs, checkpoints and commands accompany the evidence ledger.

Field	Setting / result
Independent full fit / seed	1 / 2026092401
Families / presentations per family / total	22 / 402,336 / 8,851,392
Optimizer updates / updates containing each family	2,794 / 381
Global batch / participating families / examples per family	3,168 / 3 / 1,056
Optimizer / learning rate / weight decay / clip	AdamW / 10^{-4} / 5×10^{-5} / 1.0
AdamW (β_1, β_2) / ϵ	(0.9, 0.999) / 10^{-8}
Transformer layers / width / heads / dropout	6 / 192 / 8 / 0.1
Routed experts / adapter width / graph support	3 / 64 / 254 atoms, 12 closures
Gradient and chemistry balancing	Family PCGrad; equal present-role mass
Auxiliary role / core / repeat weights	0.25 / 0.25 / 0.25
Offspring / junction / topology-chemistry weights	1.0 / 0.5 / 1.0
Hardware / precision / determinism	8 H100 / FP32 / deterministic
Prefetch workers / depth / checkpoint interval	2 / 2 / 220 updates
Execution / included preparation time	1,377.725 s / 31.801 s
Serial versus eight-GPU; kill/restart	Exact 44-step model, optimizer, RNG replay
Development sampling	352 requests; 16/family; 64 flow steps
Adaptive selected cohort	1,408 separate requests; 64/family
Final independent held-out protocol and fits	—

The 22.96-minute execution excludes allocation/image initialization and later evaluation; it does not meet a 15-minute execution target. Fixed-presentation continuation uses 31,680 additional presentations/family: batch 3,168 for 220 updates (8 GPUs; 40.859 function-GPU-minutes) versus batch 264 for 2,640 updates (1 GPU; 16.082 function-GPU-minutes). A separate paired categorical/coarse-parent pilot fixes 66 updates and 9,504 presentations/family/arm (55.706 function-GPU-minutes for the successful paired execution). Function-GPU-minutes are not billed dollar costs. Failed staging and timeout attempts remain in the run ledger; neither continuation is an independent full fit.

Training artifacts retain coordinate losses; the completed guarded-CAL diagnostic resolves clean-parent errors by family and topology. Full generation learning curves across corruption time remain unmeasured. Checkpoint comparisons use fixed development requests and report exact yield, structural assessment and diversity alongside gradients and clipping. We distinguish molecular presentations, family appearances in updates and optimizer updates. Matched-exposure batch/update comparisons and exposure-scaling comparisons isolate different effects. The original full fit ends at its fixed 2,794-update budget. Both continuation diagnostics use their fixed final endpoints; saved intermediate checkpoints are descriptive, and no best-checkpoint selection is performed.

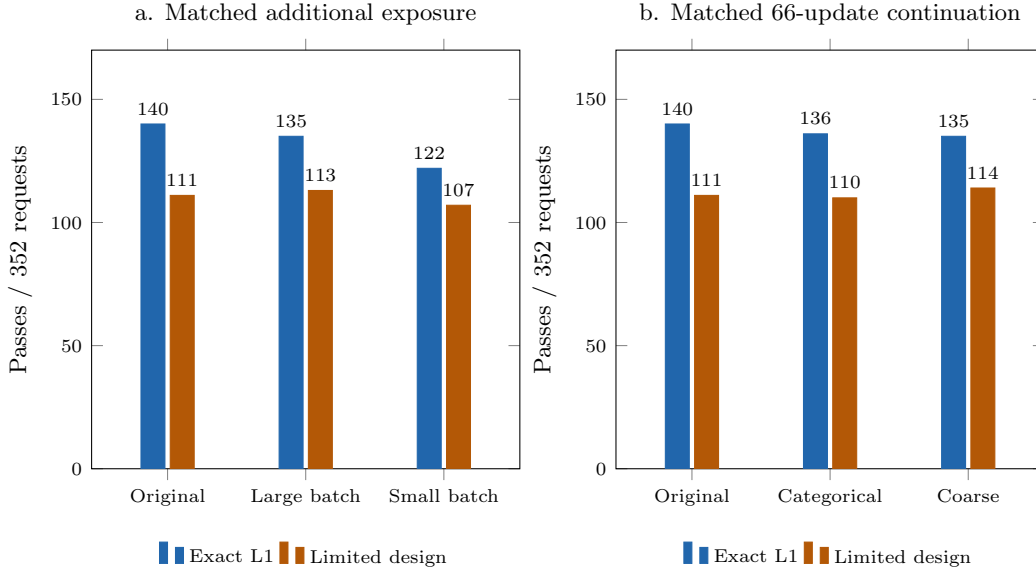


Figure 2: **Fixed-endpoint continuation diagnostics on 352 development requests.** The matched-presentation large/small-update pilot has D1 exact counts 135/122 and design counts 113/107, versus original 140/111. The separate categorical/coarse66 pilot gives 136/135 exact and 110/114 design, versus the same original 140/111. These are paired continuations of one fit, not independent training seeds; no sampling uncertainty bars or held-out claim is attached.

B.3 EXACT-L1 DECOMPOSITION AND FAILURE ACCOUNTING

We report exact assembly outcomes for each family, registered program, training seed and inference arm. Each request contributes to the denominator whether generation, decomposition or verification succeeds. Alongside exact yield, we count valid and connected graphs, distinct exact products, unique exact precursor roots and checked decomposition traces. Candidate-search coverage is the fraction of requests for which a decomposition candidate is found; verified decomposition coverage requires exact replay. Candidate precision counts passing checked traces over all checked traces, with ambiguity and search censoring explicit. The replay rate among already accepted exact traces is an implementation invariant, not an independent precision estimate. We distinguish family-local product counts from counts deduplicated across all families. Equal-family macro averages, pooled yields and worst-family outcomes use separate definitions. Independent training-seed variability remains unmeasured; the reported development outcomes derive from one full fit.

Table 6: **Attempt-level readout outcomes on the fixed 352-request development cohort.** One original checkpoint; 16 TRAIN-derived requests per family. Raw is the flow endpoint, terminal is the $t = 1$ argmax, and D1 is constrained readout. Every failed request remains in N .

Family	N	Raw exact	Terminal exact	D1 exact	D1 valid + connected	D1 distinct exact	D1 nonexact
A3 amine-aldehyde-alkyne	16	2	2	11	13	11	5
Acid-epoxide diester	16	6	9	14	16	14	2
AEMA	16	4	4	4	16	4	12
Aldehyde Ugi 3-CR	16	11	12	14	16	14	2
Aldehyde Ugi 4-CR	16	7	9	14	16	14	2
α -Isocyanoester dihydroimidazole	16	7	7	10	11	10	6
Amine alkylation	16	0	0	0	16	0	16
Amine-epoxide	16	0	0	0	15	0	16
Aryl reductive amination	16	0	0	0	16	0	16
Aza-Michael acrylamide	16	0	0	0	15	0	16
Aza-Michael acrylate	16	1	1	0	15	0	16
Disulfide Michael	16	0	0	0	13	0	16
Epoxide O-acylation	16	1	1	4	16	4	12
iPhos	16	9	9	13	15	13	3
Ketone-isocyanide amide	16	6	7	14	14	14	2
Ketone Ugi 4-CR	16	1	0	5	10	5	11
Maleate	16	0	0	0	11	0	16
O-esterification	16	7	8	8	16	8	8
Passerini	16	9	10	15	16	15	1
Preassembled thiol-yne	16	0	0	1	15	1	15
Reductive amination	16	0	0	1	14	1	15
Thiolactone	16	9	12	12	14	12	4
All requests	352	80	91	140	319	140	212

These outputs precede the larger proposal/selection pipeline used for the separate 1,408-product cohort. A failed exact check is not experimental evidence of unmakeability. Ambiguity, checked-trace precision and independent-seed failure distributions are retained as unmeasured endpoints (—); replay among accepted exact traces is a verifier invariant, not a precision estimate.

Table 7: **Current single-fit result and missing independent replications.** The first column is the adaptively selected 1,408-request cohort from seed 2026092401, not a fresh held-out evaluation. Two further independent fits and their matched inference remain unrun. Sampling seeds and continuations do not supply training-seed replication.

Family	Existing fit exact / N	Independent fit B	Independent fit C	Mean $\pm$ sample SD
A3 amine–aldehyde–alkyne	61/64	—	—	—
Acid–epoxide diester	63/64	—	—	—
AEMA	63/64	—	—	—
Aldehyde Ugi 3-CR	63/64	—	—	—
Aldehyde Ugi 4-CR	64/64	—	—	—
α -Isocyanoester dihydroimidazole	64/64	—	—	—
Amine alkylation	62/64	—	—	—
Amine–epoxide	64/64	—	—	—
Aryl reductive amination	58/64	—	—	—
Aza-Michael acrylamide	64/64	—	—	—
Aza-Michael acrylate	64/64	—	—	—
Disulfide Michael	63/64	—	—	—
Epoxide O-acylation	63/64	—	—	—
iPhos	64/64	—	—	—
Ketone–isocyanide amide	64/64	—	—	—
Ketone Ugi 4-CR	62/64	—	—	—
Maleate	64/64	—	—	—
O-esterification	64/64	—	—	—
Passerini	64/64	—	—	—
Preassembled thiol-yne	64/64	—	—	—
Reductive amination	61/64	—	—	—
Thiolactone	64/64	—	—	—

The existing selected cohort has 1,387/1,408 exact products. No multi-fit mean, sample standard deviation or confidence interval is estimated from this single fit. The prospective matched held-out values remain —.

B.4 MATCHED CONTROLS, NATIVE BASELINES AND THE THREE-FAMILY COMPARISON

Table 8: Paired inference-only cyclic program-ID control. The same checkpoint, requests and random draws receive the true or cyclically reassigned program ID. True structural context and source verification remain fixed. Counts use D1; design denotes the limited structural/design checks.

Family	<i>N</i>	True ID exact	Cyclic ID exact	True ID design	Cyclic ID design
A3 amine–aldehyde–alkyne	16	11	9	7	5
Acid–epoxide diester	16	14	12	12	9
AEMA	16	4	3	4	2
Aldehyde Ugi 3-CR	16	14	11	6	8
Aldehyde Ugi 4-CR	16	14	13	6	5
α -Isocyanoester dihydroimidazole	16	10	5	10	5
Amine alkylation	16	0	0	0	0
Amine–epoxide	16	0	0	0	0
Aryl reductive amination	16	0	0	0	0
Aza-Michael acrylamide	16	0	3	0	3
Aza-Michael acrylate	16	0	0	0	0
Disulfide Michael	16	0	0	0	0
Epoxide O-acylation	16	4	1	4	1
iPhos	16	13	11	13	11
Ketone–isocyanide amide	16	14	10	14	10
Ketone Ugi 4-CR	16	5	0	4	0
Maleate	16	0	0	0	0
O-esterification	16	8	7	8	7
Passerini	16	15	10	14	10
Preassembled thiol-yne	16	1	1	0	1
Reductive amination	16	1	0	1	0
Thiolactone	16	12	9	8	5
All requests	352	140	105	111	82

On all 352 requests, raw exact counts are 80 versus 71 and terminal-argmax counts are 91 versus 85. The intervention tests the incremental program-ID signal with correct core/role/layout/depth information. A consistent training/test label bijection would instead be a recoding. Independently trained true shared-null, finite-catalogue selector, matched/generous FACT, qualified native external baselines and the matched three-family bridge remain —. Shared-null implementation tests are not trained-null performance.

The completed control comparisons use paired request distributions from one original fit. Independent training-seed ranges and uncertainty remain unmeasured. Training exposure, checkpoint selection, decoder opportunity and source/planner budgets define the matched comparison. We distinguish supported native chemistry from unavailable baseline coverage. A wrong reaction label supplied alongside correct roles and reaction cores tests the incremental information in that label; a retrained null model additionally removes the corresponding training signal.

B.5 DECODER AND STRUCTURAL-REPAIR ATTRIBUTION

Table 9: **Readout, saved-candidate selection and repair attribution.** The 352-request readouts and 192-request repair study have distinct populations. They do not estimate a model improvement or independent realism.

Arm	<i>N</i>	Exact	Design	Attribution
Original raw endpoint	352	80	61	Same saved trajectories
Original terminal argmax	352	91	74	Same saved trajectories
Original D1	352	140	111	Same saved trajectories
Original verified fallback	352	145	116	No new model/proposal calls
Categorical66 verified fallback	352	143	116	No new model/proposal calls
Coarse66 verified fallback	352	142	120	No new model/proposal calls
Three-family repair identity control	192	187	140	Separate construction diagnostic
Ring-scheduled proposals	192	187	146	Separate construction diagnostic
Anchor-preserving proposals	192	187	146	Separate construction diagnostic
Equal-cost / no TRAIN-motif-prior control	—	—	—	—

Verified fallback retains an exact D1 candidate, otherwise takes an already exact raw endpoint, otherwise an already exact terminal argmax, otherwise retains the D1 failure. It rescues 5/7/7 original/categorical/coarse requests, with no new candidates or exact/design losses. The categorical phosphate-tail Shannon count falls from 9.1180 to 9.0757; cross-checkpoint diversity floors still fail. The repair study includes all 64 A3, aldehyde-Ugi4 and ketone-Ugi4 requests: design counts change 51/50/39 to 55/51/40. Both repair arms change the same six requests and preserve the declared integer diversity/novelty floors; three resulting structures differ between the ring and anchor arms without an additional design-pass gain. Equal proposal caps do not mean equal measured cost; the full study uses 7.209 CPU seconds, 84 source checks and 84 design checks. The changed molecules have no transferred routing verdict, and the official 1,408-product cohort is unchanged. Individual decoder-rule and matched-cost effects remain —.

The inference analysis separates readout, component completion, local chemical proposals, ring-system proposals and population selection. We retain original, proposed and selected graphs, with atom, bond and topology edits, repeated-arm identities, rejected proposals and search limits. Every design predicate is evaluated on the final graph because a later construction can undo an earlier correction. Cost includes duplicate and failed proposals, source checks and selection. TRAIN local motifs used as proposal priors are identified separately from independent evaluation references.

B.6 CURRENT STRUCTURAL QUALITY AND DESIGN-CONDITION EVIDENCE

The selected development cohort contains 1,408 requests, 64 per family, from one trained checkpoint. All selected identities match the current routing cohort. These counts describe an adaptively inspected TRAIN-derived population, not independent-seed estimates or held-out quality certification. Saved denotes the preceding selection, not unconstrained raw generation; selected denotes the context-preserving selection. Subsequent ring-receipt completion changes assessment availability, not molecular identity. Unmeasured endpoints retain placeholders.

Table 10: **Structural outcomes on the current selected development cohort.**

Family	Valid/conn.	Exact L1	Design	Alert	Context	Design, no context
A3	64/64	61/64	51/64	0/64	7/64	47/64
Acid-epoxide	64/64	63/64	63/64	0/64	0/64	63/64
AEMA	64/64	63/64	63/64	0/64	0/64	63/64
Aldehyde Ugi3	64/64	63/64	63/64	0/64	3/64	60/64
Aldehyde Ugi4	64/64	64/64	50/64	0/64	1/64	49/64
α -Isocyanoester	64/64	64/64	64/64	0/64	1/64	63/64
Amine alkylation	64/64	62/64	61/64	0/64	0/64	61/64
Amine-epoxide	64/64	64/64	58/64	0/64	6/64	52/64
Aryl reductive amination	64/64	58/64	58/64	0/64	3/64	56/64
Aza-Michael acrylamide	64/64	64/64	64/64	0/64	0/64	64/64
Aza-Michael acrylate	64/64	64/64	64/64	0/64	1/64	63/64
Disulfide Michael	64/64	63/64	62/64	0/64	2/64	60/64
Epoxide/O-acylation	64/64	63/64	60/64	0/64	0/64	60/64
iPhos	64/64	64/64	64/64	0/64	1/64	63/64
Ketone-isocyanide	64/64	64/64	64/64	0/64	0/64	64/64

Table 10, continued

Family	Valid/conn.	Exact L1	Design	Alert	Context	Design, no context
Ketone Ugi4	64/64	62/64	39/64	0/64	10/64	33/64
Maleate	64/64	64/64	60/64	0/64	2/64	59/64
O-esterification	64/64	64/64	64/64	0/64	0/64	64/64
Passerini	64/64	64/64	64/64	0/64	1/64	63/64
Preassembled thiol-yne	64/64	64/64	64/64	0/64	0/64	64/64
Reductive amination	64/64	61/64	61/64	0/64	1/64	60/64
Thiolactone	64/64	64/64	63/64	0/64	2/64	61/64
All requests	1408/1408	1387/1408	1324/1408	0/1408	41/1408	1292/1408

Design requires exact L1 and all applicable qualified chemical, head and ring checks. Zero narrow-alert failures do not establish stability. Context flags overlap: hydrated/hemiacetal carbon 24, peroxide bond 8, carbon with at least three oxygens 6, other allene 5, small unsaturated ring 1 and heteroatom cumulene 1. The source-library screen passes 29/29 controls from two studies and two families; 20 families lack controls. Ten controls are component-disjoint; none supplies individual-product characterization in this control set. Independent blinded structural review remains [blinded structural review].

Table 11: Scoped head, ring and TRAIN-feature diagnostics.

Family	Head P/F/U	Ring P/F/U	Atom-env. coverage	Ring-system support	Exact + observed/N
A3	Outside scope	52/12/0	—	—	14/64
Acid-epoxide	Outside scope	64/0/0	—	—	38/64
AEMA	Outside scope	64/0/0	—	—	63/64
Aldehyde Ugi3	Outside scope	64/0/0	—	—	31/64
Aldehyde Ugi4	59/5/0	50/14/0	—	—	20/64
α -Isocyanoester	Outside scope	64/0/0	—	—	50/64
Amine alkylation	Outside scope	63/1/0	—	—	51/64
Amine-epoxide	Outside scope	58/6/0	—	—	47/64
Aryl reductive amination	Outside scope	63/1/0	—	—	6/64
Aza-Michael acrylamide	Outside scope	64/0/0	—	—	63/64
Aza-Michael acrylate	Outside scope	64/0/0	—	—	61/64
Disulfide Michael	Outside scope	63/1/0	—	—	36/64
Epoxide/O-acylation	Outside scope	61/3/0	—	—	51/64
iPhos	Outside scope	64/0/0	—	—	56/64
Ketone-isocyanide	Outside scope	64/0/0	—	—	63/64
Ketone Ugi4	Outside scope	39/25/0	—	—	7/64
Maleate	Outside scope	60/4/0	—	—	48/64
O-esterification	Outside scope	64/0/0	—	—	55/64
Passerini	Outside scope	64/0/0	—	—	63/64
Preassembled thiol-yne	Outside scope	64/0/0	—	—	55/64
Reductive amination	Outside scope	63/1/0	—	—	45/64
Thiolactone	Outside scope	63/1/0	—	—	51/64
All requests	59/5/0; 1344 outside	1339/69/0	—	—	974/1408

P/F/U denotes pass/fail/unassessed. Only the scoped neutral aldehyde-Ugi4 head predicate is qualified for this generated cohort; outside-scope cases are not passes. Ring status combines the applicable allocation/size checks. Separate all-request allocation and comparable-cycle size rates remain [separate ring-rate denominators]. Current-cohort atom-environment mean coverage and ring-system-support aggregates remain unmeasured. Exact-plus-observed requires exact, unambiguous decomposition and all product/component local features observed in the appropriate TRAIN references; its 974 successes do not establish independent lipid realism.

B.7 DISTRIBUTIONAL AGREEMENT WITH SYNTHETIC CAL REFERENCES

The same 1,821 exact-source-qualified synthetic CAL references are used across arms. They predominantly share TRAIN components; only one reference is component-disjoint. This is a development distribution diagnostic. The available broad empirical reference and control panels contain 1,024 unique structures each, share all 12 study groups and contain 32 and 36 exact TRAIN overlaps, respectively. Source ancestry remains insufficient to admit an independent empirical panel.

Table 12: Fingerprint agreement and internal diversity, by family and saved arm.

Arm	Unique G/R	Prec. (%)	Cov. (%)	Recall (%)	Distinct/ N (%)	Nearest	Diversity
A3; $N = 64$ per arm							
Saved	64/27	64.062	70.370	92.593	64.062	0.419	0.728
Selected	64/27	56.250	70.370	92.593	56.250	0.401	0.729
TRAIN control	64/27	98.438	81.481	96.296	98.438	0.535	0.639
Acid-epoxide; $N = 64$ per arm							
Saved	64/95	43.750	41.053	73.684	43.750	0.692	0.496
Selected	64/95	42.188	43.158	84.211	42.188	0.683	0.510
TRAIN control	64/95	81.250	56.842	96.842	81.250	0.740	0.536
AEMA; $N = 64$ per arm							
Saved	64/21	85.938	95.238	80.952	85.938	0.826	0.336
Selected	64/21	85.938	95.238	80.952	85.938	0.823	0.339
TRAIN control	64/21	70.312	76.190	71.429	70.312	0.768	0.416
Aldehyde Ugi3; $N = 64$ per arm							
Saved	64/12	68.750	100.000	75.000	68.750	0.715	0.441
Selected	64/12	73.438	100.000	83.333	73.438	0.707	0.451
TRAIN control	64/12	96.875	100.000	75.000	96.875	0.834	0.344
Aldehyde Ugi4; $N = 64$ per arm							
Saved	64/61	82.812	80.328	77.049	82.812	0.599	0.624
Selected	64/61	84.375	78.689	73.770	84.375	0.588	0.636
TRAIN control	64/61	98.438	90.164	88.525	98.438	0.612	0.686
α-Isocyanoeester; $N = 64$ per arm							
Saved	64/103	29.688	8.738	48.544	29.688	0.732	0.457
Selected	64/103	32.812	8.738	47.573	32.812	0.737	0.452
TRAIN control	64/103	39.062	12.621	48.544	39.062	0.780	0.431
Amine alkylation; $N = 64$ per arm							
Saved	64/171	40.625	15.205	85.965	40.625	0.600	0.526
Selected	64/171	42.188	15.789	86.550	42.188	0.599	0.522
TRAIN control	64/171	84.375	40.351	93.567	84.375	0.711	0.541
Amine-epoxide; $N = 64$ per arm							
Saved	64/111	68.750	40.541	92.793	68.750	0.595	0.657
Selected	64/111	70.312	43.243	92.793	70.312	0.605	0.654
TRAIN control	64/111	76.562	43.243	90.991	76.562	0.650	0.640
Aryl reductive amination; $N = 64$ per arm							
Saved	63/79	6.349	24.051	98.734	6.250	0.547	0.597
Selected	63/79	3.175	15.190	100.000	3.125	0.539	0.599
TRAIN control	64/79	39.062	50.633	98.734	39.062	0.794	0.466
Aza-Michael acrylamide; $N = 64$ per arm							
Saved	64/33	75.000	60.606	90.909	75.000	0.686	0.530
Selected	64/33	75.000	60.606	90.909	75.000	0.677	0.526
TRAIN control	64/33	70.312	48.485	87.879	70.312	0.631	0.505
Aza-Michael acrylate; $N = 64$ per arm							
Saved	64/11	100.000	100.000	72.727	100.000	0.536	0.539
Selected	64/11	100.000	100.000	72.727	100.000	0.536	0.539
TRAIN control	64/11	100.000	100.000	81.818	100.000	0.561	0.494
Disulfide Michael; $N = 64$ per arm							
Saved	64/128	18.750	14.062	61.719	18.750	0.573	0.589
Selected	64/128	15.625	14.062	61.719	15.625	0.570	0.593
TRAIN control	64/128	50.000	24.219	78.906	50.000	0.727	0.452
Epoxide/O-acylation; $N = 64$ per arm							
Saved	64/70	71.875	60.000	70.000	71.875	0.722	0.492
Selected	64/70	68.750	60.000	70.000	68.750	0.715	0.498
TRAIN control	64/70	87.500	84.286	90.000	87.500	0.742	0.526
iPhos; $N = 64$ per arm							
Saved	64/62	78.125	32.258	93.548	78.125	0.672	0.535
Selected	64/62	78.125	32.258	93.548	78.125	0.673	0.536
TRAIN control	64/62	98.438	67.742	95.161	98.438	0.746	0.505
Ketone-isocyanide; $N = 64$ per arm							
Saved	64/29	87.500	72.414	96.552	87.500	0.718	0.516
Selected	64/29	87.500	72.414	96.552	87.500	0.724	0.515
TRAIN control	64/29	76.562	55.172	100.000	76.562	0.703	0.534
Ketone Ugi4; $N = 64$ per arm							
Saved	64/29	1.562	3.448	86.207	1.562	0.405	0.695
Selected	64/29	1.562	3.448	86.207	1.562	0.405	0.695
TRAIN control	64/29	87.500	100.000	96.552	87.500	0.648	0.530
Maleate; $N = 64$ per arm							
Saved	64/49	81.250	59.184	85.714	81.250	0.631	0.627
Selected	64/49	81.250	59.184	85.714	81.250	0.631	0.627
TRAIN control	64/49	89.062	65.306	83.673	89.062	0.638	0.626
O-esterification; $N = 64$ per arm							

Table 12, continued

Arm	Unique G/R	Prec. (%)	Cov. (%)	Recall (%)	Distinct/ N (%)	Nearest	Diversity
Saved	64/43	51.562	79.070	69.767	51.562	0.575	0.661
Selected	64/43	51.562	79.070	69.767	51.562	0.577	0.663
TRAIN control	64/43	48.438	53.488	95.349	48.438	0.506	0.653
Passerini; $N = 64$ per arm							
Saved	64/68	87.500	55.882	89.706	87.500	0.759	0.487
Selected	64/68	87.500	55.882	89.706	87.500	0.757	0.488
TRAIN control	64/68	87.500	50.000	97.059	87.500	0.713	0.534
Preassembled thiol-yne; $N = 64$ per arm							
Saved	64/379	20.312	2.111	70.712	20.312	0.617	0.590
Selected	64/379	23.438	2.639	72.296	23.438	0.620	0.588
TRAIN control	64/379	34.375	3.958	91.557	34.375	0.604	0.625
Reductive amination; $N = 64$ per arm							
Saved	64/144	39.062	22.222	86.806	39.062	0.664	0.563
Selected	64/144	39.062	22.222	86.806	39.062	0.664	0.563
TRAIN control	64/144	76.562	54.861	99.306	76.562	0.796	0.539
Thiolactone; $N = 64$ per arm							
Saved	64/96	53.125	48.958	90.625	53.125	0.810	0.404
Selected	64/96	51.562	47.917	88.542	51.562	0.806	0.411
TRAIN control	64/96	60.938	39.583	88.542	60.938	0.817	0.413

Fingerprint precision is reference-neighborhood membership, not the probability that a molecule is chemically realistic. Large generated neighborhoods can inflate recall. TRAIN controls are not layout-matched independent controls. Independent empirical descriptor precision/coverage remain [independent descriptor precision/coverage]; source-grouped C2ST AUC and uncertainty remain [independent C2ST].

B.8 DEFINITIONS AND INTERPRETATION OF QUALITY METRICS

Table 13: Metric definitions and admitted scope.

Metric	Computation and limitation
Valid molecules / all requests (%)	RDKit parsing/sanitization; failure remains in all-request denominator.
Connected molecules / all requests (%)	Exactly one connected molecular graph; distinct from validity.
Exact L1 + all qualified design checks / requests (%)	Source-exact final graph and every applicable qualified axis assessed/pass. Unknowns and abstentions do not pass.
Molecules with a justified chemical alert / requests (%)	Frozen narrow chemical-alert motifs; also report per-rule counts and unassessed requests. Not a complete stability screen.
Molecules requiring chemistry-context review / requests (%)	Frozen context flags are descriptive, not automatic failures.
Required head retained / family-scope requests (%)	Pass / prespecified family-scope requests; include charged or ambiguous-origin abstentions. Only neutral aldehyde-Ugi4 has a qualified predicate; other families remain unassessed.
Correct role-local cycle allocation / requests (%)	Match declared variable closures per role, including zero; cross-origin edges must also match. All requests remain in denominator; report unassessed cases.
Requested ring-size agreement / comparable requests (%)	Each observed fundamental-cycle length must belong to its role’s allowed-size set, under a comparable tree basis. Not a size-multiset, SSSR or 3D-geometry match. Report inapplicable and unassessed counts.
TRAIN atom-environment coverage (%)	Fraction of atoms with reference-observed local environments, averaged over assessed products. Report component-role summaries separately.
Ring-system TRAIN support / assessed products (%)	Among valid connected products with a family TRAIN reference, fraction with no unknown connected ring-bond signature. Ring-free products pass vacuously; report them and missing/invalid cases separately. Component roles are separate.
Exact L1 + product/component features observed / requests (%)	Exact, unambiguously decomposed product and every precursor must have all atom environments and connected ring-bond signatures observed in their family product or family/source-role TRAIN reference. Divide by all requests; absent references never pass.
Fingerprint precision (%)	Fraction of unique generated molecules inside the union of reference k-nearest-neighbor balls in ECFP4/Tanimoto distance.
Fingerprint coverage (%)	Fraction of reference-centered neighborhoods hit by at least one unique generated molecule.
Fingerprint recall (%)	Fraction of references inside generated-centered neighborhoods. Distinct from coverage; small generated sets can inflate neighborhood radii.
Distinct reference-supported products / requests (%)	Unique generated identities inside reference neighborhoods divided by all requests; exposes invalidity and duplication.
Nearest-reference Tanimoto similarity	Mean nearest-reference similarity over unique generated identities; descriptive, not a novelty or quality objective.
Internal ECFP4 diversity	Mean Tanimoto distance between distinct generated identities; compare alongside validity, realism and component diversity.

Table 13, continued

Metric	Computation and limitation
Descriptor distribution distance, per feature	One-dimensional Wasserstein distance / reference IQR; near-zero fallback: max(reference population SD, 1). Connected-request observations retain duplicates. Smaller means closer to reference, not better chemistry.
Descriptor-space precision and coverage	Robust-scaled descriptor neighborhood metrics from the historical pipeline; new-model source-held-out reference/estimator contract and results remain pending.
Source-grouped two-sample classifier AUC	Distinguish generated and reference molecules with held-out source groups; classifier, sampling, leakage controls and final analysis are not yet qualified for the new model.

Fingerprints are 2,048-bit Morgan fingerprints of radius two (ECFP4). Reference neighborhoods have radii equal to the fifth nonself reference-neighbor Tanimoto distance. Internal diversity is mean Tanimoto distance over distinct generated identity pairs. Descriptor W_1/s_R divides one-dimensional Wasserstein distance by reference IQR; for IQR at most 10^{-12} , $s_R = \max(\text{SD}_{\text{population}}, 1)$. Connected-request descriptor observations retain duplicates. No distribution distance is assigned an acceptability threshold.

Table 14: All 24 descriptor comparisons for every family.

Descriptor	μ_R	μ_S	μ_F	μ_T	W_S/s_R	W_F/s_R	W_T/s_R
A3 ; $n_R = 27, n_S = n_F = n_T = 64$							
Heavy atoms	57.000	67.453	67.453	68.453	0.636	0.636	0.704
Molecular weight	789.498	942.287	942.195	951.521	0.680	0.679	0.729
Calculated logP	14.708	18.205	18.347	18.104	1.078	1.133	1.026
Topological polar surface area	43.113	55.977	54.450	58.942	0.627	0.562	0.685
Hydrogen-bond donors	0.667	0.703	0.641	0.656	0.483	0.452	0.124
Hydrogen-bond acceptors	4.259	5.156	5.094	5.172	0.624	0.591	0.614
Rotatable bonds	36.000	46.719	46.578	46.547	1.707	1.658	1.666
Rings	1.926	1.422	1.422	1.609	0.409	0.409	0.372
Aromatic rings	1.148	0.141	0.141	1.203	0.504	0.504	0.249
Fraction sp^3 carbons	0.788	0.870	0.870	0.813	0.529	0.528	0.222
Net formal charge	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Sum of absolute formal charges	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Carbon fraction	0.924	0.919	0.919	0.918	0.243	0.246	0.280
Nitrogen atoms	1.815	1.656	1.656	1.938	0.221	0.221	0.214
Oxygen atoms	2.444	3.625	3.562	3.609	0.787	0.745	0.777
Sulfur atoms	0.037	0.000	0.000	0.000	0.037	0.037	0.037
Phosphorus atoms	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Branching heavy atoms	6.815	6.500	6.625	8.297	0.288	0.279	0.296
Nonaromatic unsaturated bonds	2.815	4.250	4.219	3.219	1.435	1.404	0.404
Maximum polar-root distance	19.296	22.734	22.797	18.891	0.875	0.891	0.178
Heuristic head fraction	0.229	0.215	0.226	0.216	0.293	0.255	0.276
Heuristic interface fraction	0.082	0.087	0.084	0.096	0.210	0.231	0.263
Heuristic tail fraction	0.689	0.698	0.691	0.688	0.238	0.221	0.206
Heuristic tail branching atoms	1.481	1.047	1.047	1.453	0.362	0.362	0.259
Acid-epoxide ; $n_R = 95, n_S = n_F = n_T = 64$							
Heavy atoms	42.695	44.922	44.922	45.094	0.290	0.290	0.299
Molecular weight	603.717	634.633	634.582	636.490	0.291	0.291	0.299
Calculated logP	10.204	10.087	10.029	10.091	0.229	0.231	0.182
Topological polar surface area	63.868	74.759	75.790	75.176	0.414	0.453	0.430
Hydrogen-bond donors	0.000	0.031	0.047	0.000	0.031	0.047	0.000
Hydrogen-bond acceptors	5.611	6.422	6.500	6.484	0.406	0.445	0.437
Rotatable bonds	30.253	32.312	32.047	30.984	0.299	0.261	0.177
Rings	0.463	0.516	0.516	0.578	0.052	0.052	0.115
Aromatic rings	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Fraction sp^3 carbons	0.891	0.860	0.854	0.845	0.673	0.786	0.939
Net formal charge	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Sum of absolute formal charges	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Carbon fraction	0.866	0.856	0.854	0.856	0.196	0.213	0.149
Nitrogen atoms	1.000	1.047	1.062	1.000	0.047	0.062	0.000
Oxygen atoms	4.611	5.406	5.484	5.484	0.398	0.437	0.437
Sulfur atoms	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Phosphorus atoms	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Branching heavy atoms	5.168	4.938	5.000	5.453	0.292	0.266	0.438
Nonaromatic unsaturated bonds	3.137	4.047	4.156	4.312	0.910	1.019	1.176
Maximum polar-root distance	21.295	23.625	23.406	22.547	1.165	1.056	0.806
Heuristic head fraction	0.151	0.155	0.156	0.158	0.161	0.195	0.219
Heuristic interface fraction	0.186	0.203	0.206	0.200	0.237	0.257	0.179
Heuristic tail fraction	0.662	0.642	0.638	0.642	0.277	0.322	0.271

Table 14, continued

Descriptor	μ_R	μ_S	μ_F	μ_T	W_S/s_R	W_F/s_R	W_T/s_R
Heuristic tail branching atoms	0.737	0.188	0.188	0.797	0.549	0.549	0.250
AEMA ; $n_R = 21$, $n_S = n_F = n_T = 64$							
Heavy atoms	61.190	69.234	69.234	76.062	0.605	0.605	1.072
Molecular weight	904.249	1019.597	1019.597	1121.404	0.647	0.647	1.153
Calculated logP	10.288	13.409	13.403	13.699	0.445	0.444	0.487
Topological polar surface area	118.896	129.558	129.558	147.961	2.913	2.913	4.705
Hydrogen-bond donors	0.095	0.078	0.078	0.047	0.078	0.078	0.048
Hydrogen-bond acceptors	12.714	13.203	13.203	15.250	2.533	2.533	4.004
Rotatable bonds	41.952	51.484	51.453	54.781	0.590	0.588	0.813
Rings	0.524	0.344	0.344	0.422	0.180	0.180	0.102
Aromatic rings	0.048	0.047	0.047	0.016	0.001	0.001	0.032
Fraction sp^3 carbons	0.864	0.900	0.900	0.885	0.539	0.539	0.419
Net formal charge	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Sum of absolute formal charges	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Carbon fraction	0.786	0.808	0.808	0.800	0.323	0.323	0.228
Nitrogen atoms	2.286	1.875	1.875	2.234	0.411	0.411	0.261
Oxygen atoms	8.429	9.312	9.312	10.672	2.009	2.009	3.337
Sulfur atoms	2.048	2.062	2.062	2.359	0.359	0.359	0.624
Phosphorus atoms	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Branching heavy atoms	9.619	8.891	8.938	11.031	1.129	1.170	2.068
Nonaromatic unsaturated bonds	5.286	5.016	5.016	6.172	0.551	0.551	0.596
Maximum polar-root distance	24.190	26.438	26.375	25.328	0.225	0.221	0.148
Heuristic head fraction	0.159	0.136	0.135	0.134	0.290	0.305	0.321
Heuristic interface fraction	0.256	0.234	0.234	0.245	0.304	0.301	0.203
Heuristic tail fraction	0.584	0.630	0.631	0.621	0.294	0.300	0.302
Heuristic tail branching atoms	3.190	2.391	2.438	3.359	0.348	0.329	0.408
Aldehyde Ugi3 ; $n_R = 12$, $n_S = n_F = n_T = 64$							
Heavy atoms	55.083	45.891	45.891	47.328	6.128	6.128	5.170
Molecular weight	776.644	649.100	649.101	669.387	4.502	4.502	3.786
Calculated logP	13.015	10.124	10.114	10.517	1.930	1.937	1.668
Topological polar surface area	78.799	73.951	73.687	74.696	0.524	0.578	0.388
Hydrogen-bond donors	2.167	2.094	2.094	2.094	0.073	0.073	0.073
Hydrogen-bond acceptors	5.500	5.203	5.188	5.312	0.484	0.594	0.188
Rotatable bonds	42.333	34.859	35.016	36.172	1.759	1.722	1.450
Rings	1.167	0.734	0.734	0.734	0.432	0.432	0.432
Aromatic rings	0.083	0.031	0.031	0.047	0.052	0.052	0.036
Fraction sp^3 carbons	0.924	0.943	0.944	0.944	0.317	0.312	0.328
Net formal charge	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Sum of absolute formal charges	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Carbon fraction	0.880	0.864	0.864	0.865	0.937	0.921	0.871
Nitrogen atoms	3.333	3.141	3.141	3.266	0.193	0.193	0.068
Oxygen atoms	3.250	3.094	3.078	3.094	1.375	1.688	0.625
Sulfur atoms	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Phosphorus atoms	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Branching heavy atoms	4.750	4.172	4.172	4.172	0.289	0.289	0.289
Nonaromatic unsaturated bonds	2.667	2.062	2.047	2.047	0.302	0.310	0.310
Maximum polar-root distance	29.583	23.047	23.156	23.484	3.268	3.214	3.049
Heuristic head fraction	0.155	0.182	0.180	0.170	1.569	1.401	0.919
Heuristic interface fraction	0.130	0.149	0.149	0.145	1.471	1.439	1.015
Heuristic tail fraction	0.715	0.669	0.671	0.686	4.023	3.811	2.648
Heuristic tail branching atoms	0.000	0.047	0.031	0.000	0.047	0.031	0.000
Aldehyde Ugi4 ; $n_R = 61$, $n_S = n_F = n_T = 64$							
Heavy atoms	47.705	46.484	46.484	45.484	0.163	0.163	0.181
Molecular weight	679.510	660.117	658.214	639.613	0.161	0.170	0.207
Calculated logP	9.265	9.572	9.486	8.667	0.099	0.088	0.143
Topological polar surface area	83.729	72.315	73.072	83.994	0.218	0.207	0.139
Hydrogen-bond donors	1.393	1.453	1.594	1.672	0.060	0.200	0.278
Hydrogen-bond acceptors	5.082	4.500	4.531	4.859	0.203	0.184	0.158
Rotatable bonds	28.066	28.828	28.828	26.000	0.125	0.134	0.258
Rings	1.951	1.984	2.016	2.281	0.093	0.123	0.439
Aromatic rings	0.459	0.125	0.125	0.750	0.334	0.334	0.291
Fraction sp^3 carbons	0.868	0.906	0.902	0.842	0.346	0.317	0.227
Net formal charge	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Sum of absolute formal charges	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Carbon fraction	0.841	0.855	0.854	0.836	0.218	0.207	0.155
Nitrogen atoms	3.492	3.531	3.594	3.969	0.274	0.199	0.477
Oxygen atoms	3.754	3.000	2.969	3.203	0.251	0.262	0.184
Sulfur atoms	0.115	0.031	0.047	0.078	0.084	0.068	0.037
Phosphorus atoms	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Branching heavy atoms	8.787	7.438	7.438	8.734	0.337	0.337	0.120

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Table 14, continued

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Descriptor	μ_R	μ_S	μ_F	μ_T	W_S/s_R	W_F/s_R	W_T/s_R
Nonaromatic unsaturated bonds	3.180	2.797	2.844	2.844	0.192	0.168	0.168
Maximum polar-root distance	18.459	18.922	19.078	17.562	0.172	0.157	0.155
Heuristic head fraction	0.179	0.184	0.178	0.179	0.156	0.147	0.145
Heuristic interface fraction	0.208	0.185	0.186	0.207	0.483	0.471	0.101
Heuristic tail fraction	0.613	0.631	0.636	0.614	0.240	0.298	0.141
Heuristic tail branching atoms	1.639	1.141	1.141	1.547	0.207	0.207	0.072
α-Isocyanoester; $n_R = 103, n_S = n_F = n_T = 64$							
Heavy atoms	38.573	40.422	40.422	40.234	0.357	0.357	0.340
Molecular weight	542.886	569.510	569.840	565.588	0.357	0.359	0.332
Calculated logP	7.479	8.200	8.231	8.124	0.318	0.324	0.322
Topological polar surface area	60.667	60.012	59.381	59.300	0.086	0.093	0.042
Hydrogen-bond donors	0.000	0.250	0.203	0.000	0.250	0.203	0.000
Hydrogen-bond acceptors	6.204	6.078	6.062	6.109	0.100	0.112	0.046
Rotatable bonds	21.495	22.078	22.094	21.188	0.248	0.250	0.330
Rings	1.767	1.672	1.672	1.641	0.095	0.095	0.126
Aromatic rings	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Fraction sp^3 carbons	0.895	0.907	0.910	0.882	0.234	0.256	0.191
Net formal charge	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Sum of absolute formal charges	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Carbon fraction	0.835	0.850	0.850	0.849	0.222	0.224	0.194
Nitrogen atoms	3.000	3.000	3.016	3.000	0.000	0.016	0.000
Oxygen atoms	3.204	3.078	3.047	3.109	0.100	0.116	0.046
Sulfur atoms	0.000	0.016	0.031	0.000	0.016	0.031	0.000
Phosphorus atoms	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Branching heavy atoms	6.485	6.922	6.922	7.094	0.229	0.229	0.306
Nonaromatic unsaturated bonds	2.932	2.719	2.688	3.266	0.138	0.154	0.167
Maximum polar-root distance	10.417	13.312	13.344	10.328	0.724	0.732	0.123
Heuristic head fraction	0.336	0.308	0.308	0.318	0.351	0.351	0.251
Heuristic interface fraction	0.174	0.165	0.164	0.169	0.182	0.191	0.117
Heuristic tail fraction	0.490	0.527	0.528	0.513	0.302	0.308	0.288
Heuristic tail branching atoms	0.000	0.125	0.125	0.000	0.125	0.125	0.000
Amine alkylation; $n_R = 171, n_S = n_F = n_T = 64$							
Heavy atoms	50.825	47.781	47.781	48.938	0.447	0.447	0.419
Molecular weight	715.903	675.129	675.019	689.989	0.443	0.444	0.424
Calculated logP	11.269	11.154	11.113	11.221	0.221	0.230	0.241
Topological polar surface area	64.281	59.754	59.809	60.671	0.490	0.485	0.391
Hydrogen-bond donors	0.070	0.031	0.031	0.031	0.039	0.039	0.039
Hydrogen-bond acceptors	6.491	6.000	6.016	6.125	0.491	0.476	0.366
Rotatable bonds	35.801	35.766	35.703	35.891	0.203	0.203	0.237
Rings	0.883	0.484	0.484	0.484	0.399	0.399	0.399
Aromatic rings	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Fraction sp^3 carbons	0.858	0.904	0.902	0.869	1.761	1.725	0.720
Net formal charge	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Sum of absolute formal charges	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Carbon fraction	0.868	0.872	0.872	0.873	0.231	0.235	0.252
Nitrogen atoms	2.018	1.953	1.969	1.984	0.080	0.064	0.111
Oxygen atoms	4.474	4.047	4.047	4.141	0.427	0.427	0.333
Sulfur atoms	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Phosphorus atoms	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Branching heavy atoms	5.947	5.078	5.016	5.328	1.163	1.131	0.619
Nonaromatic unsaturated bonds	4.000	3.000	3.062	3.781	1.312	1.375	0.219
Maximum polar-root distance	15.357	16.938	16.984	16.547	0.527	0.543	0.397
Heuristic head fraction	0.200	0.211	0.212	0.208	0.604	0.581	0.491
Heuristic interface fraction	0.163	0.156	0.156	0.152	0.207	0.202	0.277
Heuristic tail fraction	0.637	0.633	0.632	0.641	0.213	0.209	0.222
Heuristic tail branching atoms	1.193	0.641	0.641	1.031	0.339	0.339	0.081
Amine-epoxide; $n_R = 111, n_S = n_F = n_T = 64$							
Heavy atoms	54.856	56.078	56.078	54.281	0.565	0.565	0.431
Molecular weight	778.470	793.699	793.584	767.246	0.612	0.613	0.463
Calculated logP	9.041	12.229	12.121	10.437	0.477	0.461	0.225
Topological polar surface area	122.390	90.050	91.530	101.483	0.551	0.531	0.390
Hydrogen-bond donors	3.550	3.078	3.062	3.109	0.895	0.879	0.645
Hydrogen-bond acceptors	9.000	6.703	6.812	7.500	0.459	0.438	0.330
Rotatable bonds	38.829	44.078	43.891	39.188	0.846	0.837	0.489
Rings	0.378	0.375	0.375	0.266	0.003	0.003	0.113
Aromatic rings	0.009	0.062	0.062	0.016	0.053	0.053	0.007
Fraction sp^3 carbons	0.872	0.891	0.887	0.844	0.198	0.190	0.241
Net formal charge	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Sum of absolute formal charges	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Carbon fraction	0.832	0.871	0.870	0.859	0.426	0.407	0.304

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Table 14, continued

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Descriptor	μ_R	μ_S	μ_F	μ_T	W_S/s_R	W_F/s_R	W_T/s_R
Nitrogen atoms	2.072	2.062	2.078	1.953	0.269	0.285	0.244
Oxygen atoms	6.937	4.703	4.797	5.562	0.447	0.428	0.307
Sulfur atoms	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Phosphorus atoms	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Branching heavy atoms	8.441	6.328	6.406	7.484	0.325	0.313	0.186
Nonaromatic unsaturated bonds	3.937	2.594	2.750	4.203	0.336	0.297	0.226
Maximum polar-root distance	15.324	17.766	17.844	16.000	0.488	0.504	0.160
Heuristic head fraction	0.215	0.209	0.209	0.220	0.373	0.368	0.305
Heuristic interface fraction	0.197	0.133	0.135	0.152	0.417	0.404	0.292
Heuristic tail fraction	0.589	0.658	0.656	0.628	0.489	0.472	0.275
Heuristic tail branching atoms	0.541	0.656	0.656	1.172	0.171	0.171	0.548
Aryl reductive amination; $n_R = 79, n_S = n_F = n_T = 64$							
Heavy atoms	61.468	63.094	63.094	63.312	0.329	0.329	0.321
Molecular weight	853.082	880.870	880.931	880.683	0.340	0.341	0.331
Calculated logP	14.686	13.541	13.519	14.608	1.224	1.233	1.123
Topological polar surface area	78.340	99.908	100.318	86.130	0.730	0.744	0.264
Hydrogen-bond donors	1.000	1.172	1.172	1.000	0.172	0.172	0.000
Hydrogen-bond acceptors	7.063	9.062	9.094	7.812	0.666	0.677	0.250
Rotatable bonds	40.608	40.641	40.578	40.797	0.908	0.917	0.837
Rings	1.354	1.766	1.766	1.562	0.411	0.411	0.208
Aromatic rings	1.000	0.984	0.984	1.000	0.016	0.016	0.000
Fraction sp^3 carbons	0.677	0.714	0.714	0.694	4.579	4.554	3.747
Net formal charge	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Sum of absolute formal charges	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Carbon fraction	0.885	0.853	0.853	0.874	1.787	1.826	0.658
Nitrogen atoms	2.354	2.656	2.656	2.562	0.302	0.302	0.208
Oxygen atoms	4.709	6.406	6.438	5.250	0.849	0.864	0.271
Sulfur atoms	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Phosphorus atoms	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Branching heavy atoms	7.190	9.250	9.281	8.797	0.687	0.697	0.536
Nonaromatic unsaturated bonds	6.937	5.906	5.906	6.500	0.472	0.472	0.146
Maximum polar-root distance	26.848	26.266	26.203	23.812	1.037	1.047	1.055
Heuristic head fraction	0.115	0.139	0.139	0.126	1.730	1.730	1.099
Heuristic interface fraction	0.142	0.179	0.181	0.164	0.927	0.970	0.544
Heuristic tail fraction	0.743	0.682	0.680	0.710	1.064	1.094	0.560
Heuristic tail branching atoms	1.772	2.484	2.469	2.172	0.931	0.915	0.722
Aza-Michael acrylamide; $n_R = 33, n_S = n_F = n_T = 64$							
Heavy atoms	56.485	50.984	50.984	51.516	0.281	0.281	0.256
Molecular weight	796.183	718.038	717.959	723.960	0.283	0.283	0.259
Calculated logP	12.303	10.952	10.947	10.880	0.273	0.271	0.267
Topological polar surface area	91.703	76.298	76.495	77.410	0.800	0.790	0.764
Hydrogen-bond donors	3.182	2.516	2.516	2.484	0.666	0.666	0.697
Hydrogen-bond acceptors	4.758	4.312	4.328	4.312	0.601	0.586	0.633
Rotatable bonds	44.030	38.453	38.469	37.672	0.299	0.300	0.293
Rings	0.091	0.375	0.375	0.391	0.284	0.284	0.300
Aromatic rings	0.000	0.031	0.031	0.141	0.031	0.031	0.141
Fraction sp^3 carbons	0.882	0.883	0.883	0.852	0.206	0.169	0.340
Net formal charge	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Sum of absolute formal charges	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Carbon fraction	0.865	0.872	0.872	0.871	0.222	0.216	0.236
Nitrogen atoms	3.879	3.828	3.844	4.000	0.274	0.266	0.235
Oxygen atoms	3.303	2.609	2.609	2.562	0.694	0.694	0.741
Sulfur atoms	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Phosphorus atoms	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Branching heavy atoms	4.818	4.984	4.875	5.719	0.746	0.605	0.994
Nonaromatic unsaturated bonds	4.121	3.562	3.609	4.125	0.140	0.128	0.316
Maximum polar-root distance	19.636	18.703	18.938	18.500	0.187	0.195	0.250
Heuristic head fraction	0.161	0.175	0.175	0.172	0.230	0.231	0.221
Heuristic interface fraction	0.172	0.158	0.156	0.161	0.250	0.265	0.225
Heuristic tail fraction	0.668	0.667	0.669	0.666	0.236	0.243	0.245
Heuristic tail branching atoms	0.000	0.203	0.188	0.688	0.203	0.188	0.688
Aza-Michael acrylate; $n_R = 11, n_S = n_F = n_T = 64$							
Heavy atoms	45.909	50.938	50.938	51.828	11.040	11.040	12.406
Molecular weight	649.262	719.685	719.622	731.559	8.991	8.969	10.113
Calculated logP	8.442	11.660	11.653	11.660	1.551	1.548	1.572
Topological polar surface area	83.094	73.323	73.323	74.899	0.466	0.466	0.387
Hydrogen-bond donors	0.273	0.516	0.516	0.562	0.486	0.486	0.580
Hydrogen-bond acceptors	7.636	6.547	6.547	6.672	0.726	0.726	0.643
Rotatable bonds	31.000	38.188	38.172	38.109	2.888	2.881	2.844
Rings	0.273	0.297	0.297	0.297	0.048	0.048	0.048

1944	Table 14, continued							
1945	Descriptor	μ_R	μ_S	μ_F	μ_T	W_S/s_R	W_F/s_R	W_T/s_R
1946	Aromatic rings	0.000	0.031	0.031	0.031	0.031	0.031	0.031
1947	Fraction sp ³ carbons	0.825	0.882	0.880	0.863	0.816	0.798	0.609
1948	Net formal charge	0.000	0.000	0.000	0.000	0.000	0.000	0.000
1949	Sum of absolute formal charges	0.000	0.000	0.000	0.000	0.000	0.000	0.000
1950	Carbon fraction	0.834	0.869	0.869	0.869	1.683	1.683	1.701
1951	Nitrogen atoms	1.727	1.703	1.703	1.719	0.173	0.173	0.142
1952	Oxygen atoms	5.909	4.875	4.875	4.984	2.068	2.068	1.849
1953	Sulfur atoms	0.000	0.000	0.000	0.000	0.000	0.000	0.000
1954	Phosphorus atoms	0.000	0.000	0.000	0.000	0.000	0.000	0.000
1955	Branching heavy atoms	6.636	5.219	5.234	5.797	0.504	0.499	0.378
1956	Nonaromatic unsaturated bonds	4.727	3.672	3.703	4.094	0.575	0.559	0.493
1957	Maximum polar-root distance	14.909	18.484	18.438	17.281	1.022	1.008	0.678
1958	Heuristic head fraction	0.185	0.175	0.173	0.169	0.316	0.336	0.350
1959	Heuristic interface fraction	0.213	0.164	0.165	0.171	1.459	1.452	1.258
1960	Heuristic tail fraction	0.601	0.661	0.662	0.660	4.837	4.916	4.580
1961	Heuristic tail branching atoms	1.091	0.312	0.312	0.422	0.420	0.420	0.335
1962	Disulfide Michael ; $n_R = 128$, $n_S = n_F = n_T = 64$							
1963	Heavy atoms	74.820	88.672	88.672	83.828	1.798	1.798	1.421
1964	Molecular weight	1169.589	1342.853	1341.749	1311.117	1.840	1.819	1.640
1965	Calculated logP	14.594	20.048	20.283	16.331	1.086	1.133	0.390
1966	Topological polar surface area	149.533	132.604	129.758	152.348	0.770	0.891	0.776
1967	Hydrogen-bond donors	1.148	0.547	0.562	0.359	0.426	0.418	0.395
1968	Hydrogen-bond acceptors	17.766	16.250	16.000	19.859	2.234	2.359	2.609
1969	Rotatable bonds	56.539	70.719	70.719	62.516	1.719	1.719	0.999
1970	Rings	0.203	0.375	0.375	0.359	0.203	0.203	0.250
1971	Aromatic rings	0.125	0.062	0.031	0.109	0.062	0.094	0.078
1972	Fraction sp ³ carbons	0.855	0.872	0.874	0.839	0.310	0.308	0.276
1973	Net formal charge	0.000	0.000	0.000	0.000	0.000	0.000	0.000
1974	Sum of absolute formal charges	0.000	0.000	0.000	0.000	0.000	0.000	0.000
1975	Carbon fraction	0.760	0.806	0.808	0.760	0.842	0.890	0.127
1976	Nitrogen atoms	1.164	2.016	2.000	2.172	0.852	0.836	1.008
1977	Oxygen atoms	10.617	9.375	9.203	10.797	0.849	0.936	1.091
1978	Sulfur atoms	6.047	5.031	4.969	6.984	2.109	2.016	0.938
1979	Phosphorus atoms	0.000	0.000	0.000	0.000	0.000	0.000	0.000
1980	Branching heavy atoms	8.609	8.297	8.172	10.203	0.273	0.320	0.797
1981	Nonaromatic unsaturated bonds	6.047	6.719	6.562	7.453	1.255	1.147	1.214
1982	Maximum polar-root distance	29.836	32.062	31.984	29.906	0.325	0.318	0.171
1983	Heuristic head fraction	0.115	0.102	0.105	0.109	0.361	0.281	0.250
1984	Heuristic interface fraction	0.285	0.232	0.227	0.284	0.732	0.800	0.126
1985	Heuristic tail fraction	0.600	0.666	0.668	0.606	0.625	0.641	0.163
1986	Heuristic tail branching atoms	2.297	1.891	1.859	2.797	0.750	0.750	0.500
1987	Epoxide/O-acylation ; $n_R = 70$, $n_S = n_F = n_T = 64$							
1988	Heavy atoms	61.443	75.484	75.484	72.266	2.006	2.006	1.546
1989	Molecular weight	864.903	1063.155	1063.202	1015.782	1.884	1.885	1.434
1990	Calculated logP	16.081	21.883	21.869	20.097	2.031	2.026	1.406
1991	Topological polar surface area	62.028	59.160	59.353	60.118	0.515	0.483	0.357
1992	Hydrogen-bond donors	0.000	0.016	0.016	0.000	0.016	0.016	0.000
1993	Hydrogen-bond acceptors	6.143	5.984	6.000	6.109	0.228	0.213	0.103
1994	Rotatable bonds	46.329	61.094	61.047	55.938	2.109	2.103	1.373
1995	Rings	0.643	0.469	0.469	0.516	0.174	0.174	0.127
1996	Aromatic rings	0.100	0.000	0.000	0.047	0.100	0.100	0.053
1997	Fraction sp ³ carbons	0.883	0.934	0.935	0.895	0.708	0.718	0.306
1998	Net formal charge	0.000	0.000	0.000	0.000	0.000	0.000	0.000
1999	Sum of absolute formal charges	0.000	0.000	0.000	0.000	0.000	0.000	0.000
2000	Carbon fraction	0.894	0.920	0.920	0.913	1.204	1.195	0.853
2001	Nitrogen atoms	2.071	1.984	2.000	2.156	0.250	0.242	0.164
2002	Oxygen atoms	4.171	4.000	4.000	4.000	0.171	0.171	0.171
2003	Sulfur atoms	0.000	0.000	0.000	0.000	0.000	0.000	0.000
2004	Phosphorus atoms	0.000	0.000	0.000	0.000	0.000	0.000	0.000
2005	Branching heavy atoms	7.143	7.312	7.375	8.266	0.254	0.241	0.561
2006	Nonaromatic unsaturated bonds	3.743	3.266	3.250	4.312	0.301	0.309	0.285
2007	Maximum polar-root distance	15.486	19.594	19.656	17.375	0.587	0.596	0.270
2008	Heuristic head fraction	0.173	0.137	0.137	0.147	0.393	0.395	0.297
2009	Heuristic interface fraction	0.136	0.103	0.105	0.123	0.945	0.911	0.442
2010	Heuristic tail fraction	0.690	0.759	0.758	0.731	0.754	0.743	0.444
2011	Heuristic tail branching atoms	0.186	0.344	0.344	0.422	0.207	0.207	0.236
2012	iPhos ; $n_R = 62$, $n_S = n_F = n_T = 64$							
2013	Heavy atoms	44.742	49.766	49.766	51.234	3.522	3.522	3.330
2014	Molecular weight	657.213	729.860	729.890	751.014	3.569	3.568	3.418
2015	Calculated logP	8.758	12.750	12.730	12.318	2.865	2.856	2.433

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Table 14, continued

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Descriptor	μ_R	μ_S	μ_F	μ_T	W_S/s_R	W_F/s_R	W_T/s_R
Topological polar surface area	109.007	85.245	85.390	96.907	0.968	0.964	0.957
Hydrogen-bond donors	0.161	0.484	0.484	0.531	0.323	0.323	0.370
Hydrogen-bond acceptors	8.387	6.375	6.391	7.203	0.993	0.988	0.988
Rotatable bonds	33.548	41.688	41.672	40.859	3.406	3.411	2.772
Rings	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Aromatic rings	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Fraction sp^3 carbons	0.845	0.943	0.942	0.902	6.751	6.727	3.877
Net formal charge	-0.935	-0.938	-0.938	-1.016	0.211	0.211	0.258
Sum of absolute formal charges	1.258	1.469	1.469	1.516	0.211	0.211	0.258
Carbon fraction	0.780	0.836	0.835	0.828	3.526	3.500	3.078
Nitrogen atoms	1.097	1.219	1.219	1.266	0.122	0.122	0.169
Oxygen atoms	7.452	5.422	5.438	6.188	1.132	1.125	1.123
Sulfur atoms	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Phosphorus atoms	1.097	1.203	1.203	1.266	0.106	0.106	0.169
Branching heavy atoms	4.016	2.656	2.672	3.859	1.025	1.013	0.798
Nonaromatic unsaturated bonds	4.532	2.578	2.578	3.656	1.487	1.487	1.107
Maximum polar-root distance	15.000	19.750	19.719	17.125	1.594	1.583	0.718
Heuristic head fraction	0.202	0.195	0.195	0.187	0.867	0.867	0.813
Heuristic interface fraction	0.232	0.147	0.147	0.160	4.896	4.872	4.098
Heuristic tail fraction	0.566	0.659	0.658	0.653	1.298	1.292	1.220
Heuristic tail branching atoms	0.113	0.047	0.047	0.281	0.066	0.066	0.168
Ketone-isocyanide ; $n_R = 29$, $n_S = n_F = n_T = 64$							
Heavy atoms	35.517	33.672	33.672	34.266	0.365	0.365	0.293
Molecular weight	498.826	474.605	474.652	482.682	0.358	0.358	0.289
Calculated logP	7.921	7.351	7.343	7.489	0.337	0.341	0.287
Topological polar surface area	44.370	44.366	44.558	44.370	0.004	0.188	0.000
Hydrogen-bond donors	2.000	2.000	2.016	2.000	0.000	0.016	0.000
Hydrogen-bond acceptors	3.000	3.000	3.016	3.000	0.000	0.016	0.000
Rotatable bonds	21.966	22.797	22.828	22.562	0.344	0.340	0.308
Rings	1.379	0.484	0.484	0.609	0.895	0.895	0.770
Aromatic rings	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Fraction sp^3 carbons	0.915	0.911	0.912	0.911	0.230	0.222	0.210
Net formal charge	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Sum of absolute formal charges	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Carbon fraction	0.884	0.880	0.880	0.882	0.334	0.340	0.265
Nitrogen atoms	3.000	3.000	3.016	3.000	0.000	0.016	0.000
Oxygen atoms	1.000	1.000	1.000	1.000	0.000	0.000	0.000
Sulfur atoms	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Phosphorus atoms	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Branching heavy atoms	3.966	3.766	3.750	4.250	0.399	0.384	0.665
Nonaromatic unsaturated bonds	1.759	1.828	1.812	1.828	0.311	0.295	0.342
Maximum polar-root distance	14.379	14.047	14.062	13.688	0.357	0.337	0.357
Heuristic head fraction	0.229	0.201	0.201	0.202	0.551	0.535	0.525
Heuristic interface fraction	0.117	0.125	0.125	0.134	0.332	0.325	0.552
Heuristic tail fraction	0.654	0.674	0.673	0.664	0.331	0.311	0.253
Heuristic tail branching atoms	0.931	0.594	0.578	0.703	0.369	0.353	0.496
Ketone Ugi4 ; $n_R = 29$, $n_S = n_F = n_T = 64$							
Heavy atoms	65.690	72.781	72.781	71.984	0.483	0.483	0.436
Molecular weight	917.762	1020.966	1020.966	1006.236	0.477	0.477	0.416
Calculated logP	11.939	15.606	15.606	13.763	0.610	0.610	0.351
Topological polar surface area	118.749	109.870	109.870	123.891	0.553	0.553	0.208
Hydrogen-bond donors	1.241	1.688	1.688	1.094	0.446	0.446	0.148
Hydrogen-bond acceptors	8.000	7.188	7.188	8.531	0.637	0.637	0.266
Rotatable bonds	37.724	48.641	48.641	42.234	0.646	0.646	0.298
Rings	1.552	2.234	2.234	2.016	0.357	0.357	0.233
Aromatic rings	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Fraction sp^3 carbons	0.782	0.879	0.879	0.809	1.082	1.082	0.298
Net formal charge	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Sum of absolute formal charges	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Carbon fraction	0.845	0.871	0.871	0.851	0.702	0.702	0.215
Nitrogen atoms	3.138	3.266	3.266	3.203	0.128	0.128	0.065
Oxygen atoms	6.862	6.016	6.016	7.328	0.642	0.642	0.233
Sulfur atoms	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Phosphorus atoms	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Branching heavy atoms	9.931	10.219	10.219	10.969	0.291	0.291	0.274
Nonaromatic unsaturated bonds	8.138	5.656	5.656	8.094	1.241	1.241	0.140
Maximum polar-root distance	22.207	27.781	27.781	22.172	0.557	0.557	0.180
Heuristic head fraction	0.117	0.113	0.113	0.123	0.143	0.143	0.219
Heuristic interface fraction	0.225	0.179	0.179	0.212	0.694	0.694	0.243
Heuristic tail fraction	0.658	0.708	0.708	0.665	0.583	0.583	0.211

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Table 14, continued

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Descriptor	μ_R	μ_S	μ_F	μ_T	W_S/s_R	W_F/s_R	W_T/s_R
Heuristic tail branching atoms	1.862	3.656	3.656	2.625	0.600	0.600	0.254
Maleate ; $n_R = 49, n_S = n_F = n_T = 64$							
Heavy atoms	53.184	80.609	80.609	74.266	2.748	2.748	2.146
Molecular weight	754.986	1136.892	1136.892	1048.998	2.751	2.751	2.151
Calculated logP	10.219	18.488	18.488	14.621	1.760	1.760	0.939
Topological polar surface area	102.466	128.366	128.366	139.873	0.448	0.448	0.658
Hydrogen-bond donors	1.245	1.156	1.156	1.219	0.139	0.139	0.139
Hydrogen-bond acceptors	8.592	10.750	10.750	11.594	0.436	0.436	0.600
Rotatable bonds	38.061	61.734	61.734	52.859	6.072	6.072	4.384
Rings	0.367	0.406	0.406	0.531	0.039	0.039	0.164
Aromatic rings	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Fraction sp^3 carbons	0.864	0.861	0.861	0.833	0.238	0.238	0.497
Net formal charge	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Sum of absolute formal charges	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Carbon fraction	0.837	0.861	0.861	0.841	0.283	0.283	0.119
Nitrogen atoms	2.102	2.516	2.516	2.703	0.289	0.289	0.301
Oxygen atoms	6.490	8.250	8.250	9.094	0.445	0.445	0.688
Sulfur atoms	0.122	0.016	0.016	0.047	0.107	0.107	0.076
Phosphorus atoms	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Branching heavy atoms	7.224	8.328	8.328	10.156	0.344	0.344	0.733
Nonaromatic unsaturated bonds	4.653	6.969	6.969	7.500	0.579	0.579	0.712
Maximum polar-root distance	17.265	22.828	22.828	19.906	0.695	0.695	0.344
Heuristic head fraction	0.146	0.124	0.124	0.127	1.069	1.069	0.938
Heuristic interface fraction	0.219	0.171	0.171	0.213	0.522	0.522	0.161
Heuristic tail fraction	0.635	0.706	0.706	0.660	0.569	0.569	0.212
Heuristic tail branching atoms	0.571	0.938	0.938	1.344	0.320	0.320	0.676
O-esterification ; $n_R = 43, n_S = n_F = n_T = 64$							
Heavy atoms	33.581	36.688	36.688	35.812	0.571	0.571	0.539
Molecular weight	475.067	515.698	515.872	503.140	0.579	0.577	0.550
Calculated logP	6.520	8.807	8.817	8.224	1.189	1.203	0.951
Topological polar surface area	63.481	55.021	54.664	55.028	0.322	0.335	0.321
Hydrogen-bond donors	0.000	0.172	0.141	0.125	0.172	0.141	0.125
Hydrogen-bond acceptors	5.651	4.484	4.469	4.484	0.583	0.591	0.583
Rotatable bonds	22.279	25.250	25.219	23.641	0.563	0.552	0.519
Rings	0.093	0.562	0.562	0.594	0.469	0.469	0.501
Aromatic rings	0.000	0.328	0.328	0.391	0.328	0.328	0.391
Fraction sp^3 carbons	0.778	0.801	0.810	0.784	0.432	0.481	0.521
Net formal charge	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Sum of absolute formal charges	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Carbon fraction	0.826	0.861	0.861	0.856	2.625	2.690	2.348
Nitrogen atoms	1.093	1.438	1.422	1.500	0.344	0.329	0.407
Oxygen atoms	4.558	3.375	3.375	3.375	0.592	0.592	0.592
Sulfur atoms	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Phosphorus atoms	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Branching heavy atoms	4.326	4.344	4.312	4.828	0.532	0.485	0.251
Nonaromatic unsaturated bonds	3.907	3.250	3.156	3.266	2.198	2.323	1.541
Maximum polar-root distance	12.256	16.672	16.516	15.219	2.944	2.840	2.006
Heuristic head fraction	0.206	0.237	0.237	0.241	1.215	1.217	1.385
Heuristic interface fraction	0.235	0.160	0.160	0.177	1.051	1.047	0.811
Heuristic tail fraction	0.559	0.603	0.603	0.582	0.659	0.657	0.532
Heuristic tail branching atoms	0.000	0.125	0.125	0.219	0.125	0.125	0.219
Passerini ; $n_R = 68, n_S = n_F = n_T = 64$							
Heavy atoms	46.779	46.016	46.016	44.000	0.376	0.376	0.406
Molecular weight	656.738	646.808	646.840	618.064	0.369	0.369	0.398
Calculated logP	11.123	11.393	11.394	10.530	0.207	0.205	0.234
Topological polar surface area	64.383	60.911	60.911	60.745	1.267	1.267	1.154
Hydrogen-bond donors	1.000	1.000	1.000	1.000	0.000	0.000	0.000
Hydrogen-bond acceptors	4.603	4.203	4.203	4.172	0.462	0.462	0.462
Rotatable bonds	32.118	33.484	33.500	31.062	0.234	0.231	0.230
Rings	1.074	0.625	0.625	0.469	0.449	0.449	0.605
Aromatic rings	0.294	0.125	0.125	0.109	0.169	0.169	0.185
Fraction sp^3 carbons	0.858	0.863	0.864	0.836	0.144	0.143	0.218
Net formal charge	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Sum of absolute formal charges	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Carbon fraction	0.879	0.886	0.886	0.881	0.211	0.211	0.099
Nitrogen atoms	2.221	2.016	2.016	2.016	0.205	0.205	0.205
Oxygen atoms	3.382	3.188	3.188	3.156	0.226	0.226	0.226
Sulfur atoms	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Phosphorus atoms	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Branching heavy atoms	5.603	4.703	4.719	5.172	0.450	0.442	0.216

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Table 14, continued

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Descriptor	μ_R	μ_S	μ_F	μ_T	W_S/s_R	W_F/s_R	W_T/s_R
Nonaromatic unsaturated bonds	3.162	3.438	3.422	3.844	0.138	0.130	0.341
Maximum polar-root distance	22.029	23.750	23.719	21.125	0.423	0.416	0.234
Heuristic head fraction	0.183	0.166	0.166	0.159	0.365	0.365	0.478
Heuristic interface fraction	0.165	0.154	0.154	0.168	0.227	0.227	0.111
Heuristic tail fraction	0.652	0.680	0.680	0.673	0.413	0.413	0.317
Heuristic tail branching atoms	0.529	0.547	0.562	0.812	0.058	0.042	0.283
Preassembled thiol-yne; $n_R = 379, n_S = n_F = n_T = 64$							
Heavy atoms	45.153	47.250	47.250	45.781	0.233	0.233	0.163
Molecular weight	670.309	698.470	698.549	677.574	0.236	0.237	0.165
Calculated logP	10.970	12.893	12.883	11.817	0.385	0.383	0.207
Topological polar surface area	37.218	30.360	29.861	31.857	0.617	0.604	0.410
Hydrogen-bond donors	0.491	0.734	0.672	0.578	0.244	0.181	0.087
Hydrogen-bond acceptors	5.042	4.109	4.125	4.422	0.933	0.917	0.620
Rotatable bonds	31.079	34.641	34.641	32.094	0.324	0.324	0.160
Rings	1.459	0.797	0.797	0.781	0.662	0.662	0.678
Aromatic rings	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Fraction sp^3 carbons	0.931	0.910	0.911	0.886	0.249	0.239	0.528
Net formal charge	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Sum of absolute formal charges	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Carbon fraction	0.863	0.891	0.891	0.880	0.525	0.520	0.327
Nitrogen atoms	2.348	2.109	2.125	2.234	0.239	0.223	0.114
Oxygen atoms	1.694	1.000	1.000	1.188	0.694	0.694	0.506
Sulfur atoms	2.000	2.000	2.000	2.000	0.000	0.000	0.000
Phosphorus atoms	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Branching heavy atoms	5.546	4.953	5.094	6.016	0.312	0.242	0.317
Nonaromatic unsaturated bonds	1.918	2.406	2.375	2.812	0.244	0.228	0.447
Maximum polar-root distance	22.325	23.719	23.672	21.562	0.235	0.230	0.214
Heuristic head fraction	0.222	0.179	0.178	0.188	0.588	0.591	0.456
Heuristic interface fraction	0.137	0.110	0.110	0.121	0.564	0.564	0.351
Heuristic tail fraction	0.641	0.712	0.712	0.691	0.669	0.671	0.468
Heuristic tail branching atoms	1.470	1.891	1.922	2.734	0.210	0.226	0.632
Reductive amination; $n_R = 144, n_S = n_F = n_T = 64$							
Heavy atoms	55.681	60.703	60.703	60.016	0.390	0.390	0.345
Molecular weight	785.244	863.134	863.134	845.407	0.427	0.427	0.347
Calculated logP	11.628	10.618	10.618	12.630	0.395	0.395	0.610
Topological polar surface area	96.990	143.462	143.462	106.306	5.668	5.668	1.140
Hydrogen-bond donors	1.458	4.000	4.000	1.672	2.542	2.542	0.214
Hydrogen-bond acceptors	7.896	10.078	10.078	8.328	2.214	2.214	0.488
Rotatable bonds	39.264	45.125	45.125	42.891	0.652	0.652	0.446
Rings	0.847	0.766	0.766	0.844	0.314	0.314	0.361
Aromatic rings	0.410	0.062	0.062	0.297	0.347	0.347	0.113
Fraction sp^3 carbons	0.856	0.915	0.915	0.834	0.562	0.562	0.252
Net formal charge	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Sum of absolute formal charges	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Carbon fraction	0.856	0.832	0.832	0.856	0.679	0.679	0.308
Nitrogen atoms	2.278	1.984	1.984	2.203	0.293	0.293	0.075
Oxygen atoms	5.674	8.141	8.141	6.328	1.802	1.802	0.466
Sulfur atoms	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Phosphorus atoms	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Branching heavy atoms	7.632	7.734	7.734	7.094	0.293	0.293	0.538
Nonaromatic unsaturated bonds	3.556	3.250	3.250	4.969	0.224	0.224	0.707
Maximum polar-root distance	22.160	27.609	27.609	26.125	1.093	1.093	0.797
Heuristic head fraction	0.177	0.153	0.153	0.158	0.560	0.560	0.501
Heuristic interface fraction	0.160	0.191	0.191	0.172	0.581	0.581	0.230
Heuristic tail fraction	0.663	0.655	0.655	0.670	0.187	0.187	0.504
Heuristic tail branching atoms	1.306	1.719	1.719	0.750	0.207	0.207	0.278
Thiolactone; $n_R = 96, n_S = n_F = n_T = 64$							
Heavy atoms	49.271	52.141	52.141	53.000	0.302	0.302	0.393
Molecular weight	713.537	752.610	752.814	764.816	0.297	0.298	0.388
Calculated logP	9.720	10.953	10.935	11.011	0.481	0.474	0.504
Topological polar surface area	89.150	88.147	88.345	90.397	0.562	0.503	0.385
Hydrogen-bond donors	1.906	1.953	1.953	2.000	0.047	0.047	0.094
Hydrogen-bond acceptors	6.271	6.109	6.141	6.250	0.161	0.130	0.021
Rotatable bonds	35.323	37.672	37.625	37.922	0.261	0.256	0.289
Rings	0.625	0.594	0.594	0.719	0.031	0.031	0.094
Aromatic rings	0.042	0.000	0.000	0.062	0.042	0.042	0.021
Fraction sp^3 carbons	0.886	0.874	0.877	0.878	0.311	0.245	0.251
Net formal charge	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Sum of absolute formal charges	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Carbon fraction	0.828	0.843	0.843	0.842	0.741	0.713	0.663

Table 14, continued

Descriptor	μ_R	μ_S	μ_F	μ_T	W_S/s_R	W_F/s_R	W_T/s_R
Nitrogen atoms	3.188	3.031	3.047	3.156	0.156	0.141	0.031
Oxygen atoms	4.125	4.078	4.094	4.156	0.047	0.031	0.031
Sulfur atoms	1.000	1.000	1.000	1.000	0.000	0.000	0.000
Phosphorus atoms	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Branching heavy atoms	5.688	5.812	5.891	6.250	0.156	0.203	0.562
Nonaromatic unsaturated bonds	3.719	4.297	4.219	4.109	0.578	0.500	0.391
Maximum polar-root distance	25.833	27.031	26.859	26.391	0.299	0.257	0.181
Heuristic head fraction	0.151	0.147	0.147	0.147	0.195	0.212	0.224
Heuristic interface fraction	0.228	0.207	0.208	0.206	0.509	0.478	0.533
Heuristic tail fraction	0.621	0.646	0.644	0.647	0.341	0.325	0.337
Heuristic tail branching atoms	0.562	1.047	1.031	1.484	0.484	0.469	0.922

R is synthetic CAL reference, S saved predecessor selection, F current selection and T TRAIN control. Molecular weight is in Da and TPSA in Å²; other descriptors retain their count, fraction or calculated-logP units. Means and distances are point estimates. Descriptor medians, SDs, IQRs and tails were not exported by this computation and remain [descriptor distribution quantiles and tails]. Head/interface/tail fractions and polar-root distance are graph heuristics, not precursor-origin or ionization measurements.

B.9 PRODUCT AND PRECURSOR DIVERSITY AND TRAIN-RELATIVE NOVELTY

Table 15: Current product novelty and exact-component diversity.

Family / role	N /obs.	Unique exact	Novel product	Novel-comp. products	D_1	D_2	Novel unique comp.
A3	64	61	63	60	64.000	64.000	142
aldehyde	61	48	—	42 obs.	40.008	29.768	41
alkyne	61	55	—	48 obs.	51.085	43.776	47
amine_head	61	60	—	55 obs.	59.629	59.063	54
Acid-epoxide	64	63	64	61	64.000	64.000	97
amine_acid_head	63	35	—	32 obs.	28.835	23.211	23
epoxide	63	48	—	26 obs.	42.273	35.757	26
hydrophobic_acid	63	60	—	50 obs.	58.975	57.522	48
AEMA	64	63	29	12	64.000	64.000	13
AEMA	63	1	—	0 obs.	1.000	1.000	0
amine_core	63	26	—	3 obs.	23.596	21.689	3
thiol_periphery	63	45	—	11 obs.	35.882	25.606	10
Aldehyde Ugi3	64	63	63	58	64.000	64.000	71
amine_head	63	34	—	26 obs.	24.846	17.640	22
isocyanide_tail	63	5	—	0 obs.	4.814	4.653	0
oxoester_aldehyde_body_tail	63	60	—	50 obs.	58.975	57.522	49
Aldehyde Ugi4	64	64	64	60	64.000	64.000	89
amine_head	64	56	—	49 obs.	52.235	46.545	48
carboxylic_acid_tail	64	47	—	39 obs.	38.746	29.681	32
isocyanide_tail	64	11	—	2 obs.	9.090	8.127	2
oxoester_aldehyde_body_tail	64	28	—	7 obs.	19.524	13.932	7
α-Isocynoester	64	64	62	62	64.000	64.000	69
amine_head	64	27	—	9 obs.	22.385	19.321	8
coupled_ketone	64	64	—	61 obs.	64.000	64.000	61
isocyanide	64	1	—	0 obs.	1.000	1.000	0
Amine alkylation	64	62	49	39	64.000	64.000	44
amine_head	62	35	—	22 obs.	30.225	26.329	18
bromoester_arm	62	60	—	26 obs.	59.288	58.242	26
Amine-epoxide	64	64	51	42	64.000	64.000	54
amine_head	64	42	—	25 obs.	35.558	29.681	21
epoxide_tail	64	61	—	34 obs.	59.973	58.514	33
Aryl reductive amination	64	57	61	55	62.629	62.061	72
amine_head	58	20	—	45 obs.	11.071	7.410	19
coupled_aldehyde	58	57	—	54 obs.	56.630	56.067	53
Aza-Michael acrylamide	64	64	43	35	64.000	64.000	30
acceptor_tail	64	56	—	27 obs.	53.379	49.951	22
amine_head	64	28	—	13 obs.	23.303	20.277	8
Aza-Michael acrylate	64	64	48	38	64.000	64.000	36
acceptor_tail	64	57	—	25 obs.	54.104	49.951	24
amine_head	64	31	—	17 obs.	25.446	21.113	12

Table 15, continued

Family / role	N /obs.	Unique exact	Novel product	Novel-comp. products	D_1	D_2	Novel unique comp.
Disulfide Michael	64	63	61	60	64.000	64.000	88
acceptor_tail	63	63	—	59 obs.	63.000	63.000	59
amine_head	63	47	—	32 obs.	41.697	35.757	29
Epoxide/O-acylation	64	63	56	35	64.000	64.000	38
acyl_tail	63	36	—	8 obs.	22.539	13.186	8
amine_head	63	29	—	27 obs.	22.539	17.959	19
epoxide_tail	63	34	—	11 obs.	22.224	13.829	11
iPhos	64	64	55	50	64.000	64.000	56
amine_head	64	61	—	48 obs.	59.485	56.889	48
phosphate_tail	64	31	—	8 obs.	22.271	16.254	8
Ketone-isocyanide	64	64	43	33	64.000	64.000	32
amine_head	64	3	—	1 obs.	2.085	1.952	1
coupled_ketone	64	58	—	33 obs.	56.200	53.895	31
isocyanide	64	6	—	0 obs.	5.735	5.476	0
Ketone Ugi4	64	62	64	62	64.000	64.000	167
amine_head	62	47	—	39 obs.	42.038	36.264	34
carboxylic_acid	62	57	—	49 obs.	54.216	49.282	49
coupled_ketone	62	62	—	62 obs.	62.000	62.000	62
isocyanide	62	25	—	30 obs.	14.322	9.422	22
Maleate	64	64	56	55	64.000	64.000	76
amine_head	63	56	—	47 obs.	53.560	50.241	43
maleate	64	56	—	34 obs.	53.379	49.951	32
thiol_head	1	1	—	1 obs.	1.000	1.000	1
O-esterification	64	64	46	35	64.000	64.000	36
acid_tail	64	60	—	30 obs.	58.210	55.351	27
aminoalcohol_head	64	18	—	13 obs.	13.720	11.636	9
Passerini	64	64	64	36	64.000	64.000	38
aldehyde_body	64	50	—	22 obs.	44.886	38.642	18
carboxylic_acid_tail	64	29	—	10 obs.	23.619	19.883	9
isocyanide_tail	64	48	—	14 obs.	40.621	32.000	11
Preassembled thiol-yne	64	64	54	53	64.000	64.000	64
alkynoic_linker	64	4	—	0 obs.	3.917	3.835	0
amine_head	64	56	—	46 obs.	53.379	49.951	42
thiol_first	64	58	—	22 obs.	55.742	52.513	22
thiol_second	64	58	—	22 obs.	55.742	52.513	22
Reductive amination	64	61	64	61	64.000	64.000	72
amine_head	61	32	—	23 obs.	25.638	20.558	20
coupled_aldehyde	61	53	—	60 obs.	47.719	39.168	52
Thiolactone	64	64	56	48	64.000	64.000	62
acrylate_tail	64	39	—	11 obs.	25.389	14.222	11
amine_head	64	41	—	31 obs.	34.023	27.307	26
thiolactone_region	64	55	—	27 obs.	51.536	46.545	25
Global products	1408	1386	1216	1050	1406.614	1406.003	1350

Family rows report products: unique exact identities, TRAIN-novel product observations and exact products containing a globally TRAIN-novel component. Product D_1 , D_2 use all connected outputs. Indented role rows instead report exact component observations, unique components, novel-component observations and role-specific effective counts. Global deduplication gives 1,407 products; 1,216 TRAIN-novel observations represent 1,215 distinct products. Novel components occur in 1,050 exact products (1,049 distinct), or 1,081 products (1,080 distinct) under role-relative novelty. Of 3,723 exact component-role observations, 1,618 are globally novel, representing 1,350 distinct component constitutions. Full-reference novelty and exhaustive catalogue reachability remain [full-reference novelty and catalogue reachability].

B.10 GENERALIZATION BEYOND DEVELOPMENT REQUESTS

Table 16: Generalization endpoints requiring independent evaluation.

Family	Held-comp. yield	Held identity coverage	Component-disjoint L1	Source-disjoint L1	Size/repeat strata
A3	—	—	—	—	—
Acid-epoxide	—	—	—	—	—
AEMA	—	—	—	—	—

Table 16, continued

Family	Held-comp. yield	Held identity coverage	Component- disjoint L1	Source-disjoint L1	Size/repeat strata
Aldehyde Ugi3	—	—	—	—	—
Aldehyde Ugi4	—	—	—	—	—
α -Isocyanoester	—	—	—	—	—
Amine alkylation	—	—	—	—	—
Amine-epoxide	—	—	—	—	—
Aryl reductive amination	—	—	—	—	—
Aza-Michael acrylamide	—	—	—	—	—
Aza-Michael acrylate	—	—	—	—	—
Disulfide Michael	—	—	—	—	—
Epoxide/O-acylation	—	—	—	—	—
iPhos	—	—	—	—	—
Ketone-isocyanide	—	—	—	—	—
Ketone Ugi4	—	—	—	—	—
Maleate	—	—	—	—	—
O-esterification	—	—	—	—	—
Passerini	—	—	—	—	—
Preassembled thiol-yne	—	—	—	—	—
Reductive amination	—	—	—	—	—
Thiolactone	—	—	—	—	—

TRAIN-derived generated requests cannot fill these held-out endpoints. Source-qualified CAL distribution comparisons are not generation on independent requests. Independent training-seed uncertainty remains unmeasured. A metadata-only census identified 9,594 possible large-molecule test records across ten families, but admitted none for evaluation and computed no model scores. Held-reaction-family behavior remains a secondary stress test, not a hard gate.

B.11 UPSTREAM SYNTHESIS AND TERMINAL-EVIDENCE COMPLETENESS

Table 17: **Product-level computational makeability and strict secondary dossiers.** The fixed, adaptively developed 1,408-request selected cohort contains 64 requests per family. All columns retain all requested products, including the 21 nonexact-L1 products. Evidence is assessed as of 2026-09-26 01:56:45 UTC.

Family	<i>N</i>	Exact L1	L2 ready	L3 direct-only	Computationally makeable	Strict secondary
A3 amine–aldehyde–alkyne	64	61	50	1	12	1
Acid–epoxide diester	64	63	50	0	39	0
AEMA	64	63	58	0	47	13
Aldehyde Ugi 3-CR	64	63	61	1	38	1
Aldehyde Ugi 4-CR	64	64	47	18	32	0
α -Isocyanoeester dihydroimidazole	64	64	60	8	39	0
Amine alkylation	64	62	62	1	34	0
Amine–epoxide	64	64	48	7	40	1
Aryl reductive amination	64	58	42	0	9	0
Aza-Michael acrylamide	64	64	64	7	28	3
Aza-Michael acrylate	64	64	64	12	22	1
Disulfide Michael	64	63	62	0	6	0
Epoxide O-acylation	64	63	39	14	33	1
iPhos	64	64	56	2	31	2
Ketone–isocyanide amide	64	64	63	12	58	0
Ketone Ugi 4-CR	64	62	32	0	2	0
Maleate	64	64	61	1	27	0
O-esterification	64	64	39	22	39	3
Passerini	64	64	64	3	22	0
Preassembled thiol-yne	64	64	42	10	33	1
Reductive amination	64	61	61	0	0	0
Thiolactone	64	64	64	0	25	0
All requests	1408	1387	1189	119	616	27

Computational makeability requires every required component to be directly vendor-listed or connected through a complete admitted computational route to vendor-listed terminal materials. L2-ready counts products whose required components have a direct listing or admitted computational path; it does not require every route leaf to be listed. Direct-only requires all required components to be directly listed. Strict dossiers retain the original exact-source/current-item checks. Vendor-directory listings are not stock, prices or experimental synthesis. Missing listings remain unknown. The 3,723 required branches are distinct from 4,510 incorporated component occurrences: 3,488 branches are L2-ready and 2,787 computationally makeable. There are 792 products without demonstrated primary closure. Of the 616 makeable products, 606 pass the limited design checks and 604 additionally have no context flags. Independent-seed held-out route coverage remains —.

Route assessment follows the actual generated component identity. The primary computational criterion admits checked planner-generated and published-transform paths to vendor-listed leaves; it does not require an exact-substrate experimental precedent at every step. Strict dossiers remain a separate secondary outcome with their original checks. Neither outcome is an experimental yield, stock confirmation or synthesis-success probability. Route-search and listing unknowns are retained; the available searches are not exhaustive. Detailed source evidence is not transferred between reaction families. Changed molecules from the separate repair diagnostic require fresh route assessment and are not substituted into Table 17.

B.12 ARCHITECTURE, THEORETICAL ASSUMPTIONS AND SYNTHESIS GUIDANCE

Table 18: **Learning diagnostics and unmeasured architecture contrasts.** Matched retraining ablations remain separate from the completed continuation studies. The clean-CAL diagnostic uses all 1,885 admitted condition variants of 1,859 molecules, with coordinate counts as its denominator.

Contrast / readout	Result
Whole graph versus factorized; matched/generous capacity	—
Input-only versus layerwise conditioning	—
Role/core/repeat, routed adapters and PCGrad retraining ablations	—
Original / categorical66 / coarse66 raw exact	80 / 83 / 86 of 352
Original / categorical66 / coarse66 D1 exact	140 / 136 / 135 of 352
Original / categorical66 / coarse66 D1 design	111 / 110 / 114 of 352
D1 design without TRAIN-context flags	110 / 105 / 109 of 352
Clean-CAL nonconsecutive-parent errors	1,164 / 1,034 / 1,026 of 7,936
Large / small update pilot D1 exact; design	135 / 122 exact; 113 / 107 design (352)
Continuation advancement	Neither objective nor update regime advances

The coarse-parent objective averages categorical parent cross-entropy with a binary immediate-predecessor versus other-legal-parent loss at fixed weight one. Its eight fewer nonconsecutive-parent errors versus ordinary continuation do not establish a general architecture or final-quality improvement. All pilots retain the original model initializer; generation improvements coexist with family and diversity regressions. No checkpoint was selected using the best intermediate result.

Table 19: **Scoped correspondence between theory and implementation.** Nine assumptions are mapped to the implementation and existing counterchecks. This audit is not a new theorem proof or an empirical guarantee of synthesis success.

Obligation	Current evidence / limitation
Bounded representation	Declared 254 atoms / 12 closures; qualified vocabulary and masks. No claim of arbitrary graph support.
Full-support source and endpoint population	TRAIN-fitted priors and admitted records; conditional layout support remains narrower.
Population-optimal cross-entropy predictor	Finite Transformer and updates; auxiliary losses, balanced reductions and PCGrad. Ideal optimum unestablished.
Exact continuous-time limit	Finite capped sampling and 64-step diagnostics; no measured convergence or total-variation bound.
Program-fixed coordinates	Restoration and equality assertions; an implementation invariant, not molecular quality.
Terminal prediction and readout	Raw, terminal-argmax and D1 yields differ (Table 6); output maps are distinct.
Reference versus altered time law	Clean-state continuation is an objective intervention; its diagnostic gains did not establish generated-quality gains.
Independent attempts	Batch diversity/novelty selection couples requests; binomial independence is not assumed.
Forward replay and recursive evidence	Exact L1, computational makeability and strict dossiers use separate checks and denominators.
Final released-code correspondence and remaining proof checks	—

Table 20: **Synthesis-guidance gaps and separate negative activity-conditioning evidence.** The synthesis comparison remains unrun. The completed activity experiment adapts the frozen backbone in two assay/family contexts and cannot establish all-family guidance.

Validation / contrast	Result
22-family synthesis-guidance zero-weight identity	—
Matched unguided/post-hoc/in-trajectory synthesis comparison	—
Synthesis-guidance eligibility, abstention and total cost	—
Separate activity-conditioning data	550 HeLa/Ugi and 581 A549/aza-Michael constitutions
Activity fits and generation attempts	27 final fits, 6 capacity comparisons; 24,192 attempts
Genuine versus hidden denoising loss reduction	HeLa -1.109% ; A549 $+0.0213\%$
Genuine versus shuffled denoising loss reduction	HeLa -0.935% ; A549 $+0.0086\%$
High-target HeLa joint versus prior-only (exact / eligible)	438 / 53 versus 438 / 55, each of 576
High-target A549 joint versus prior-only (exact / eligible)	103 / 1 versus 104 / 1, each of 576
Primary potency gain / advancement	Not estimable / neither cell advances

Activity fits use three reused structural development folds and three seeds with genuine, hidden and shuffled labels. Shared precursors are allowed; study, cell and chemistry are confounded. None of the 64-attempt cohorts fills 16 distinct eligible selection slots, so the primary potency-gain comparison is not estimable. Exact-yield guardrails fail in one of nine HeLa and two of nine A549 blocks; diversity guards pass. Predictions are not new activity measurements, and these negative results do not qualify synthesis guidance or independent confirmation.

Architecture comparisons distinguish matched retraining from fixed-weight inference changes. We relate each theoretical assumption to the implemented vocabulary, graph support, objective, finite-step sampler and selection map. The ideal flow theorem does not certify a learned numerical sampler, a TRAIN-derived repair or a batch-coupled selection procedure. Synthesis-guidance comparisons separate unguided, post-hoc and in-trajectory arms at matched total budgets and require both a zero-guidance identity check and auditable L1/L2/L3 dossiers. Outcomes outside the declared evidence or applicability scope remain unassessed.

B.13 ASSEMBLY, STRUCTURAL AND SYNTHESIS-EVIDENCE OUTCOMES

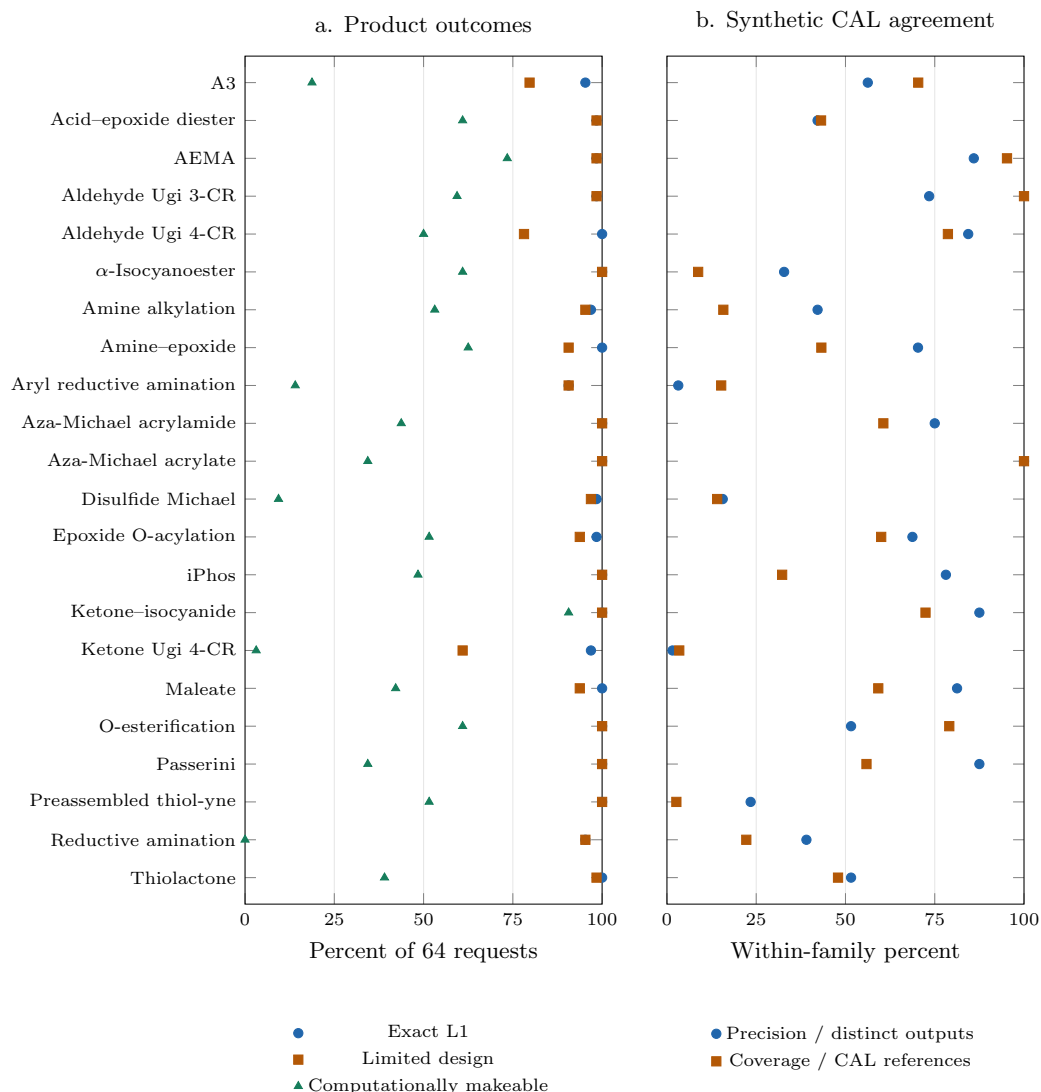


Figure 3: **Current development results across 22 reaction families.** Exact L1, limited-design and computational-makeability counts use all 64 selected requests per family. ECFP reference precision and CAL coverage use separate generated/reference denominators. The selected cohort is adaptively developed from one checkpoint; independent-seed and independent held-out confirmation remain unmeasured. Assembly consistency, limited checks, reference similarity and route evidence are separate outcomes.

B.14 REPRESENTATIVE STRUCTURES AND UNRESOLVED FAILURES

a. Seeded all-family atlas

One or more prespecified random products per family, with exact precursors and evidence depth.

[result]

b. Paired failure analysis

Original and repaired graphs; edits, remaining concerns and unchanged source checks.

[result]

Figure 4: **Structures and failure analysis across reaction families.** A prespecified all-family atlas remains unmeasured. Existing targeted failure galleries are developmental diagnostics, not a random or method-masked quality sample. Atom origins, component novelty and L1/L2/L3 evidence require separate annotations. Molecular drawings represent exact graphs and do not establish three-dimensional stability.

The structure analysis pairs original and repaired graphs and includes oversized rings, malformed cages, missing retained heads and joint atom/bond errors, alongside source-positive controls. Graph and drawing hashes distinguish molecular changes from depiction corrections. Random samples and examples selected to illustrate a specific failure are identified separately.

B.15 FAILURE ACCOUNTING AND REPRODUCIBILITY

The analysis retains failed requests and runs, unsupported chemistry, unavailable references, source conflicts, null contrasts and search censoring. Source, configuration, checkpoint, policy and split hashes link each result to its executable analysis and verification receipt. We distinguish TRAIN-derived proposal and selection diagnostics from evaluation against independent references. Passing limited structural checks does not establish complete chemical quality, stability, ionization or delivery activity; context flags and unresolved cases remain part of the reported outcome.

C THEORETICAL ANALYSIS AND PROOFS

The results establish separate properties of the construction. Bounded completeness concerns the state representation. Catalogue reachability concerns exact precursor inputs. The population argument connects a reference denoising loss to an ideal continuous-time law. Budget and route results concern computational procedures. None establishes experimental synthesis success or positive learned probability for every representable graph.

C.1 GRAPH STATES AND PROGRAM COMPATIBILITY

The graph space $\mathfrak{G}_{N,K_{\text{cl}}}$ is defined in Section 2.3, with finite atom and bond alphabets and integers $N \geq 1, K_{\text{cl}} \geq 0$. Molecular validity is an additional predicate on that space. The raw space $\bar{\mathfrak{X}}_{N,K_{\text{cl}}}$ contains padded categorical states.

Definition C.1 (Well-formed sparse state). A state belongs to $\bar{\mathfrak{X}}_{N,K_{\text{cl}}}$ when, for some $1 \leq n \leq N$, its active atoms are exactly positions $1, \dots, n$ with labels in Σ_{atom} ; position 1 is the sole root with null parent and parent-bond states; and every $2 \leq i \leq n$ has an earlier parent $1 \leq \pi_i < i$ and a bond label in Σ_{bond} . Each active closure has endpoints $1 \leq e^L < e^R \leq n$, a bond label in Σ_{bond} , and an edge distinct from every tree edge and other closure. Active closures precede unused slots and are sorted by endpoint pair. All unused atom, pointer and bond coordinates carry their designated null values.

The decoder reconstructs the labeled edges of a well-formed state and returns $\perp_{\text{dec}}$ otherwise. It is therefore a total map into $\mathfrak{G}_{N,K_{\text{cl}}} \cup \{\perp_{\text{dec}}\}$.

Lemma C.2 (Tree residual count). *For a finite nonempty connected graph $G = (V, E)$ and any spanning tree T , $|E \setminus E(T)| = |E| - |V| + 1 = \beta(G)$.*

Proof. The tree contains $|V| - 1$ edges, all in E . Subtracting this count from $|E|$ gives the result, including the single-vertex case. $\square$

Theorem C.3 (Bounded completeness of the sparse representation). *For the spaces in Definition C.1, the restriction $\text{Dec} : \bar{\mathfrak{X}}_{N,K_{\text{cl}}} \rightarrow \mathfrak{G}_{N,K_{\text{cl}}}$ is well defined and surjective. Thus every well-formed state decodes to a graph in the declared support, and every graph in that support has an encoding that decodes to its constitutional isomorphism class.*

Proof. We first reconstruct a graph from a state, then construct a state for an arbitrary graph. In a well-formed state with n active atoms, every parent pointer strictly decreases its index. Following parents must reach position 1, so the $n - 1$ parent edges form a connected tree. Adding $k \leq K_{\text{cl}}$ distinct, non-loop closure edges preserves connectedness and gives cycle rank $(n - 1 + k) - n + 1 = k$. The label and padding conditions ensure the decoded graph belongs to $\mathfrak{G}_{N,K_{\text{cl}}}$.

Conversely, choose a representative of any $G \in \mathfrak{G}_{N,K_{\text{cl}}}$ and a rooted spanning tree. Enumerate vertices in a traversal that places every parent before its children. Store their labels, parent indices and tree-bond labels. Lemma C.2 leaves exactly $\beta(G) \leq K_{\text{cl}}$ residual edges; store each with increasing endpoint indices and sort these triples. Pad all unused coordinates with null states. This state satisfies Definition C.1, and the vertex enumeration is a label-preserving isomorphism between its decoded graph and G . $\square$

This result concerns the unrestricted representation. For program-specific alphabets $\mathcal{X}_i(P)$ and fixed values $x_{P,i}$, define

$$\Omega_P = \{x \in \bar{\mathfrak{X}}_{N,K_{\text{cl}}} : x_i \in \mathcal{X}_i(P) \forall i, \quad x_i = x_{P,i} \forall i \in \mathcal{F}(P)\}. \quad (33)$$

Its nonempty well-formed intersection defines $\mathcal{P}_{\text{adm}} = \{P : \bar{\mathfrak{X}}_{N,K_{\text{cl}}} \cap \Omega_P \neq \emptyset\}$. Nonemptiness alone establishes neither chemical validity nor satisfaction of every morphology constraint. Corpus admission additionally checks the annotated encoding, program alignment and exact decode round trip. Representational completeness makes no claim that every graph is encodable under every P .

C.2 EXACT ASSEMBLY AND CATALOGUE REACHABILITY

Definition A.1 distinguishes the complete input preimage $\mathcal{E}_c^*(G)$ from recovered input-trace pairs $\widehat{\mathcal{E}}_c(G)$. Fix a registered chemistry c and role-specific catalogues $\mathcal{C} = (\mathcal{C}_r)_r$. Let $\mathfrak{C}_c(\mathcal{C}) \subseteq \mathfrak{C}_c$ contain the admissible precursor inputs whose every role-assigned component belongs to its catalogue. The reachable product set is

$$\mathcal{G}_{\text{cat}}(c, \mathcal{C}) = \bigcup_{\mathbf{C} \in \mathfrak{C}_c(\mathcal{C})} \mathcal{F}_c(\mathbf{C}). \quad (34)$$

This definition includes all registered outcomes of the admissible inputs, not only products found by a bounded search.

Proposition C.4 (Finite-component reachability). *If $\mathfrak{C}_c(\mathcal{C})$ and each $\mathcal{F}_c(\mathbf{C})$ are finite, then*

$$|\mathcal{G}_{\text{cat}}(c, \mathcal{C})| \leq \sum_{\mathbf{C} \in \mathfrak{C}_c(\mathcal{C})} |\mathcal{F}_c(\mathbf{C})| \leq m_{\text{max},c}(\mathcal{C}) |\mathfrak{C}_c(\mathcal{C})|, \quad (35)$$

where $m_{\text{max},c}(\mathcal{C})$ is the largest outcome count, or zero for an empty input set. For every $G \in \mathfrak{G}_{N,K_{\text{cl}}}$,

$$G \in \mathcal{G}_{\text{cat}}(c, \mathcal{C}) \iff \mathcal{E}_c^*(G) \cap \mathfrak{C}_c(\mathcal{C}) \neq \emptyset. \quad (36)$$

Proof. The cardinality of a finite union is at most the sum of its member-set cardinalities; bounding each summand by its maximum proves both inequalities. If there are no inputs, all three quantities are zero. Membership in the union means that some catalogue input has G as a forward product, which is exactly the stated intersection condition. $\square$

Definition C.5 (Strong catalogue escape). An exactly assemblable graph strongly escapes $\mathcal{C}$ under c when every exact input uses at least one out-of-catalogue component. Its indicator is

$$\text{Escape}^*(G; c, \mathcal{C}) = \mathbf{1}[\mathcal{E}_c^*(G) \neq \emptyset, \quad \mathcal{E}_c^*(G) \cap \mathfrak{C}_c(\mathcal{C}) = \emptyset]. \quad (37)$$

A catalogue assembler whose outputs lie in $\mathcal{G}_{\text{cat}}(c, \mathcal{C})$ therefore has zero strong-escape yield. For a verified exact-L1 product, strong escape is equivalent to $G \notin \mathcal{G}_{\text{cat}}(c, \mathcal{C})$. This can be checked against a complete canonical product set or by exhaustive exclusion of all catalogue-only decompositions. A bounded reverse search cannot establish that exclusion from one novel witness.

The reported experiments did not construct a complete common-reference reachable product set. Their component-novelty metric concerns the selected verified input. Whole-graph generation removes catalogue membership from the generative representation; that architectural property alone proves neither strong escape nor positive probability of any particular out-of-catalogue graph.

C.3 CONDITIONAL FLOW AND IDEAL POPULATION RECOVERY

We prove Theorem 3.1 for the reference objective in Equation 7. Let $X_\star = \text{Enc}_P(G, \alpha)$ denote a clean training endpoint, and distinguish it from the time-indexed process X_t . The training-path measure is

$$\nu(dG, d\alpha, dP, dt, dx) = \mu_{\text{bal}}(dG, d\alpha, dP) \mathcal{U}_{(0,1)}(dt) q_{t|1}^P(dx \mid \text{Enc}_P(G, \alpha)). \quad (38)$$

For fixed P , the encoded endpoint law q_1^P induces

$$q_t^P(x) = \sum_{x_1} q_1^P(x_1) q_{t|1}^P(x \mid x_1), \quad \rho_i(a \mid x, t, P) = \Pr_\nu[X_{\star,i} = a \mid X_t = x, t, P]. \quad (39)$$

Full source support on the finite allowed alphabets makes $q_t^P(x) > 0$ for every $x \in \Omega_P$ and $t < 1$. At $t = 0$ the path is q_0^P ; at $t = 1$ it is q_1^P .

Conditional and marginal rates. For $i \in \mathcal{V}(P)$ and $a, u, v \in \mathcal{X}_i(P)$, write $s_{t,i}^P(u | a) = (1-t)q_{0,i}^P(u) + t\mathbf{1}\{u=a\}$ and $\dot{s}_i^P(u | a) = \mathbf{1}\{u=a\} - q_{0,i}^P(u)$. The conditional rate used in the ideal analysis is the R^* construction of Qin et al. (2025), restricted to the allowed alphabet:

$$R_{t,i}^*(u, v | a, P) = \frac{[\dot{s}_i^P(v | a) - \dot{s}_i^P(u | a)]_+}{|\mathcal{X}_i(P)| s_{t,i}^P(u | a)}, \quad u \neq v, \quad 0 \leq t < 1. \quad (40)$$

Diagonal entries are the negative row sums. Denominators are positive for $t < 1$; no value at $t = 1$ is needed. Fixed coordinates have zero rates. For $x \neq y$ in Ω_P , the joint conditional and predicted rates are

$$R_t^*(x, y | x_1, P) = \sum_{i \in \mathcal{V}(P)} \mathbf{1}\{y_{-i} = x_{-i}\} R_{t,i}^*(x_i, y_i | x_{1,i}, P), \quad (41)$$

$$R_t^\theta(x, y | P) = \sum_{i \in \mathcal{V}(P)} \mathbf{1}\{y_{-i} = x_{-i}\} R_{t,i}^\theta(x_i, y_i | x, P). \quad (42)$$

Here x_{-i} omits coordinate i , and Equation 6 defines the predicted coordinate rate. Diagonal joint rates again make each row sum to zero. States differing at two or more coordinates have zero instantaneous transition rate.

Proposition C.6 (Program-fixed invariance). *Fix $P \in \mathcal{P}_{\text{adm}}$ and an endpoint in Ω_P . Corruption by Equation 5 stays in Ω_P . The ideal CTMC started from q_0^P stays there for $t < 1$ whenever its coordinate predictions are probability distributions on the allowed alphabets. Numerical sampling in Equation 9, started from q_0^P and using the allowed coordinate alphabets, satisfies $\Pr[Z_k \in \Omega_P \text{ for all } 0 \leq k \leq L_{\text{samp}}] = 1$.*

Proof. The corruption law has zero mass outside Ω_P , and the ideal generator has no transition to its complement. Each numerical coordinate draw stays in its allowed alphabet and Π_P restores the prescribed fixed values. Induction on k proves the discrete statement. These arguments do not require an accurate denoiser and do not imply molecular validity. $\square$

Proposition C.7 (Posterior minimizer of the reference loss). *Under the reference training measure ν , assume unrestricted measurable coordinate predictors and positive weights $w_i(P)$ for each variable coordinate. Every population minimizer of Equation 7 satisfies $d_{\theta,i}(a | x, t, P) = \rho_i(a | x, t, P)$ for all $a \in \mathcal{X}_i(P)$, ν -almost surely on contexts with $i \in \mathcal{V}(P)$.*

Proof. Condition on (X_t, t, P) and a variable coordinate. Since $w_i(P)$ is positive and does not depend on its clean target after this conditioning, its conditional loss is

$$w_i(P) [H(\rho_i) + D_{\text{KL}}(\rho_i \| d_{\theta,i})]. \quad (43)$$

The entropy is independent of the predictor, and the divergence is nonnegative with equality only at $d_{\theta,i} = \rho_i$. An unrestricted class can attain all these pointwise minima together. A population minimizer must attain them almost surely, since a positive-measure excess would increase the integrated loss. Fixed coordinates contribute no term. $\square$

Corollary C.8 (Population recovery of the local rate). *For the reference population minimizer in Proposition C.7, $R_t^{\theta^*}(x, y | P)$ equals the conditional expectation of $R_t^*(x, y | X_*, P)$ given $(X_t = x, t, P)$, almost surely under the path measure.*

Proof. Each local conditional rate depends on the endpoint only through its coordinate $X_{*,i}$. Substituting the posterior marginal into Equation 6 therefore computes the required conditional expectation. Summing over coordinates gives the joint rate. $\square$

Proof of Theorem 3.1. We verify the forward equation first for one coordinate, then for the conditional product path, and finally after averaging over endpoints. Fix P , x_1 and i ; suppress these indices in s_t and $\dot{s}$. For each v and $t < 1$, the net flux of Equation 40 is

$$\sum_u s_t(u) R_t^*(u, v) = \frac{1}{|\mathcal{X}_i(P)|} \sum_u ([\dot{s}(v) - \dot{s}(u)]_+ - [\dot{s}(u) - \dot{s}(v)]_+) \quad (44)$$

$$= \dot{s}(v) - \frac{1}{|\mathcal{X}_i(P)|} \sum_u \dot{s}(u) = \dot{s}(v). \quad (45)$$

The last equality uses $\sum_u \dot{s}(u) = 0$. The product rule then gives the conditional joint forward equation because the generator changes one coordinate at a time. Averaging over q_1^P and using Bayes' rule yields

$$\partial_t q_t^P(y) = \sum_{x, x_1} q_1^P(x_1) q_{t|1}^P(x | x_1) R_t^*(x, y | x_1, P) \quad (46)$$

$$= \sum_x q_t^P(x) \mathbb{E}[R_t^*(x, y | X_*, P) | X_t = x, t, P]. \quad (47)$$

Corollary C.8 identifies the last conditional expectation with the learned ideal rate for almost every t . On each interval $[0, 1 - \eta]$, $0 < \eta < 1$, source positivity bounds all denominators away from zero. The rates are bounded measurable functions on a finite state space, so the forward equation has a unique solution for initial law q_0^P . Since q_t^P is positive throughout Ω_P , almost-sure equality under the path measure implies rate equality at every state for Lebesgue-almost every t . Changes on the exceptional times do not change the integral forward equation. The exactly simulated process therefore has law q_t^P .

The finite product path converges to the endpoint point mass as $t \uparrow 1$. Summing over its finite endpoint support gives $q_t^P \rightarrow q_1^P$ in total variation. This establishes a limit of marginal laws; a pathwise value at the singular endpoint is not required. Exact decode round trip maps each endpoint to its source constitution, proving $\text{Dec}_\# q_1^P = \mu_G^P$. The assumed endpoint validity and verified L1 give the probability-one conclusion. If there are no variable coordinates, the compatible state space is a singleton and the same conclusions hold for the constant process. $\square$

Corollary C.9 (Decoder contraction and failure probability). *Let q_1^P satisfy the endpoint assumptions of Theorem 3.1. Let $\tilde{q}^P$ be any law on $\bar{\mathcal{X}}_{N, K_{\text{el}}} \cup \{\perp_{\text{term}}\}$, with q_1^P extended by zero mass at $\perp_{\text{term}}$ and $\text{Dec}(\perp_{\text{term}}) = \perp_{\text{dec}}$. Define $\mathcal{B}_P$ as states whose decode fails, is molecularly invalid or fails verified L1. Then*

$$\|\text{Dec}_\# \tilde{q}^P - \mu_G^P\|_{\text{TV}} \leq \|\tilde{q}^P - q_1^P\|_{\text{TV}}, \quad \tilde{q}^P(\mathcal{B}_P) \leq \|\tilde{q}^P - q_1^P\|_{\text{TV}}, \quad (48)$$

where $\|p - q\|_{\text{TV}} = \frac{1}{2} \sum_x |p(x) - q(x)|$.

Proof. Every decoder-output event has a preimage in the raw space, so its probability difference is bounded by the raw-state total variation. Taking the supremum over output events proves the first bound. The endpoint assumptions give $q_1^P(\mathcal{B}_P) = 0$; applying the same event bound to $\mathcal{B}_P$ proves the second. $\square$

The corollary can be applied to the law *after* numerical sampling and terminal readout. It does not estimate that law's distance from q_1^P . In particular, error before the learned terminal readout cannot be substituted without controlling the readout's effect.

C.4 ACCEPTED COUNTS UNDER A FIXED SAMPLING BUDGET

Fix a trained model, requested chemistry c and its program-sampling policy. An attempt outcome Y contains its sampled program P and decoded graph or typed failure. Let Q be the common law of these outcomes and define

$$A(Y) = \mathbf{1}\{G \text{ is valid and } \widehat{\text{L1}}(G, c) = 1\}, \quad z_Q = \mathbb{E}_Q[A(Y)], \quad (49)$$

with $A(Y) = 0$ for failures. This definition permits a fresh morphology-specific P on every attempt. A comparison with a single fixed P is the corresponding special case.

Proposition C.10 (Exact law of matched-budget post-hoc filtering). *For an integer $B \geq 0$ and independent attempts with common law Q , the accepted-attempt count $N_B = \sum_{b=1}^B A(Y_b)$ is binomial with parameters (B, z_Q) . For all $z_Q \in [0, 1]$,*

$$\mathbb{E}[N_B] = Bz_Q, \quad \text{Var}(N_B) = Bz_Q(1 - z_Q), \quad \Pr[N_B = 0] = (1 - z_Q)^B. \quad (50)$$

For $B = 0$ the last probability is one, including when $z_Q = 1$. If $z_Q > 0$, the conditional law of an accepted attempt is $Q_{\text{post}}(y) = Q(y)A(y)/z_Q$, and the number T_m of attempts needed for $m \geq 1$ acceptances satisfies $\mathbb{E}[T_m] = m/z_Q$. For two proposal laws under the same attempt budget, $\mathbb{E}[N_B^{(1)} - N_B^{(0)}] = B(z_{Q_1} - z_{Q_0})$.

Proof. Each acceptance indicator is Bernoulli with parameter z_Q , and independence makes their sum binomial. Its expectation and variance give the first two identities; all B attempts fail with probability $(1 - z_Q)^B$. For $z_Q > 0$, conditioning on acceptance gives the normalized law by Bayes’ rule. The waiting times between acceptances are independent geometric variables on $\{1, 2, \dots\}$ with mean $1/z_Q$; summing m of them proves the waiting-time formula. Subtracting the expected counts proves the comparison identity, without requiring independence between the two arms. $\square$

If $z_Q = 0$, accepted outcomes have no conditional law and $T_m = \infty$ almost surely for $m \geq 1$. The proposition counts accepted attempts, including duplicates. It does not establish that program conditioning increases acceptance; that is an empirical comparison. If requested programs are fixed in unequal strata, independent acceptance indicators generally have different probabilities and their sum is Poisson-binomial. Retraining variation is separate from both sampling laws.

C.5 ROUTE TERMINATION AND EVIDENCE COMPLETENESS

Lemma C.11 (Termination of bounded route assessment). *Fix a route contract $\mathfrak{B}$ with finite maximum depth $D_{\mathfrak{B}}$. Assume finitely many proposed expansions per molecule and reactants per expansion, a terminating evidence and forward-verification call for each proposal, and a search that assesses each proposed expansion finitely many times. For an admitted root trace, recursive assessment terminates with a dossier or typed failure in $\mathfrak{D} \cup \mathfrak{F}_{\mathfrak{B}}$.*

Proof. Induct on remaining depth. At depth zero, the terminating availability query either admits a terminal or returns a failure without further expansion. At positive remaining depth, only finitely many proposals are assessed, each using terminating local calls. Every proposed child has one fewer remaining level, so its assessment terminates by induction. A finite number of finite child assessments and proposals therefore terminates at the parent. The same argument applies if a finite outer root loop is used. Rejecting ancestor identities prevents circular dependencies from being admitted; the finite depth and no-infinite-retry assumptions establish termination. $\square$

Implementation invariant C.12 (Synthesis-dossier admission). *Under the frozen contract $\mathfrak{B}$, a successful assessor return $T \in \mathfrak{D}$ is admitted only after checking $\text{Complete}_{\mathfrak{B}}(G, T, P) = 1$. Every other outcome is recorded as a typed failure. This is the admission rule in Algorithm 1, not a theorem that bounded search finds every supported chemical route.*

The predicate checks molecular validity, a unique verified root trace, a well-formed finite tree, evidence admission and forward replay for every preparation step, and availability for every leaf. Its factors are defined in Appendix A.6. A failure establishes only that the assessor did not return a complete dossier under the recorded contract. It does not exclude a route under a larger budget, other admitted chemistry or another supplier snapshot.

D ASSEMBLY FAMILIES AND REACTION SCHEMES

The assembly schemes depict the chemistry introduced in Section 2.2. The 22-family structure atlas pairs products with precursor origins and exact assembly traces (Figure 4).

D.1 ASSEMBLY SCHEMES

The precursor roles and assembly patterns are summarized in Table 21.

Table 21: **Registered assembly families and precursor roles.** Source role names are retained; a family can contain multiple admitted program variants and role aliases. A list of role names does not specify that every role occurs in a single reaction. The atom-mapped program determines the participating components and transformation order.

Family	Registered source roles
A3 amine–aldehyde–alkyne	aldehyde, alkyne, amine head
Acid–epoxide diester	amine acid head, epoxide, hydrophobic acid
AEMA	AEMA, amine core, thiol periphery
Aldehyde Ugi 3-CR	aldehyde side, amine side, isocyanide side
Aldehyde Ugi 4-CR	aldehyde, amine, carboxylic acid, isocyanide
α -Isocyanoester dihydroimidazole	amine head, coupled ketone, isocyanide
Amine alkylation	amine head, bromoester arm
Amine–epoxide	amine head, epoxide tail
Aryl reductive amination	amine head, coupled aldehyde
Aza-Michael acrylamide	acceptor tail, amine head
Aza-Michael acrylate	acceptor tail, acrylate, alkyl acrylate, amine, amine head, amine nucleophile, amine precursor, conjugated carbonyl acceptor
Disulfide Michael	amine head, disulfide acceptor
Epoxide O-acylation	acyl tail, amine head, epoxide tail
iPhos	amine head, phosphate tail
Ketone–isocyanide amide	amine head, coupled ketone, isocyanide
Ketone Ugi 4-CR	amine, amine head, carboxylic acid, coupled ketone, isocyanide, ketone
Maleate	amine head, maleate
O-esterification	acid tail, aminoalcohol head
Passerini	aldehyde tail, amine acid head, isocyanide tail
Preassembled thiol-yne	alkynoic linker, amine head, thiol tail
Reductive amination	amine head, coupled aldehyde
Thiolactone	acrylate tail, amine head, thiolactone region

D.2 REACTION SCHEMES ACROSS THE ASSEMBLY FAMILIES

Figures 5–15 show the registered transformations for all 22 families. The companion schemes use exact-replay-verified representative assemblies. Unchanged hydrocarbon fragments are abbreviated by *R* labels local to each arrow; heteroatoms, ring systems and the connecting molecular framework are retained. Ordered stages use the preceding intermediate after the indicated repeated incorporations. Auxiliary reagents, salts, work-up and stereochemistry absent from the constitutional record are omitted. The drawings specify reaction-enumerated support without assigning experimental outcomes to the depicted combinations. Precursor-origin boundaries can differ from the functional headgroup, linker and tail regions introduced in Section 2.1. An L1 pass requires forward replay on the recovered precursors to reproduce the complete constitutional lipid graph exactly.

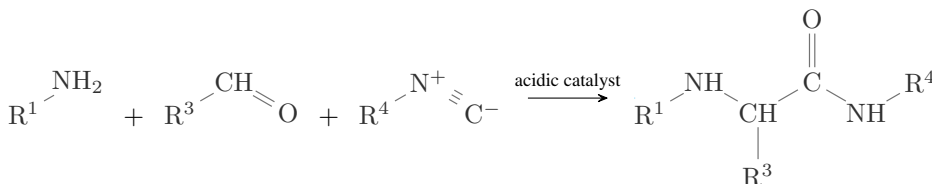
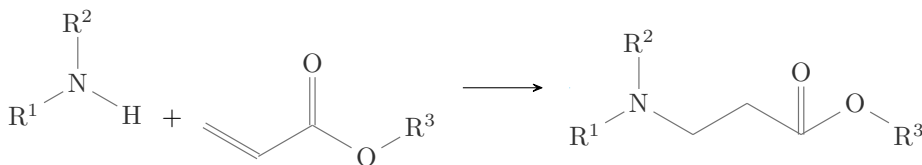
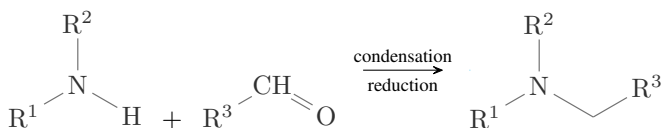
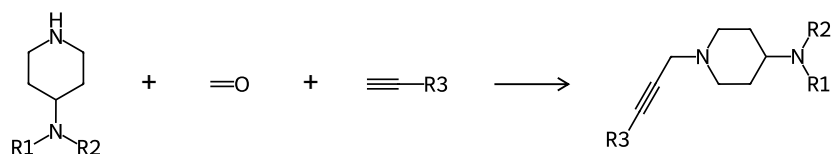
Ugi three-component reaction*amine head, aldehyde and isocyanide combine in one step; no carboxylic-acid reactant component***Repeated aza-Michael addition***one addition shown; repeat at registered N-H sites; the ester is unchanged***Repeated reductive amination***condensation and reduction form the carbon-nitrogen linkage*

Figure 5: **Representative assembly chemistries.** Monochrome skeletal drawings show the reactants and products of each registered assembly. Hydrogens are drawn only on the atoms whose substitution decides whether a product can be decomposed back into precursors: the head N-H, and the α -carbon C-H that the Ugi reaction takes from the aldehyde. R^i stand for the source-linked components of the frozen library, which are larger than the fragments drawn here. The Ugi variant has no carboxylic-acid reactant component.

a A3 amine–aldehyde–alkyne

A3 coupling

**b Aldehyde Ugi four-component assembly**

Ugi 4-CR

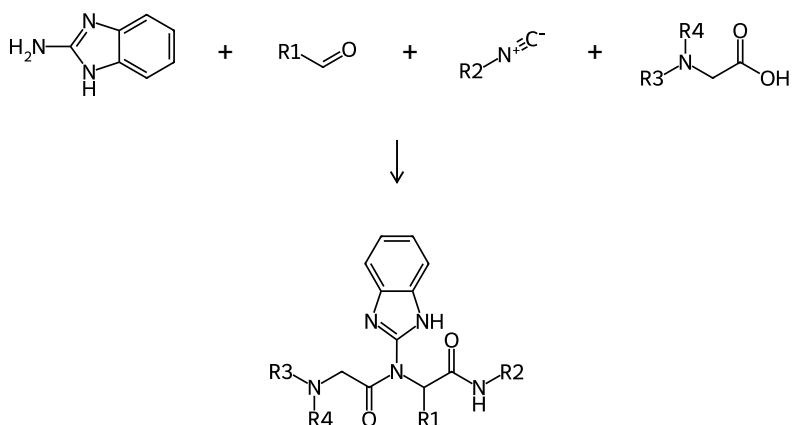


Figure 6: **Multicomponent assembly: A3 and Ugi four-component reactions.** **a**, A3 coupling creates C–N and C–C bonds while retaining the alkyne. **b**, Aldehyde Ugi 4-CR combines an amine, aldehyde, isocyanide and carboxylic acid. The acid supplies the N-acyl unit in the four-component reaction.

a Ketone Ugi four-component assembly

Ugi 4-CR

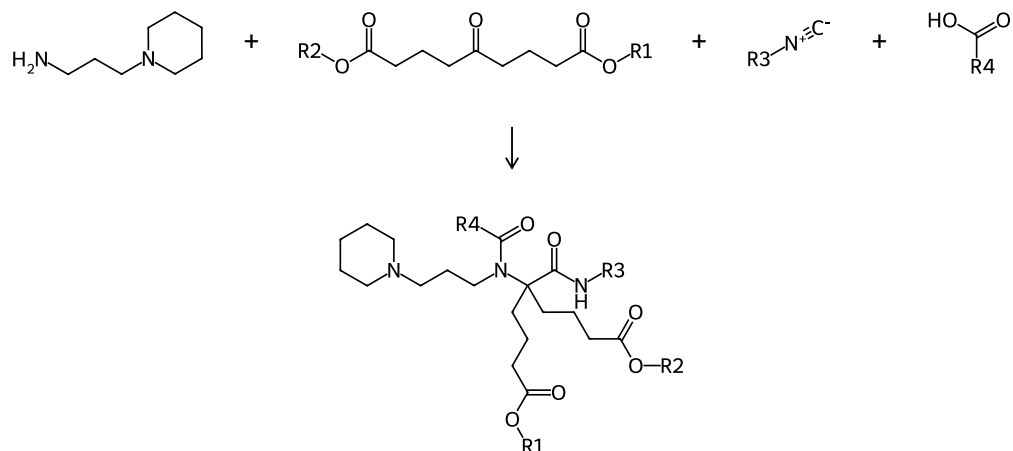
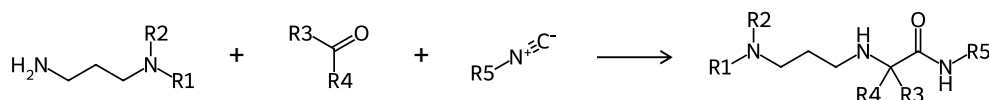


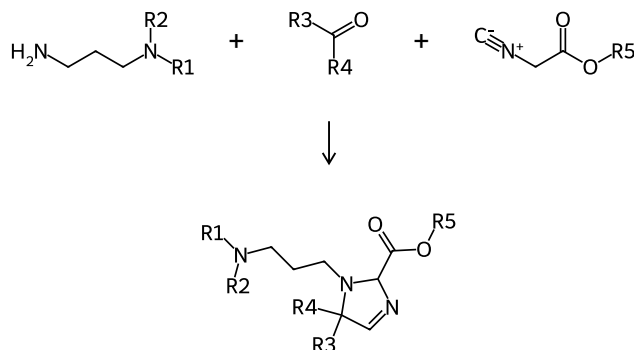
Figure 7: **Ketone Ugi four-component assembly.** The ketone reacts with an amine, isocyanide and carboxylic acid. Both carbon substituents remain attached to the ketone-derived center; the preassembled ester-bearing arms are retained, and the acid supplies the N-acyl unit.

a Ketone–isocyanide aminoamide

Three-component aminoamide assembly

**b α -Isocyanoester dihydroimidazole**

Condensation and cyclization

**c Passerini three-component assembly**

Passerini 3-CR

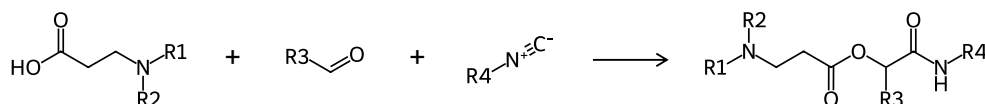
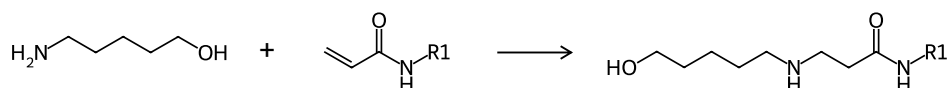


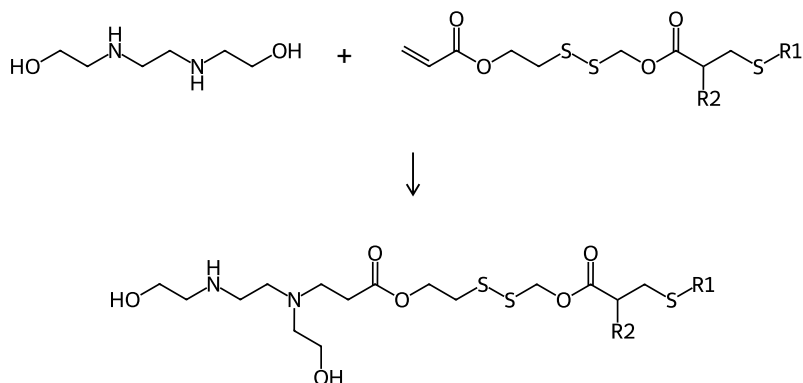
Figure 8: Multicomponent assembly: aminoamide, dihydroimidazole and Passerini reactions. **a**, The ketone–isocyanide branch forms an aminoamide from three components. **b**, An α -isocyanoester participates in dihydroimidazole ring formation, retaining its ester. **c**, Passerini assembly produces an α -acyloxy amide from an aldehyde, acid and isocyanide; the acid-derived acyl group attaches through oxygen.

a Aza-Michael addition to acrylamide

Aza-Michael addition; 2 incorporated events, first shown

**b Disulfide Michael addition**

Aza-Michael addition; S-S retained; 2 incorporated events, first shown

**c Maleate addition**

Aza-Michael addition (N-addition branch)

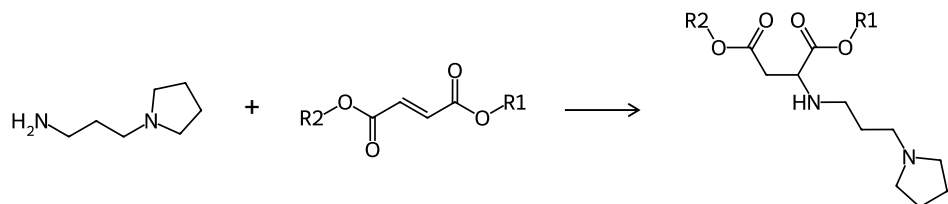
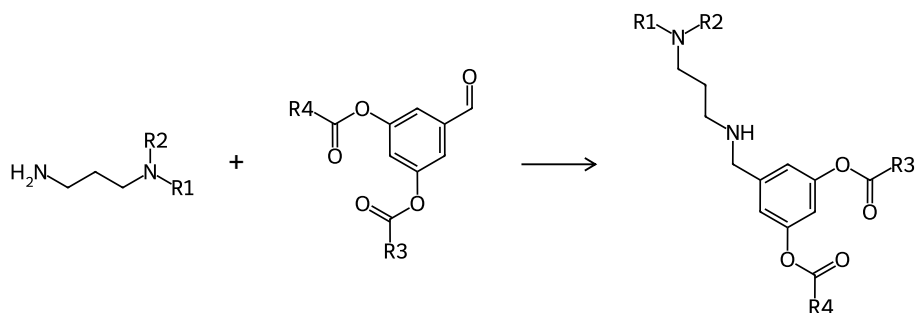


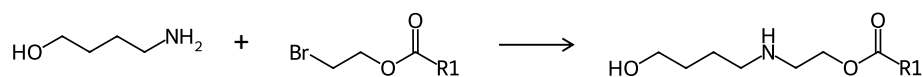
Figure 9: **Conjugate-addition families.** **a**, Amine addition to an acrylamide retains the amide interface. **b**, Addition to a disulfide-containing acrylate retains its S-S bond. **c**, The illustrated maleate program uses the N-addition branch and retains both ester groups. For repeated programs, one event is shown and the incorporated-event count is indicated. The aza-Michael acrylate reaction is shown in Figure 5.

a Aryl reductive amination

Condensation, then reduction

**b Amine alkylation**

N-alkylation; 2 incorporated events, first shown

**c Amine-epoxide opening**

Amine-epoxide ring opening; 2 incorporated events, first shown

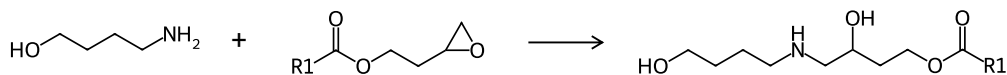
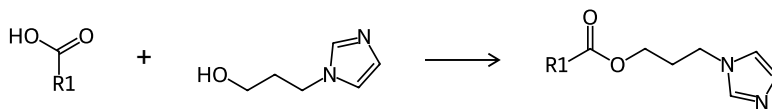


Figure 10: **Nitrogen functionalization.** **a**, Aryl reductive amination combines condensation with reduction. **b**, N-alkylation replaces the alkyl bromide leaving group. **c**, Amine attack opens an epoxide and retains the resulting alcohol. Repeated incorporations are labeled.

a O-Esterification

O-esterification

**b iPhos ring opening**

Cyclic-phosphate ring opening; deprotonated product

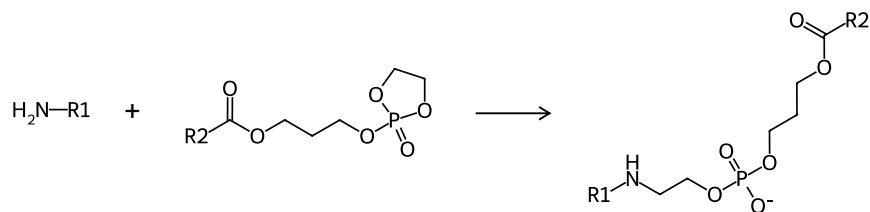
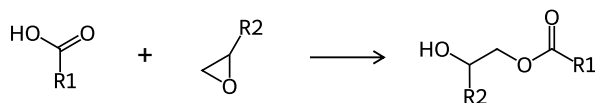


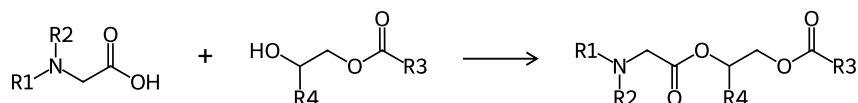
Figure 11: **Esterification and phosphate opening.** **a**, O-esterification joins an aminoalcohol to an acid while retaining the basic nitrogen. **b**, iPhos assembly opens a C–O bond of a cyclic phosphate to form the N–CH₂CH₂–O–P connection. Its deprotonated product representation is preserved.

a Acid-epoxide opening, then esterification

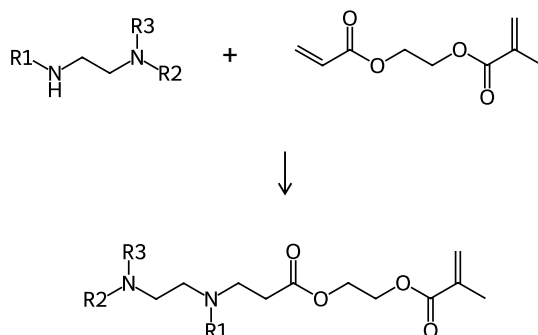
1. Acid-epoxide opening



2. O-esterification

**b AEMA: aza addition, then thiol addition**

1. Aza-Michael at acrylate



2. Thiol addition at methacrylate

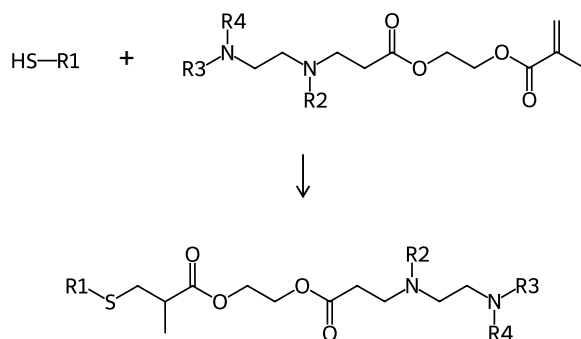
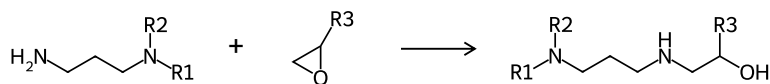


Figure 12: **Ordered assembly through hydroxy-ester and dual-acceptor intermediates.** **a**, Acid-epoxide opening first produces a hydroxy ester; subsequent esterification attaches the aminoacid head through its carboxyl group. **b**, The AEMA program consumes the acrylate in an aza-Michael addition, then the remaining methacrylate in a thiol addition. Each second-stage reactant is the corresponding first-stage intermediate; *R* abbreviations are local to each arrow.

a Amine–epoxide opening, then O-acylation

1. Amine–epoxide opening; 2 incorporated events, first shown



2. O-acylation; 2 incorporated events, first shown

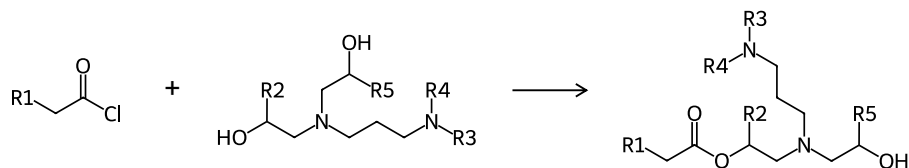
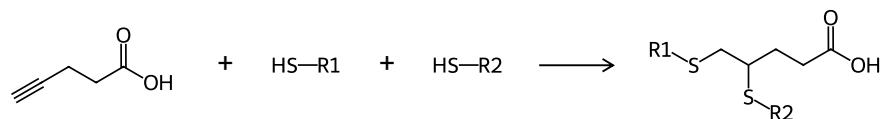


Figure 13: **Epoxide opening followed by O-acylation.** Amine–epoxide additions first form aminoalcohol arms. The resulting alcohols then react with acyl chlorides to form ester linkages. The representative program incorporates two epoxides and two acyl groups; one event from each stage is drawn, with its multiplicity indicated.

a Thiol-yne tail assembly, then amidation

1. Bis-thiol-yne addition



2. Amidation

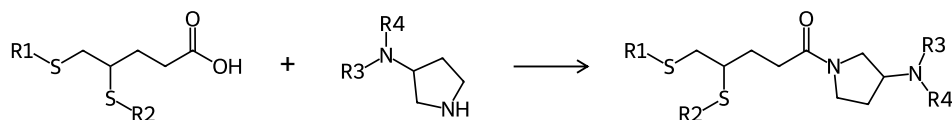
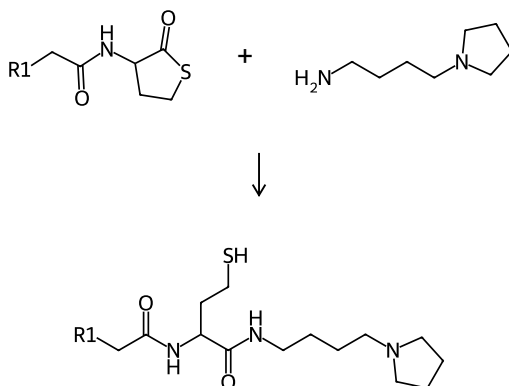


Figure 14: **Thiol-yne tail assembly followed by amidation.** Two identical thiols ($R_1 = R_2$ within each arrow) add across a terminal alkyne to form the bis(thioether) acid, which then undergoes amidation. The first arrow denotes the net double addition.

a Thiolactone aminolysis, then thiol-Michael addition

1. Thiolactone aminolysis



2. Thiol-Michael addition

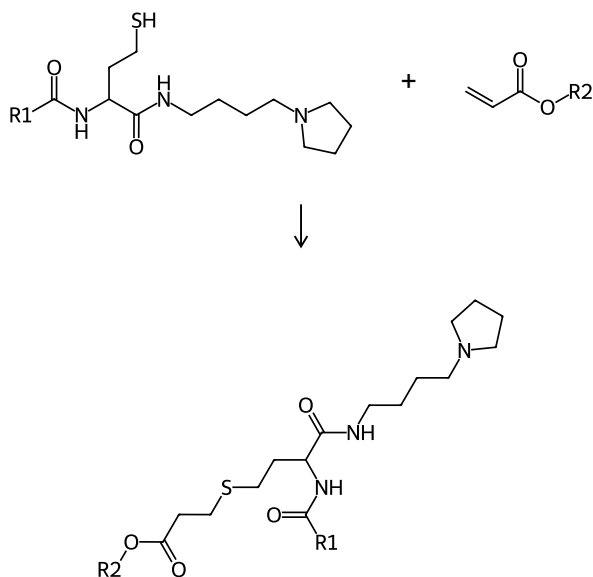


Figure 15: **Thiolactone aminolysis followed by thiol-Michael addition.** Aminolysis opens the thiolactone to form an amide and release a thiol. The subsequent addition to an acrylate forms a thioether. The retained sulfur framework distinguishes this transformation from disulfide formation.